

# EQUIVARIANT ASPECTS OF DE-COMPLETING CYCLIC HOMOLOGY

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ABSTRACT. Derived de Rham cohomology turns out to be important in  $p$ -adic geometry, following Bhatt’s discovery [Bha12] of conjugate filtration in char  $p$ , de-Hodge-completing results in [Bei12]. In [Kal18], Kaledin introduced an analogous de-completion of the periodic cyclic homology, called the polynomial periodic cyclic homology, equipped with a conjugate filtration in char  $p$ , and expected to be related to derived de Rham cohomology.

In this article, using genuine equivariant homotopy structure on Hochschild homology as in [ABG+18, BHM22], we give an equivariant description of Kaledin’s polynomial periodic cyclic homology. This leads to Morita invariance without any Noetherianness assumption as in [Kal18], and the comparison to derived de Rham cohomology becomes transparent. Moreover, this description adapts directly to “topological” analogues, which gives rise to a de-Nygaard-completion of the topological periodic cyclic homology, which admits an extension to linear categories over truncated Brown–Peterson spectra.

As an application, we establish a noncommutative crystalline–de Rham comparison, which de-completes the result in [PV19], and extends it to prime  $p = 2$ . We also compare polynomial periodic cyclic homology to topological Hochschild homology over  $\mathbb{F}_p$ , and produce a conjugate filtration in char  $p$  from our description.

## 1. INTRODUCTION

The main goal of this article is to give a genuine equivariant homotopy theoretic approach to (non-commutative) de-(Hodge-)completion of negative and periodic cyclic homology, which conceptually explains Kaledin’s *co-periodic cyclic homology* in [Kal18], giving rise to motivic filtrations. This approach also adapts to topological versions, which extends Devalapurkar–Hahn–Raksit–Yuan’s  $\mathrm{THH}_\Delta$  to DG-categories (or even linear categories over truncated Brown–Peterson spectra). As an application, we give a de-completion of noncommutative crystalline–de Rham comparison in [PV19], which works for the even prime  $p = 2$  as well. We start with backgrounds of this de-completion in algebraic geometry.

Grothendieck’s algebraic de Rham cohomology, introduced in [Gro66], turns out to be an important tool to study the cohomology of smooth schemes. However, it does not behave well beyond the smooth case. Illusie, following ideas of Quillen, introduced *derived de Rham cohomology*, along with its Hodge-completion, in [Ill72, Ch. VIII].

Hodge-completion makes derived de Rham cohomology easier to control, since the associated graded pieces are given by shifts of (derived) exterior powers of the cotangent complex. In particular, it coincides with algebraic de Rham cohomology for smooth schemes.

On the other hand, (non-Hodge-completed) derived de Rham cohomology was intractable until Bhatt’s discovery in [Bha12] of *conjugate filtration* on it in char  $p$ , whose associated graded pieces are equivalent to shifts of Frobenius twists of algebraic differential forms. He also observed the triviality of derived de Rham cohomology after rationalization. Using this new tool, he identified derived de Rham cohomology with crystalline cohomology for lci maps between  $\mathbb{Z}/p^r$ -schemes. Later on, this non-Hodge-complete version

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becomes useful in  $p$ -adic geometry. For example, Fontaine’s period rings  $A_{\text{cris}}$  and  $C_{\text{st}}$  are equipped with non-complete Hodge-filtration, and Bhatt applied this non-Hodge-complete version to prove some of Beilinson’s conjectures in [Bei12].

Periodic cyclic homology is a noncommutative counterpart of Hodge-completed derived de Rham cohomology, defined for general DG-categories. For morphisms of  $\mathbb{Q}$ -schemes, the relation is particularly simple: periodic cyclic homology is a product of shifts of derived de Rham cohomology, as recently shown by Konrad BALS in full generality in [Bal24]. This relation was firstly discovered by Loday–Quillen for smooth morphisms of  $\mathbb{Q}$ -schemes, cf. [Lod98, §5.1.12], and studied in [TV11]. For schemes beyond char 0, it was studied in [Maj96]. Breakthroughs were made in [BMS19, Ant19], which proved that, there is a complete filtration on periodic cyclic homology, whose associated graded pieces are shifts of Hodge-completed derived de Rham cohomology.

In view of usefulness of non-Hodge-completed derived de Rham cohomology in  $p$ -adic geometry, it is natural to ask whether there is a “non-Hodge-completion” of periodic cyclic homology for DG-categories, which carries a filtration with associated graded pieces being shifts of non-Hodge-completed derived de Rham cohomology? In [Kal18], following Kontsevich’s suggestion [Kon08, 2.32], Kaledin defined *polynomial periodic cyclic homology*, equipped with a conjugate filtration in char  $p$  (when  $p = 2$ , it is later constructed in [Kal17]), whose associated graded pieces are equivalent to shifts of Frobenius twisted Hochschild homology. Using this, he showed that a certain completion of polynomial periodic cyclic homology, called *co-periodic cyclic homology*, is a derived Morita invariant when the base is Noetherian. Moreover, he expected that polynomial periodic cyclic homology is closely related to derived de Rham cohomology.

As explained in [Kal18], Kaledin’s defines polynomial periodic cyclic homology and deals with it by explicit manipulations of chain complexes, which makes the arguments technical and difficult, and the homotopy-theoretic functoriality of this construction becomes opaque. A main goal of this article is to give a “more invariant treatment” as he wished in the introduction, which overcomes these difficulties.

The key to our description is the genuine equivariant homotopy structure on the usual  $\mathbb{Z}$ -linear Hochschild homology. More precisely, let  $\mathcal{C}$  be a DG-category. Then the Hochschild homology of  $\mathcal{C}$ , being a Borel  $\mathbb{T}$ -equivariant  $\mathbb{Z}$ -module spectrum, has the formula

$$\mathrm{HH}(\mathcal{C}/\mathbb{Z}) = \mathrm{THH}(\mathcal{C}) \otimes_{\mathrm{THH}(\mathbb{Z})}^{\mathbb{L}} \mathbb{Z}.$$

This admits an obvious cyclonic (à la [BG16]) structure: the  $\mathbb{T}$ -equivariant ring  $\mathbb{Z}$  is the underlying object of the constant Tambara functor  $\underline{\mathbb{Z}}$ , and the universal property of  $\mathrm{THH}$  in [ABG+18] gives rise to a map  $\mathrm{THH}(\mathbb{Z}) \rightarrow \underline{\mathbb{Z}}$  of  $\mathbb{T}$ - $\mathbb{E}_{\infty}$ -rings. This gives rise to an enhancement

$$\mathrm{HH}(\mathcal{C}/\underline{\mathbb{Z}}) = \mathrm{THH}(\mathcal{C}) \otimes_{\mathrm{THH}(\mathbb{Z})}^{\mathbb{L}} \underline{\mathbb{Z}}$$

as a  $\underline{\mathbb{Z}}$ -module in cyclonic spectra. This was generalized to a genuine version of factorization homology in [BHM22]. However, up to our knowledge, such a genuine equivariant homotopy structure on  $\mathbb{Z}$ -linear Hochschild homology does not seem to be studied in the literature. By definition, such an enhancement is a derived Morita invariant.

*Remark 1.1.* This genuine equivariant homotopy structure has other applications. In a companion paper [Mao24b], we use a “thickening” of it to define prismatic Hochschild homology. In our paper [Mao24a], we use a similar genuine equivariant structure to streamline Kaledin’s Hochschild–Witt homology, a noncommutative counterpart of de Rham–Witt complex.

It turns out that such a structure contains enough information to recover polynomial periodic cyclic homology. Recall that, in terms of explicit chain complexes, the usual Tate construction involves a product totalization in one direction, and Kaledin’s polynomial periodic cyclic homology is taking the direct sum

totalization instead, so that it has good colimit-preserving properties. Inspired by this, we introduce the following definitions for all primes:

**Definition 1.2** (Definitions 2.1 and 3.3). The  $(\mathbb{Z})$ -de-completed  $\mathbb{T}$ -Tate construction  $(-)^{\theta_{\mathbb{Z}}\mathbb{T}}$  is the filtered-colimit-preserving approximation of the composite functor

$$\mathrm{Mod}_{\mathbb{Z}}(\mathrm{Sp}^{g^{<\mathbb{T}}}) \longrightarrow \mathrm{Sp}^{B\mathbb{T}} \xrightarrow{(-)^{t\mathbb{T}}} \mathrm{Sp}.$$

The *polynomial periodic cyclic homology*  $\mathrm{HP}^{\mathrm{poly}}(\mathcal{C}/\mathbb{Z})$  of a DG-category  $\mathcal{C}$  is defined to be  $\mathrm{HH}(\mathcal{C}/\mathbb{Z})^{\theta_{\mathbb{Z}}\mathbb{T}}$ , applying the de-completed  $\mathbb{T}$ -Tate construction to Hochschild homology  $\mathrm{HH}(\mathcal{C}/\mathbb{Z})$ .

This is a point where we could explain a similar strategy does not work if we only look at the usual non-genuine equivariant Hochschild homology: the colimit-preserving approximation of the functor  $(-)^{t\mathbb{T}} : D(\mathbb{Z})^{B\mathbb{T}} \rightarrow D(\mathbb{Z})$  is zero, as its value on the compact generator  $\mathbb{Z}[\mathbb{T}]$  is, more or less by definition, zero.

The same construction works for any  $t$ -bounded (i.e. bounded with respect to the  $t$ -structure) animated ring as base in place of  $\mathbb{Z}$ , as done in the main text.

*Remark 1.3.* The de-completed Tate construction depends on the choice of base. However, in some cases, it does not quite depend on that. We will prove relevant results in Section 4.

From this description, it is immediate that polynomial periodic cyclic homology is rationally zero (Remark 2.9), since the functor  $\mathrm{Mod}_{\mathbb{Z}}(\mathrm{Sp}^{g^{<\mathbb{T}}}) \rightarrow \mathrm{Sp}^{B\mathbb{T}}$  becomes an equivalence after rationalization. With slightly more efforts, we show that

**Theorem 1.4** (Proposition 4.4). *Let  $\mathcal{C}$  be a smooth and bounded DG-category<sup>1</sup>. Then the assembly map  $\mathrm{HP}^{\mathrm{poly}}(\mathcal{C}/\mathbb{Z}) \rightarrow \mathrm{HP}(\mathcal{C}/\mathbb{Z})$  is an equivalence after profinite completion.*

Note that, for every quasicompact quasiseparated scheme  $X$ , its derived category  $D(X)$  is bounded, by [BvdB03, Cor 3.1.8]. When  $X$  is in addition smooth, then its derived category  $D(X)$  is also smooth. We refer to [Orl16] for general discussions. This proposition, along with colimit-preservation of de-completed Tate construction, implies that, on animated rings, the polynomial periodic cyclic homology is left Kan extended from polynomial rings, and thus it coincides with various ad-hoc constructions in the literature<sup>2</sup>, such as in [BMS19, AMMN22]. Consequently, we address Kaledin’s expectation in the following.

**Theorem 1.5** (Construction 4.8). *Let  $R$  be a commutative ring. Then there exists a functorial filtration  $\mathrm{Fil}_{\mathrm{HKR}}^*$  on the profinite completion  $\mathrm{HP}^{\mathrm{poly}}(R/\mathbb{Z})^\wedge$  of polynomial periodic cyclic homology with associated graded pieces equivalent to shifts of derived de Rham cohomology  $\mathrm{dR}_{R/\mathbb{Z}}$  after profinite completion.*

Our description also suggests a “topological” analogue.

**Definition 1.6** (Definition 4.9). Let  $S$  be a perfectoid ring, and  $\mathcal{C}$  a DG-category over  $S$ . Then *topological polynomial periodic cyclic homology*  $\mathrm{TP}^{\mathrm{poly}/S}(\mathcal{C})$  is defined to be  $\mathrm{THH}(\mathcal{C})^{\theta_{\mathrm{THH}(S)}\mathbb{T}}$ , defined by applying  $\mathrm{THH}(S)$ -de-completed  $\mathbb{T}$ -Tate construction to topological Hochschild homology  $\mathrm{THH}(\mathcal{C})$ .

Note that this is an arena where explicit chain complex manipulations cannot arrive. Previous results for polynomial periodic cyclic homology adapts to its topological analogue as well:

<sup>1</sup>Smooth DG-categories are necessarily modules categories over smooth  $\mathbb{E}_1$ -rings, by [Lur18, Prop 11.3.2.4]. A DG-category is *bounded* if, for every pair  $(x, y)$  of compact objects, the mapping  $\mathbb{Z}$ -module spectrum  $\mathrm{Hom}(x, y) \in D(\mathbb{Z})$  has bounded Tor-amplitude (or equivalently,  $t$ -bounded, since  $\mathbb{Z}$  is of finite flat dimension).

<sup>2</sup>Also compare with [Man24, §1].

**Theorem 1.7** (Corollary 4.12). *Let  $S$  be a perfectoid ring, and  $R$  a  $p$ -completely smooth and bounded DG-category over  $S$ . Then the assembly map  $\mathrm{TP}^{\mathrm{poly}/S}(R) \rightarrow \mathrm{TP}(R)$  is an equivalence after  $(p, \ker(\theta))$ -completion, where  $\theta : A_{\mathrm{inf}}(S) \rightarrow S$  is Fontaine's map.*

**Theorem 1.8** (Construction 4.13). *Let  $S$  be a perfectoid ring, and  $R$  a commutative  $S$ -algebra. Then there exists a functorial filtration  $\mathrm{Fil}_M^*$  on the  $(p, \ker(\theta))$ -completed topological polynomial periodic cyclic homology  $\mathrm{TP}^{\mathrm{poly}/S}(R)_{(p, \ker(\theta))}^\wedge$  with associated graded pieces equivalent to shifts of Frobenius twisted prismatic cohomology  $\varphi_A^* \Delta_{R/A}$  after  $(p, \ker(\theta))$ -completion, where  $A := A_{\mathrm{inf}}(S)$ .*

In a forthcoming work of Devalapurkar–Hahn–Raksit–Yuan, they produce de-completions of  $\mathrm{TC}^-$  and  $\mathrm{TP}$  which carry filtrations with associated graded pieces given by *absolute prismatic cohomology* and its Nygaard filtered pieces. Our techniques adapt to this situation as well, giving rise to *de-completed Borel completion*  $(-)^{\eta_{\mathrm{THH}(\mathbb{Z})}} : \mathrm{Mod}_{\mathrm{THH}(\mathbb{Z})}(\mathrm{Sp}^{g^{\leq \mathbb{T}}}) \rightarrow \mathrm{Mod}_{\mathrm{THH}(\mathbb{Z})^h}(\mathrm{Sp}^{g^{\leq \mathbb{T}}})$ .

**Theorem 1.9** (Corollary 5.10). *The functor*

$$\mathrm{THH}_\Delta : \mathrm{CAlg}_{\mathbb{Z}}^{\mathrm{an}} \rightarrow \mathrm{RMod}_{\mathrm{THH}(\mathbb{Z})^h}(\mathrm{Sp}^{g^{\leq \mathbb{T}}})_p^\wedge$$

*in Devalapurkar–Hahn–Raksit–Yuan extends to a localizing invariant over  $\mathbb{Z}$ , which is equivalent to  $(\mathrm{THH}(-)^{\eta_{\mathrm{THH}(\mathbb{Z})}})_p^\wedge$ . We can replace  $\mathbb{Z}$  by truncated Brown–Peterson spectra  $\mathrm{BP}\langle n \rangle$  for  $n \in \mathbb{N}$  as well (Remark 5.32).*

Thus this extended  $\mathrm{THH}_\Delta$  is indeed a noncommutative counterpart of (Nygaard-filtered) absolute prismatic cohomology. Actually, we can recover  $\mathrm{TP}^{\mathrm{poly}/S}$  from this new construction:

**Proposition 1.10** (Example 5.29 and Corollary 5.31). *Let  $S$  be a perfectoid ring, and  $\mathcal{C}$  a DG-category over  $S$ . Then we have equivalences*

$$\begin{aligned} ((\mathrm{THH}(\mathcal{C})^{\eta_{\mathrm{THH}(\mathbb{Z})}})^{C_{p^\infty}})_{(p, v_1)}^\wedge &\simeq \mathrm{TC}^{-, \mathrm{poly}/S}(\mathcal{C})_{(p, \xi)}^\wedge, \\ \left( ((\mathrm{THH}(\mathcal{C})^{\eta_{\mathrm{THH}(\mathbb{Z})}})^{\Phi_{C_p}})^{C_{p^\infty}/C_p} \right)_{(p, v_1)}^\wedge &\simeq \mathrm{TP}^{\mathrm{poly}/S}(\mathcal{C})_{(p, \xi)}^\wedge. \end{aligned}$$

It seems slightly surprising that cyclotomic  $\mathrm{THH}$  (with  $\mathrm{THH}(\mathbb{Z})$ -module structure) is already enough to de-Nygaard-complete topological periodic cyclic homology, but this phenomenon is demystified by Efimov's rigidity of localizing motives (Remark 4.17).

As for noncommutative geometry on its own, we first give a noncommutative crystalline–de Rham comparison. Let  $R$  be an animated ring. Then the crystalline–de Rham comparison theorem tells us that the crystalline cohomology of  $R \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$  over  $\mathbb{Z}_p$  is equivalent to the  $p$ -completion of the (non-Hodge-completed) derived de Rham of  $R$  over  $\mathbb{Z}$ . In [PV19], when  $p$  is an odd prime, for DG-categories  $\mathcal{C}$ , they give a comparison between  $\mathrm{TP}(\mathcal{C} \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p)$  with  $\mathrm{HP}(\mathcal{C}/\mathbb{Z})$ , which corresponds to such a comparison under certain completions. This result is improved by [DR25, Thm 0.5.1] as well for odd primes  $p$ . Equipped with previous constructions, adapting main ideas in [PV19] along with an observation by A. RAKSIT, we prove a comparison of de-completed constructions which corresponds to crystalline–de Rham comparison on the nose:

**Theorem 1.11** (Corollary 6.28). *Let  $\mathcal{C}$  be a DG-category. Then there exists a lax symmetric monoidal (in  $\mathcal{C}$ ) equivalence*

$$\mathrm{TP}^{\mathrm{poly}/\mathbb{F}_p}(\mathcal{C} \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p)_p^\wedge \simeq \mathrm{HP}^{\mathrm{poly}}(\mathcal{C}/\mathbb{Z})_p^\wedge.$$

We then give two proofs for the following comparison, due to Kaledin in [Kal20, Cor 11.15], but our proof is much simpler.

**Proposition 1.12** (Corollary 7.10). *Let  $\mathcal{C}$  be a DG-category over  $\mathbb{F}_p$ . Then the polynomial periodic cyclic homology  $\mathrm{HP}^{\mathrm{poly}}(\mathcal{C}/\mathbb{F}_p)$  is equivalent to  $\mathrm{THH}(\mathcal{C})[\sigma^{-1}]$  as  $\mathbb{Z}^{t\mathbb{T}}$ -module spectra.*

We also produce a conjugate filtration on polynomial periodic cyclic homology in char  $p$  in Section 8, and prove that

**Proposition 1.13** (Corollary 8.21). *Let  $k$  be a commutative  $\mathbb{F}_p$ -algebra, and  $\mathcal{C}$  a DG-category over  $k$ . Then the conjugate filtration on  $\mathrm{HP}^{\mathrm{poly}}(\mathcal{C}/k)$  is complete in the following two cases:*

- (1)  $\mathcal{C} = D(R)$  for some  $(-1)$ -connective  $\mathbb{E}_1$ - $k$ -algebra  $R$  (which includes all associative  $k$ -algebras  $R$ );
- (2)  $\mathcal{C} = D(X)$  for a quasicompact quasiseparated  $k$ -scheme  $X$ .

*Remark 1.14.* In comparison to Efimov’s refined negative cyclic homology and its continuation in Scholze’s ongoing work<sup>3</sup> on refined  $\mathrm{TC}^-$ , as mentioned above, in Remark 4.17, we explain that topological Hochschild homology as a cyclotomic spectrum already sees “all”  $p$ -adic formal information. However, their versions capture rigid analytic information. For example, for smooth  $\mathbb{F}_p$ -schemes, their versions see rigid cohomology. It might be worth understanding whether equivariant homotopy theory could say something for their versions as well.

**Notation 1.15.** *Let  $G$  be a finite group. We denote by  $\mathrm{Sp}^{gG}$  the symmetric monoidal  $\infty$ -category of genuine  $G$ -spectra, by  $\mathrm{Mack}_G^{\mathrm{coh}}(k)$  the abelian category of  $k$ -linear cohomological  $G$ -Mackey functors (we omit  $k$  when  $k = \mathbb{Z}$ ), and by  $\mathrm{Sp}^{g<\mathbb{T}}$  (resp.  $\mathrm{Sp}^{g_p\mathbb{T}}$ ) the symmetric monoidal  $\infty$ -category of cyclonic (resp.  $p$ -cyclonic) spectra as in [BG16].*

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## 2. A DE-COMPLETION OF TATE CONSTRUCTION

Let  $k$  be a commutative ring, and  $G$  a finite group or  $\mathbb{T}$ . Recall that the  $G$ -Tate construction  $(-)^{tG} : D(k)^{BG} \rightarrow D(k)$  does not preserve filtered colimits. In this section, we try to “de-complete” it when the input is further equipped with a genuine equivariant structure. When  $G = C_p$ , we will show that it can be expressed in terms of the geometric fixed points. We will also establish a de-completed version of the Tate orbit lemma.

**Definition 2.1.** Let  $G$  be a finite group (resp.  $\mathbb{T}$ ), and  $A$  an  $\mathbb{E}_1$ -algebra in the symmetric monoidal  $\infty$ -category  $\mathrm{Sp}^{gG}$  (resp.  $\mathrm{Sp}^{g<\mathbb{T}}$ ) of  $G$ -spectra (resp. cyclonic spectra). Then

- The *(A-)de-completed homotopy  $G$ -fixed points*  $(-)^{\eta_{AG}} : \mathrm{RMod}_A(\mathrm{Sp}^{gG}) \rightarrow \mathrm{Sp}$  (resp.  $\mathrm{RMod}_A(\mathrm{Sp}^{g<\mathbb{T}}) \rightarrow \mathrm{Sp}$ ) is the filtered-colimit-preserving approximation of the homotopy  $G$ -fixed points  $(-)^{hG} : \mathrm{RMod}_A(\mathrm{Sp}^{gG}) \rightarrow \mathrm{Sp}$  (resp.  $\mathrm{RMod}_A(\mathrm{Sp}^{g<\mathbb{T}}) \rightarrow \mathrm{Sp}$ ), equipped with an *assembly map*  $(-)^{\eta_{AG}} \rightarrow (-)^{hG}$ . We omit “ $A$ ” when the context is clear.
- The *(A-)de-completed  $G$ -Tate construction*  $(-)^{\theta_{AG}} : \mathrm{RMod}_A(\mathrm{Sp}^{gG}) \rightarrow \mathrm{Sp}$  (resp.  $\mathrm{RMod}_A(\mathrm{Sp}^{g<\mathbb{T}}) \rightarrow \mathrm{Sp}$ ) is the filtered-colimit-preserving approximation of the  $G$ -Tate construction  $(-)^{tG} :$

<sup>3</sup>An abstract, along with recordings, of the talk can be found at <https://www.mpim-bonn.mpg.de/node/13359>.

$\mathrm{RMod}_A(\mathrm{Sp}^{gG}) \rightarrow \mathrm{Sp}$  (resp.  $\mathrm{RMod}_A(\mathrm{Sp}^{g<\mathbb{T}}) \rightarrow \mathrm{Sp}$ ), equipped with an *assembly map*  $(-)^{\theta_A G} \rightarrow (-)^{tG}$  which canonically fits into a commutative diagram

$$(1) \quad \begin{array}{ccc} (-)^{\eta_A G} & \longrightarrow & (-)^{hG} \\ \downarrow & & \downarrow \\ (-)^{\theta_A G} & \longrightarrow & (-)^{tG} \end{array}.$$

*Remark 2.2.* Let  $G$  be a finite cyclic group (resp.  $\mathbb{T}$ ), and  $A$  a commutative algebra in  $\mathcal{E} := \mathrm{Sp}^{gG}$  (resp.  $\mathcal{E} := \mathrm{Sp}^{g<\mathbb{T}}$ ). Then the diagram (1) is Cartesian in the  $\infty$ -category  $\mathrm{Fun}^{\mathrm{Ex}}(\mathcal{E}, \mathrm{Sp})$ : the fiber of the canonical natural transformation  $(-)^{hG} \rightarrow (-)^{tG}$  is  $(-)^{hG}[d] \in \mathrm{Fun}^{\mathrm{Ex}}(\mathcal{E}, \mathrm{Sp})$  (where  $d = 0$  if  $G$  is a finite cyclic group, and  $d = 1$  when  $G = \mathbb{T}$ ), which preserves filtered colimits as well, and consequently, the diagram (1) induces an equivalence on fibers of vertical arrows.

*Remark 2.3.* Let  $G$  be a finite group (resp.  $\mathbb{T}$ ), and  $A \rightarrow B$  a map of  $\mathbb{E}_1$ -algebras in the symmetric monoidal  $\infty$ -category  $\mathcal{E} := \mathrm{Sp}^{gG}$  (resp.  $\mathcal{E} := \mathrm{Sp}^{g<\mathbb{T}}$ ) of  $G$ -spectra (resp. cyclonic spectra). Then the assembly maps on compact objects of  $\mathrm{RMod}_B(\mathcal{E})$  induce “relative” assembly maps  $(-)^{\eta_A G} \rightarrow (-)^{\eta_B G}$  and  $(-)^{\theta_A G} \rightarrow (-)^{\theta_B G}$ , which fit into a commutative diagram

$$\begin{array}{ccc} (-)^{\eta_A G} & \longrightarrow & (-)^{\eta_B G} \\ \downarrow & & \downarrow \\ (-)^{\theta_A G} & \longrightarrow & (-)^{\theta_B G} \end{array}$$

which is Cartesian by Remark 2.2.

*Remark 2.4.* Let  $G$  be a finite group (resp.  $\mathbb{T}$ ), and  $A$  an  $\mathbb{E}_1$ -algebra in the symmetric monoidal  $\infty$ -category  $\mathcal{E} := \mathrm{Sp}^{gG}$  (resp.  $\mathcal{E} := \mathrm{Sp}^{g<\mathbb{T}}$ ) of  $G$ -spectra (resp. cyclonic spectra). Let  $A \rightarrow A^h$  denote the Borel completion of  $A$ . Then it follows immediately from the definitions that the de-completed homotopy  $G$ -fixed points  $(-)^{\eta_A G}$  factors as

$$\mathrm{RMod}_A(\mathcal{E}) \xrightarrow{(-) \otimes_A^{\mathrm{L}} A^h} \mathrm{RMod}_{A^h}(\mathcal{E}) \xrightarrow{(-)^{\eta_{A^h} G}} \mathrm{Sp}$$

and similarly for the de-completed  $G$ -Tate construction. Roughly speaking, it does not hurt to replace all genuine equivariant bases by their Borel completions. However, sometimes it seems to be convenient to consider genuine equivariant bases.

*Remark 2.5.* Let  $G$  be a finite group, and  $A$  an  $\mathbb{E}_{\infty}$ -algebra in  $\mathrm{Sp}^{gG}$ . Then the lax symmetric monoidal structure on the homotopy fixed points  $(-)^{hG}$  (resp. the Tate construction  $(-)^{tG}$ ) gives rise<sup>4</sup> to a lax symmetric monoidal structure on the de-completed homotopy fixed points  $(-)^{\eta_A G}$  (resp. the de-completed Tate construction  $(-)^{\theta_A G}$ ). The assembly maps are equipped with a lax symmetric monoidal structure as well. In particular, the objects  $M^{\eta_A G}$  (resp.  $M^{\theta_A G}$ ) carries a canonical  $A^{hG}$ -(resp.  $A^{tG}$ -)module structure, which is functorial in  $M \in \mathrm{Mod}_A(\mathrm{Sp}^{gG})$ .

*Remark 2.6.* In desirable situations, the  $A$ -de-completed homotopy  $G$ -fixed points (resp.  $G$ -Tate construction) does not quite depend on  $A$ . We discuss some independences of this form in Section 4.

*Remark 2.7.* Let  $G$  be a finite group, and  $A$  an  $\mathbb{E}_{\infty}$ -algebra in  $\mathrm{Sp}^{gG}$ . Recall that the symmetric monoidal  $\infty$ -category  $\mathrm{Sp}^{gG}$  is rigid, thus the forgetful functor  $\mathrm{Mod}_A(\mathrm{Sp}^{gG}) \rightarrow \mathrm{Mod}_A(\mathrm{Sp}^{BG})$  factors through the rigidification  $(\mathrm{Mod}_A(\mathrm{Sp}^{BG}))^{\mathrm{rig}}$  of the target. Then the  $A$ -de-completed  $G$ -Tate construction  $(-)^{\theta_A G}$  coincides with the composite

$$\mathrm{Mod}_A(\mathrm{Sp}^{gG}) \longrightarrow (\mathrm{Mod}_A(\mathrm{Sp}^{BG}))^{\mathrm{rig}} \xrightarrow{(-)^{tG, \mathrm{rig}}} \mathrm{Sp},$$

<sup>4</sup>Here the same argument does not work for  $G = \mathbb{T}$ , as in this case, the object  $A \in \mathrm{Mod}_A(\mathrm{Sp}^{gG})$  is not compact.

where  $(-)^{tG, \text{rig}}$  is the composite functor

$$(\text{Mod}_A(\text{Sp}^{BG}))^{\text{rig}} \hookrightarrow \text{Ind}(\text{Mod}_A(\text{Sp}^{BG})) \xrightarrow{\text{Ind}((-)^{tG})} \text{Ind}(\text{Sp}) \xrightarrow{\text{colim}} \text{Sp}.$$

This can be checked by restricting to compact objects of  $\text{Mod}_A(\text{Sp}^{gG})$ . The same holds for de-completed homotopy  $G$ -fixed points.

*Remark 2.8.* Let  $A$  be an  $\mathbb{E}_\infty$ -algebra in  $D(\mathbb{Z}) \otimes \text{Sp}^{g^{\leq \mathbb{T}}}$ . Recall that, for every  $A$ -module  $M$  in  $\text{Sp}^{g^{\leq \mathbb{T}}}$  and every positive integer  $n \in \mathbb{N}_{>0}$ , the canonical maps

$$M^{t\mathbb{T}}/n \longleftarrow M^{t\mathbb{T}} \otimes_{A^{t\mathbb{T}}} A^{tC_n} \longrightarrow M^{tC_n}$$

are equivalences, [NS18, Lem IV.4.12]. It follows that, the canonical maps

$$M^{\theta_A \mathbb{T}}/n \longleftarrow M^{\theta_A \mathbb{T}} \otimes_{A^{t\mathbb{T}}} A^{tC_n} \longrightarrow M^{\theta_A C_n}$$

are equivalences as well. Consequently, up to profinite completion, the  $A$ -de-completed  $\mathbb{T}$ -Tate construction  $M^{\theta_A \mathbb{T}}$  can be recovered from de-completed  $C_n$ -Tate constructions, where  $n$  runs through all positive integers. This gives rise to a lax symmetric monoidal structure on the profinitely completed de-completed  $\mathbb{T}$ -Tate construction  $(M^{\theta_A \mathbb{T}})^\wedge$ .

*Remark 2.9.* Let  $k$  be a commutative ring. Then the constant Green functor  $\underline{k}$  can be viewed as an object of  $D(\mathbb{Z}) \otimes \text{Sp}^{g^{\leq \mathbb{T}}}$ . Then for every  $n \in \mathbb{N}_{>0}$ , we have

$$(\underline{k} \otimes \Sigma_{\mathbb{T}}^\infty [\mathbb{T}/C_n]_+)^{t\mathbb{T}} \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{Q} \simeq 0.$$

Consequently, the rationalized de-completed  $\mathbb{T}$ -Tate construction  $(-)^{\theta_{\underline{k}} \mathbb{T}}$  vanishes. Combining with Remark 2.8, we see that the de-completed  $\mathbb{T}$ -Tate construction  $(-)^{\theta_{\underline{k}} \mathbb{T}}$  can be completely recovered from de-completed  $C_n$ -Tate constructions, where  $n$  runs through all positive integers.

The Tate orbit lemma admits a de-completion. We first observe that, since  $C_p$  is a simple group<sup>5</sup>, the de-complete  $C_p$ -Tate construction has a fairly simple formula:

**Lemma 2.10.** *Let  $A$  be an  $\mathbb{E}_1$ -algebra in the symmetric monoidal category  $\text{Sp}^{gC_p}$  of  $C_p$ -spectra. Then the  $A$ -de-completed  $C_p$ -Tate construction  $(-)^{\theta_A C_p}$  coincides with the composite functor*

$$\text{RMod}_A(\text{Sp}^{gC_p}) \xrightarrow{(-)^{\Phi C_p}} D(A^{\Phi C_p}) \longrightarrow D(A^{tC_p})$$

where the second functor is the base change along the map  $A^{\Phi C_p} \rightarrow A^{tC_p}$  of  $\mathbb{E}_1$ -rings. Moreover, when  $A$  is an  $\mathbb{E}_\infty$ -algebra in  $\text{Sp}^{gC_p}$ , this identification is (lax) symmetric monoidal.

*Proof.* It suffices to restrict to compact objects of  $\text{RMod}_A(\text{Sp}^{gC_p})$ . For compact objects, there are many ways to see this. For example, the functor  $(-)^{tC_p}$  vanishes on induced  $C_p$ -spectra, thus it canonically factors through  $D(A^{\Phi C_p})$  in  $\text{Pr}^L$  (here we use the fact that  $C_p$  is a simple group), cf. [AMR21, §5], and then to see that the resulting functor  $D(A^{\Phi C_p}) \rightarrow D(A^{tC_p})$  in  $\text{Pr}^L$  coincides with the base change, it suffices to check on the generator, which is straightforward. The symmetric monoidal structure follows from a similar argument.  $\square$

**Corollary 2.11.** *Let  $A$  be an  $\mathbb{E}_1$ -algebra in the symmetric monoidal category  $\text{Sp}^{gC_p}$  of  $C_p$ -spectra. Then the  $A$ -de-completed  $C_p$ -Tate construction  $(-)^{\theta_A C_p} : \text{RMod}_A(\text{Sp}^{gC_p}) \rightarrow D(A^{tC_p})$  is strongly continuous<sup>6</sup>.*

<sup>5</sup>In this article, we only consider cyclic group actions, but the argument works for any simple group.

<sup>6</sup>A functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  in  $\text{Pr}_{\text{St}}^L$  is *strongly continuous* if its right adjoint  $F^R : \mathcal{D} \rightarrow \mathcal{C}$  preserves filtered colimits. When  $\mathcal{C}$  is compactly generated, it is equivalent to  $F$  preserving compact objects.

**Corollary 2.12.** *Let  $A$  be an  $\mathbb{E}_\infty$ -algebra in the symmetric monoidal category  $\mathrm{Sp}^{gC_p}$  of  $C_p$ -spectra. Then the  $A$ -de-completed  $C_p$ -Tate construction  $(-)^{\theta_A C_p}$  is symmetric monoidal.*

**Lemma 2.13.** *Let  $A$  be a bounded below  $\mathbb{E}_1$ -algebra in cyclonic spectra. Then the  $p$ -completed  $A$ -de-completed  $\mathbb{T}$ -Tate construction<sup>7</sup>*

$$((-)^{\theta_A \mathbb{T}})_p^\wedge : \mathrm{RMod}_A(\mathrm{Sp}^{g<\mathbb{T}}) \longrightarrow D(A^{t\mathbb{T}})_p^\wedge,$$

*factors as*<sup>8</sup>

$$\mathrm{RMod}_A(\mathrm{Sp}^{g<\mathbb{T}}) \xrightarrow{(-)^{\theta_A C_p}} \mathrm{RMod}_{A^{tC_p}}(\mathrm{Sp}^{g_p(\mathbb{T}/C_p)}) \xrightarrow{((-)^{\eta_{A^{tC_p}}(\mathbb{T}/C_p)})_p^\wedge} D(A^{t\mathbb{T}})_p^\wedge$$

*in  $\mathrm{Pr}_{\mathrm{St}}^L$ .*

*Proof.* Since every functor preserves filtered colimits, it suffices to check on compact objects in  $\mathrm{RMod}_A(\mathrm{Sp}^{g<\mathbb{T}})$ , and this follows from Corollary 2.11 and the Tate orbit lemma (since  $A$  is bounded below, so is  $A \otimes \Sigma_{\mathbb{T}}^\infty[\mathbb{T}/C_m]_+$  for every  $m \in \mathbb{N}_{>0}$ ).  $\square$

The same argument works for finite cyclic groups (for which we can further keep track of the lax symmetric monoidal structure on the de-completed Tate construction).

**Lemma 2.14.** *Let  $r \in \mathbb{N}_{>0}$ , and  $A$  a bounded below  $\mathbb{E}_1$ -(resp.  $\mathbb{E}_\infty$ -)algebra in  $C_{p^r}$ -spectra. Then the  $A$ -de-completed  $C_{p^r}$ -Tate construction*

$$(-)^{\theta_A C_p} : \mathrm{Mod}_A(\mathrm{Sp}^{C_{p^r}}) \longrightarrow D(A^{tC_{p^r}}),$$

*as a presentable (resp. lax symmetric monoidal) functor, factors as*

$$\mathrm{Mod}_A(\mathrm{Sp}^{gC_{p^r}}) \xrightarrow{(-)^{\theta_A C_p}} \mathrm{Mod}_{A^{tC_p}}(\mathrm{Sp}^{g_p(C_{p^r}/C_p)}) \xrightarrow{((-)^{\eta_{A^{tC_p}}(C_{p^r}/C_p)})} D(A^{tC_{p^r}}).$$

*Remark 2.15.* Let  $k$  be a commutative ring. By Remarks 2.8 and 2.9 and Lemma 2.14, we can also keep track of the lax symmetric monoidal structure on the  $p$ -completed de-completed  $\mathbb{T}$ -Tate construction  $((-)^{\theta_k \mathbb{T}})_p^\wedge$ , showing that, as an exact lax symmetric monoidal functor, it factors through the composite exact symmetric monoidal functor

$$\mathrm{Mod}_k(\mathrm{Sp}^{g<\mathbb{T}}) \xrightarrow{(-)^{\Phi C_p}} \mathrm{Mod}_{k^{\Phi C_p}}(\mathrm{Sp}^{g<(\mathbb{T}/C_p)}) \longrightarrow \mathrm{Mod}_{k^{tC_p}}(\mathrm{Sp}^{g<(\mathbb{T}/C_p)}),$$

where the remaining functor  $\mathrm{Mod}_{k^{tC_p}}(\mathrm{Sp}^{g<(\mathbb{T}/C_p)}) \rightarrow D(k^{t\mathbb{T}})_p^\wedge$  is the limit of de-completed homotopy  $C_{p^r}$ -fixed points along  $r \in \mathbb{N}$  (which at least a priori does not necessarily preserve filtered colimits).

### 3. POLYNOMIAL PERIODIC AND NEGATIVE CYCLIC HOMOLOGY

We briefly review Kaledin's *polynomial periodic cyclic homology* of cyclic objects, and then describe it in terms of the de-completed  $\mathbb{T}$ -Tate construction (Definition 3.3), informally explaining why it coincides with Kaledin's original construction.

Let  $k$  be a commutative ring, and  $X_\bullet : \Lambda^{\mathrm{op}} \rightarrow \mathrm{Mod}_k$  a cyclic objects in  $k$ -modules. Recall that, for every  $[n]_\Lambda \in \Lambda^{\mathrm{op}}$ , the  $k$ -module  $X_n := X_\bullet([n]_\Lambda)$  carries an  $k$ -linear  $C_n$  action, which gives rise to a 2-periodic complex

$$\begin{array}{ccccccc} \cdots & \xleftarrow{1-\sigma_n} & X_n & \xleftarrow{N_n} & X_n & \xleftarrow{1-\sigma_n} & X_n & \xleftarrow{N_n} & \cdots \\ \text{weight} & & -1 & & 0 & & 1 & & \end{array}$$

<sup>7</sup>Note that the forgetful functor  $\mathrm{Sp}^{g<\mathbb{T}} \rightarrow \mathrm{Sp}^{g_p \mathbb{T}}$  induces an equivalence on  $p$ -complete objects, thus we could work  $p$ -typically throughout.

<sup>8</sup>Thanks to Remark 2.4, it does not matter what cyclonic structure we put on  $A^{tC_p}$ .

of  $k$ -modules, where  $\sigma_n : X_n \rightarrow X_n$  is the generator of  $C_n$ , and  $N_n := 1 + \sigma_n + \cdots + \sigma_n^{n-1}$  is the  $C_n$ -norm. This complex represents the shifted Tate construction  $X_n^{tC_n}[-1]$ . These complexes compile into a double complex

$$\begin{array}{ccccccc}
 & & \vdots & & \vdots & & \vdots \\
 & & \downarrow & & \downarrow & & \downarrow \\
 \cdots & \xleftarrow{1-\sigma} & X_1 & \xleftarrow{N} & X_1 & \xleftarrow{1-\sigma} & X_1 \xleftarrow{N} \cdots \\
 & & \downarrow & & \downarrow & & \downarrow \\
 \cdots & \xleftarrow{1-\sigma} & X_0 & \xleftarrow{N} & X_0 & \xleftarrow{1-\sigma} & X_0 \xleftarrow{N} \cdots
 \end{array}$$

where we suppress the subscripts of  $N$  and  $\sigma$ , and vertical differentials are appropriately 2-periodically given (for details, see [Kal18, §3.1]). Then

- The periodic cyclic homology  $\mathrm{HP}(X_\bullet/k)$  of the cyclic object  $X_\bullet$ , is the object in the derived category  $D(k)$  of  $k$ -modules represented by the product totalization of this double complex.
- The polynomial periodic cyclic homology  $\mathrm{HP}^{\mathrm{poly}}(X_\bullet/k)$  of the cyclic object  $X_\bullet$  is the object in the derived category  $D(k)$  of  $k$ -modules represented by the direct sum totalization of this double complex.

We now give an alternative, more conceptual description of this double complex and the polynomial periodic cyclic homology. Recall that the geometric realization  $|X_\bullet|_\Lambda \in D(k)^{B\mathbb{T}}$  can be rewritten as a geometric realization

$$\mathrm{colim}_{[n] \in \Delta^{\mathrm{op}}} \mathrm{Ind}_{C_n}^{\mathbb{T}}(X_n) \in D(k)^{B\mathbb{T}}.$$

If we apply the  $\mathbb{T}$ -Tate construction to it, and *incorrectly* interchange  $(-)^{t\mathbb{T}}$  with  $\mathrm{colim}_{[n]}$ , we get

$$\mathrm{colim}_{[n] \in \Delta^{\mathrm{op}}} (\mathrm{Ind}_{C_n}^{\mathbb{T}}(X_n))^{t\mathbb{T}} \in D(k)$$

where  $(\mathrm{Ind}_{C_n}^{\mathbb{T}}(X_n))^{t\mathbb{T}} \simeq X_n^{tC_n}[-1]$  [HN20, Prop 3], thus we see that this computes the polynomial periodic cyclic homology. Note that every  $G$ -module  $M$  gives rise to a cohomological  $G$ -Mackey functor  $\underline{M} : [G/H] \mapsto M^H$ , we can endow  $C_n$ -module  $X_n$  with a cohomological  $C_n$ -Mackey functor structure, and thus realize the geometric realization  $|X_\bullet|_\Lambda$  as an object of  $\mathrm{Mod}_{\underline{k}}(\mathrm{Sp}^{\mathbb{T}})$ , and apply the de-completed  $\mathbb{T}$ -Tate construction  $(-)^{\theta_{\underline{k}}\mathbb{T}}$  to it, obtaining polynomial periodic cyclic homology.

*Remark 3.1.* This procedure can be made more rigorous by considering cohomological Mackey functors over Connes' cyclic category  $\Lambda^{\mathrm{op}}$ . Since we do not depend on Kaledin's original construction, we skip such a development. However, a toy version of this is explained in Section A.

*Question 1.* This comparison does not compare the lax symmetric monoidal structure of Kaledin's polynomial periodic cyclic homology and ours. How do we compare the ring structure on the two?

Now we give our formal definitions, and explain why it corresponds to the construction above for associative flat  $k$ -algebras.

**Construction 3.2.** *Let  $k$  be a  $t$ -bounded<sup>9</sup> animated ring. The universal property of  $\mathrm{THH}$  as in [ABG+18] gives rise to a map  $\mathrm{THH}(k) \rightarrow \underline{k}$  of  $\mathbb{T}\text{-}\mathbb{E}_{\infty}$ -rings. Thus for every  $\mathrm{THH}(k)$ -module  $M$  in cyclonic spectra, we get an object  $M \otimes_{\mathrm{THH}(k)}^{\mathbb{L}} \underline{k} \in \mathrm{Mod}_{\underline{k}}(\mathrm{Sp}^{g^{<\mathbb{T}}})$ . In particular, let  $\mathcal{C}$  be a dualizable presentable stable  $k$ -linear*

<sup>9</sup>This means that it is bounded with respect to the canonical  $t$ -structure. We add the prefix “ $t$ -” to avoid confusion with boundedness of  $p$ -power torsion (which we do not use in this article anyways).

$\infty$ -category, we have a canonical genuine equivariant enhancement of the  $k$ -linear Hochschild homology of  $\mathcal{C}$ , denoted by  $\mathrm{HH}(\mathcal{C}/\underline{k}) \in \mathrm{Mod}_{\underline{k}}(\mathrm{Sp}^{g^{<\mathbb{T}}})$ .

**Definition 3.3.** Let  $k$  be a  $t$ -bounded animated ring, and  $\mathcal{C}$  a dualizable presentable stable  $k$ -linear  $\infty$ -category. Then the *polynomial periodic cyclic homology*  $\mathrm{HP}^{\mathrm{poly}}(\mathcal{C}/k)$  (resp. the *polynomial negative cyclic homology*  $\mathrm{HC}^{-,\mathrm{poly}}(\mathcal{C}/k)$ ) is defined to be the  $\underline{k}$ -de-completed  $\mathbb{T}$ -Tate construction  $\mathrm{HH}(\mathcal{C}/\underline{k})^{\theta_{\underline{k}}\mathbb{T}}$  (resp. the  $\underline{k}$ -de-completed homotopy  $\mathbb{T}$ -fixed points  $\mathrm{HH}(\mathcal{C}/\underline{k})^{\eta_{\underline{k}}\mathbb{T}}$ ).

Let  $k$  be a commutative ring, and  $R$  an associative flat  $k$ -algebra. Recall that, as in [ABG+18, §6], the Hochschild homology  $\mathrm{HH}(R/\underline{k})$  relative to the constant Tambara functor  $\underline{k}$  is informally given by the geometric realization of relative norms  $[n] \mapsto R^{\otimes_{\underline{k}}^L C_n}$ . In [Mao24a], we formally identify these norms with the naive tensor powers equipped with the obvious cyclic action. Thus the previous discussion basically identifies the polynomial periodic cyclic homology  $\mathrm{HP}^{\mathrm{poly}}(R/k)$  with Kaledin's original one.

#### 4. DE RHAM AND PRISMATIC COMPARISON

In this section, we will first establish a comparison result (Proposition 4.4), which says that, for smooth commutative algebras, polynomial periodic cyclic homology is the same as periodic cyclic homology. As a consequence, polynomial periodic cyclic homology of commutative algebras acquires a filtration whose associated graded pieces are equivalent to shifts of (non-Hodge-completed) derived de Rham cohomology. We then define a “topological” analogue (Definition 4.9), and for smooth algebras over perfectoid rings, it acquires a *motivic filtration* (Construction 4.13) whose associated graded pieces are equivalent to shifts of (non-Nygaard-completed) prismatic cohomology.

We first give some sufficient conditions for assembly maps being equivalences.

**Lemma 4.1.** *Let  $A$  be a commutative algebra in cyclonic spectra (resp.  $G$ -spectra for a finite group  $G$ ) whose underlying Borel equivariant spectrum, denoted by  $A^h$ , is  $t$ -bounded. Then the assembly map  $(-)^{\theta_{A^h}G} \rightarrow (-)^{tG}$  (resp.  $(-)^{\eta_{A^h}G} \rightarrow (-)^{hG}$ ) is an equivalence on the idempotent-complete stable subcategory of  $\mathrm{Mod}_A(\mathrm{Sp}^{g^{<\mathbb{T}}})$  (resp.  $\mathrm{Mod}_A(\mathrm{Sp}^{g^G})$ ) generated by  $A$ -modules of the form  $\bigoplus_{i \in I} A \otimes [\mathbb{T}/H_i]$  (resp.  $\bigoplus_{i \in I} A \otimes [G/H_i]$ ) for an indexed family  $(H_i)_{i \in I}$  of finite cyclic groups  $H_i$  (resp. finite groups  $H_i \subseteq G$ ).*

*Proof.* We write the argument for the finite group case. The cyclonic case is similar.

- By construction, the assembly map is an equivalence on  $A \otimes [G/H]$  for finite groups  $H \subseteq G$ .
- The family  $\{A^h \otimes [G/H] \in \mathrm{Sp}^{BG} \mid (H \subseteq G \text{ is a finite subgroup})\}$  is uniformly  $t$ -bounded, thus the canonical map

$$\bigoplus_i (A \otimes [G/H_i])^{hG} \longrightarrow \left( \bigoplus_i A \otimes [G/H_i] \right)^{hG}$$

is an equivalence for an indexed family  $(H_i \subseteq G)_i$  of finite subgroups of  $G$ . It follows that the assembly map is an equivalence on  $\bigoplus_i A \otimes [G/H_i]$ .

- The result follows from the fact that the functors  $(-)^{\theta_{A^h}G}$ ,  $(-)^{\eta_{A^h}G}$ ,  $(-)^{tG}$  and  $(-)^{hG}$  are exact.

□

When  $G$  is a finite cyclic group, the situation is particularly simple.

**Corollary 4.2.** *Let  $G$  be a finite cyclic group. Then the assembly map  $(-)^{\theta_{\mathbb{Z}}G} \rightarrow (-)^{tG}$  (resp.  $(-)^{\eta_{\mathbb{Z}}G} \rightarrow (-)^{hG}$ ) is an equivalence on  $t$ -bounded objects in  $\mathrm{Mod}_{\mathbb{Z}}(\mathrm{Sp}^{g^G})$ .*

*Proof.* Since the abelian category  $\text{Mack}_G^{\text{coh}}$  has finite projective dimension when  $G$  is finite cyclic [BSW17, Cor 7.2], every  $t$ -bounded object is represented by a finite complex of projective objects. Thus the assembly map  $(-)^{\theta_G} \rightarrow (-)^{t_G}$  is an equivalence on these objects by Lemma 4.1.  $\square$

**Corollary 4.3.** *Let  $G$  be a finite cyclic group, and  $A$  a commutative algebra in  $D^b \text{Mack}_G^{\text{coh}} \subseteq \text{Mod}_{\mathbb{Z}}(\text{Sp}^{gG})$ . Then the assembly map  $(-)^{\theta_{\mathbb{Z}}G} \rightarrow (-)^{\theta_AG}$  (resp.  $(-)^{\eta_{\mathbb{Z}}G} \rightarrow (-)^{\eta_AG}$ ) is an equivalence. Consequently, the  $A$ -de-completed  $G$ -Tate construction  $(-)^{\theta_AG}$  (resp. the  $A$ -de-completed homotopy  $G$ -fixed points  $(-)^{\eta_AG}$ ) coincides with the composite functor*

$$\text{Mod}_A(\text{Sp}^{gG}) \longrightarrow \text{Mod}_{\mathbb{Z}}(\text{Sp}^{gG}) \xrightarrow{(-)^{\theta_{\mathbb{Z}}G} \text{ or } (-)^{\eta_{\mathbb{Z}}G}} D(\mathbb{Z}).$$

*Proof.* Note that the  $\infty$ -category  $\text{Mod}_A(\text{Sp}^{gG})$  is generated by objects of the form  $M \otimes_{\mathbb{Z}}^L A$  for finite permutation  $G$ -modules  $M$ , which is  $t$ -bounded since  $M$  is  $\mathbb{Z}$ -flat, and the result follows from Corollary 4.2.  $\square$

We are ready to establish the comparison between the polynomial periodic (resp. negative) cyclic homology and the periodic (resp. negative) cyclic homology on smooth algebras:

**Proposition 4.4.** *Let  $k$  be a  $t$ -bounded animated ring, and  $R$  an  $\mathbb{E}_1$ - $k$ -algebra. Then the commutative diagram*

$$\begin{array}{ccc} \text{HC}^{-, \text{poly}}(R/k) & \longrightarrow & \text{HC}^{-}(R/k) \\ \downarrow & & \downarrow \\ \text{HP}^{\text{poly}}(R/k) & \longrightarrow & \text{HP}(R/k) \end{array}$$

as an instance of (1) is Cartesian. If  $R$  is  $p$ -completely smooth as an  $\mathbb{E}_1$ - $k$ -algebra, and has bounded Tor-amplitude in  $D(k)_p^{\wedge}$ , then the horizontal assembly maps are equivalences after  $p$ -completion.

We first give a proof in the special case of  $R$  being a  $p$ -completely smooth animated  $k$ -algebra (i.e. with commutativity), since the general case needs knowledge on polygonic spectra in [KMN23], and the commutative case is sufficient for this section.

**Proof of Proposition 4.4 with commutativity** By Remark 2.2, this commutative diagram is Cartesian, thus for  $p$ -completely smooth animated  $k$ -algebras  $R$ , it suffices to check that the map  $\text{HP}^{\text{poly}}(R/k) \rightarrow \text{HP}(R/k)$  is an equivalence after modulo  $p$ , which is equivalent to base change along  $k \rightarrow k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$ , thus we may assume that  $k$  is a  $t$ -bounded animated  $\mathbb{F}_p$ -algebra. We can further check it after modulo  $p$  again. By Remark 2.8, it reduces to check that the assembly map

$$\text{HH}(R/\underline{k})^{\theta_{\underline{k}}C_p} \longrightarrow \text{HH}(R/\underline{k})^{tC_p}$$

is an equivalence. By Corollaries 4.2 and 4.3, it suffices to check that the cyclonic spectrum  $\text{HH}(R/\underline{k}) \in \text{Mod}_{\underline{k}}(\text{Sp}^{g^{<\mathbb{T}}})$  is  $t$ -bounded (thus so after forgetting to  $\text{Sp}^{gC_p}$ ). We check in two steps.

**Polynomial case:** When  $R = P \otimes_{\mathbb{F}_p}^{\mathbb{L}} k$  where  $P$  is a (finite) polynomial  $\mathbb{F}_p$ -algebra. Then we have  $\text{HH}(R/\underline{k}) = \text{HH}(P/\underline{\mathbb{F}_p}) \otimes_{\mathbb{F}_p}^{\mathbb{L}} k$ , and by  $t$ -boundedness of  $k$ , it suffices to show that  $\text{HH}(P/\underline{\mathbb{F}_p})$  is  $t$ -bounded. This follows from [Hes96, 2.2.4 & 2.2.5].

**General case:** By passing to a Zariski cover, we may assume that there exists an étale map  $S \rightarrow R$  where  $S$  is a (finite) polynomial  $k$ -algebra. Then by [HLL20, Add 3.2] (along with [Bor11, Thm B], which is used in their proof), the map  $\text{THH}(S) \rightarrow \text{THH}(R)$  is flat (even étale) in  $\text{CAlg}(\text{Sp}^{g^{<\mathbb{T}}})$ , thus so is the map  $\text{HH}(S/\underline{k}) \rightarrow \text{HH}(R/\underline{k})$ , and the result follows.  $\square$

Now we give the proof for Proposition 4.4 in full generality. As in the the first proof of Proposition 4.4, it reduces to check that the assembly map

$$\mathrm{HH}(R/\underline{k})^{\theta_{\underline{k}} C_p} \longrightarrow \mathrm{HH}(R/k)^{t C_p}$$

is an equivalence when  $k$  is a  $t$ -bounded animated  $\mathbb{F}_p$ -algebra, and  $R$  is  $p$ -completely smooth as an  $\mathbb{E}_1$ - $k$ -algebra with bounded Tor-amplitude in  $D(k)$ . We prove a generalized version with coefficients.

Let  $R$  be an  $\mathbb{E}_1$ -ring spectrum, and  $M$  an  $R$ - $R$ -bimodule in  $\mathrm{Sp}$ . Then by [KM23, §6], the topological Hochschild homology  $\mathrm{THH}(R; M)$  carries a canonical  $p$ -polygonic structure, given by the sequence  $\left( \mathrm{THH}(R; M^{\otimes_{R^p}^{\mathbb{L}}}) \in \mathrm{Sp}^{BC_{p^r}} \right)_{r \in \mathbb{N}}$ , along with polygonic Frobenius maps

$$\mathrm{THH}(R; M^{\otimes_{R^p}^{\mathbb{L}}}) \longrightarrow \mathrm{THH}(R; M^{\otimes_{R^p}^{\mathbb{L}}})^{t C_p}$$

in  $\mathrm{Sp}^{BC_{p^r}}$ . Moreover, when all  $\mathrm{THH}(R; M^{\otimes_{R^p}^{\mathbb{L}}})$  in question are bounded below (this is the case when  $M$  is a perfect  $R$ - $R$ -bimodule in  $\mathrm{Sp}$ ), we get a sequence  $\left( \mathrm{THH}(R; M^{\otimes_{R^p}^{\mathbb{L}}}) \in \mathrm{Sp}^{gC_{p^r}} \right)_{r \in \mathbb{N}}$  with equivalences

$$\mathrm{THH}(R; M^{\otimes_{R^p}^{\mathbb{L}}})^{\Phi C_p} \xrightarrow{\sim} \mathrm{THH}(R; M^{\otimes_{R^p}^{\mathbb{L}}})$$

of genuine  $C_{p^r}$ -spectra. This construction extends to the case without bounded-below-ness, by a forthcoming work by Harpaz–Nikolaus–Saunier. All of these constructions are functorial in  $(R; M)$ .

There is also a forgetful functor from cyclotomic spectra to polygonic spectra, and  $\mathrm{THH}(R; R)$  as a polygonic spectrum is the same as the underlying polygonic spectrum of the cyclotomic spectrum  $\mathrm{THH}(R)$ .

**Construction 4.5.** *Let  $k$  be an animated ring,  $R$  an  $\mathbb{E}_1$ - $k$ -algebra, and  $M$  an  $R$ - $R$ -bimodule in  $D(k)$ . Then for every  $r \in \mathbb{N}$ , we get a genuine equivariant enhancement  $\mathrm{HH}((R; M)/\underline{k})$  of Hochschild homology  $\mathrm{HH}((R; M^{\otimes_{R^p}^{\mathbb{L}}})/k)$  with coefficients given by*

$$\mathrm{HH}((R; M^{\otimes_{R^p}^{\mathbb{L}}})/\underline{k}) := \mathrm{THH}(R; M^{\otimes_{R^p}^{\mathbb{L}}}) \otimes_{\mathrm{THH}(k)}^{\mathbb{L}} \underline{k}.$$

**Lemma 4.6.** *Let  $k$  be a  $t$ -bounded animated ring,  $R$  an  $\mathbb{E}_1$ - $k$ -algebra with bounded Tor-amplitude in  $D(k)$ , and  $M$  a perfect  $R$ - $R$ -bimodule in  $D(k)$ . Then the assembly map*

$$\mathrm{HH}((R; M^{\otimes_{R^p}^{\mathbb{L}}})/\underline{k})^{\theta_{\underline{k}} C_p} \longrightarrow \mathrm{HH}((R; M^{\otimes_{R^p}^{\mathbb{L}}})/k)^{t C_p}$$

*is an equivalence.*

*Proof.* The proof of [NS18, Prop III.1.1] (or [Lur11, Prop 2.2.3]) implies that the functor  $\mathrm{HH}((R; (-)^{\otimes_{R^p}^{\mathbb{L}}})/k)^{t C_p}$  is exact. By Lemma 2.10, the symmetric monoidal structure on  $(-)^{\Phi C_p}$ , and the equivalence  $\mathrm{THH}(R; M^{\otimes_{R^p}^{\mathbb{L}}})^{\Phi C_p} \xrightarrow{\sim} \mathrm{THH}(R; M)$ , we see that the functor  $\mathrm{HH}((R; (-)^{\otimes_{R^p}^{\mathbb{L}}})/k)^{\Phi C_p}$  is exact as well. Thus, to see that the assembly map in question is an equivalence, it suffices to show that it is an equivalence when  $M = R \otimes_k^{\mathbb{L}} R$ , the free  $R$ - $R$ -bimodule in  $D(k)$  of rank 1. In this case, for every  $r \in \mathbb{N}$ , we have an equivalence

$$\mathrm{THH}(R; (R \otimes_k^{\mathbb{L}} R)^{\otimes_{R^p}^{\mathbb{L}}}) \simeq \mathrm{THH}(k; R^{\otimes_k^{\mathbb{L}} p^r})$$

in  $\mathrm{Mod}_{\mathrm{THH}(k)}(\mathrm{Sp}^{gC_{p^r}})$ , and after base change along  $\mathrm{THH}(k) \rightarrow \underline{k}$ , it becomes the relative  $C_{p^r}$ -norm  $R^{\otimes_k^{\mathbb{L}} C_{p^r}}$ . Under this identification, the assembly map in question becomes the assembly map

$$(R^{\otimes_k^{\mathbb{L}} C_p})^{\theta_{\underline{k}} C_p} \longrightarrow (R^{\otimes_k^{\mathbb{L}} C_p})^{t C_p},$$

which is an equivalence by Lemma 4.7. □

**Lemma 4.7.** *Let  $k$  be a  $t$ -bounded animated ring, and  $M$  a  $k$ -module spectrum of bounded Tor-amplitude in  $D(k)$ . Then the assembly map*

$$(M^{\otimes_k^{\perp} C_p})^{\theta_k C_p} \longrightarrow (M^{\otimes_k^{\perp} C_p})^{t C_p}$$

*is an equivalence.*

*Proof.* Since both sides are exact in  $M$ , we may assume that  $M$  is a flat  $k$ -module spectrum, which is a filtered colimit  $\text{colim}_i M_i$  of finite free  $k$ -module spectra by Lazard's theorem [Lur17, Thm 7.2.2.15]. In this case, each derived cohomological  $C_{p^r}$ -Mackey functor  $M_i^{\otimes_k^{\perp} C_p}$  is  $t$ -bounded by the  $t$ -boundedness of  $k$ , thus so is their filtered colimit  $M^{\otimes_k^{\perp} C_p}$ . The result follows from Corollaries 4.2 and 4.3.  $\square$

**Proof of Proposition 4.4 in general** It follows from Lemma 4.6 by setting  $M = R$ .  $\square$

*Question 2.* Is there any categorical generalization of Lemma 4.6, namely, replacing  $R$  by dualizable presentable stable  $k$ -linear  $\infty$ -category  $\mathcal{C}$  which is smooth with bounded Tor-amplitude<sup>10</sup>, and replacing  $M$  by an colimit-preserving  $k$ -linear endofunctor  $\mathcal{C} \rightarrow \mathcal{C}$ ?

Now we construct an HKR filtration on the polynomial periodic (resp. negative) cyclic homology whose associated graded pieces are shifts of (non-Hodge-completed) derived de Rham cohomology (resp. its Hodge-filtered pieces), Hodge-de-completing [BMS19, Thm 1.17] and [Ant19].

**Construction 4.8. (HKR filtration)** *Let  $k$  be a  $t$ -bounded animated ring. Then by [Rak20] (which generalizes [Ant19]), for every smooth  $k$ -algebra  $R$ , there is an exhaustive filtration  $\text{Fil}_{\text{HKR}}^*$ , functorial in  $R$ , on the canonical  $p$ -completed map*

$$\text{HC}^-(R/k)_p^\wedge \longrightarrow \text{HP}(R/k)_p^\wedge$$

*whose  $i$ -th associated graded piece  $\text{gr}_{\text{HKR}}^i$  is given by the canonical map*

$$\text{Fil}_H^i \text{dR}_{R/k}[2i]_p^\wedge \rightarrow \text{dR}_{R/k}[2i]_p^\wedge.$$

*By Proposition 4.4 and sifted-colimit-preservation of the functors  $\text{CAlg}_k^{\text{an}} \rightarrow D(k)$ ,  $R \mapsto \text{HP}^{\text{poly}}(R/k)$  (resp.  $R \mapsto \text{HC}^{-, \text{poly}}(R/k)$ ), for every animated  $k$ -algebra  $R$ , we get an exhaustive filtration  $\text{Fil}_{\text{HKR}}^*$  on the Cartesian square*

$$\begin{array}{ccc} \text{HC}^{-, \text{poly}}(R/k)_p^\wedge & \longrightarrow & \text{HC}^-(R/k)_p^\wedge \\ \downarrow & & \downarrow \\ \text{HP}^{\text{poly}}(R/k)_p^\wedge & \longrightarrow & \text{HP}(R/k)_p^\wedge \end{array}$$

*whose  $i$ -th associated graded piece  $\text{gr}_{\text{HKR}}^i$  is given by*

$$\begin{array}{ccc} \text{Fil}_H^i \text{dR}_{R/k}[2i]_p^\wedge & \longrightarrow & \text{Fil}_H^i \widehat{\text{dR}}_{R/k}[2i]_p^\wedge \\ \downarrow & & \downarrow \\ \text{dR}_{R/k}[2i]_p^\wedge & \longrightarrow & \widehat{\text{dR}}_{R/k}[2i]_p^\wedge \end{array}$$

*where  $\widehat{\text{dR}}_{R/k}$  is the Hodge-completed derived de Rham cohomology of  $R/k$ .*

*Question 3.* Is the HKR filtration in Construction 4.8 complete?

It is very natural to extend our definition to de-Nygaard-complete topological periodic cyclic homology.

<sup>10</sup>An attempt for this definition: a dualizable presentable stable  $k$ -linear  $\infty$ -category  $\mathcal{C}$  has bounded Tor-amplitude if the coevaluation functor  $\mathcal{C}^\vee \otimes_{D(k)} \mathcal{C} \rightarrow D(k)$  sends compact objects to objects of bounded Tor-amplitude in  $D(k)$ .

**Definition 4.9.** Let  $k$  be a  $t$ -bounded  $\mathbb{E}_\infty$ -ring, and  $\mathcal{C}$  a dualizable presentable stable  $k$ -linear  $\infty$ -category. Then the *topological  $k$ -polynomial periodic cyclic homology*  $\mathrm{TP}^{\mathrm{poly}/k}(\mathcal{C})$  (resp. the *topological  $k$ -polynomial negative cyclic homology*  $\mathrm{TC}^{-,\mathrm{poly}/k}(\mathcal{C})$ ) is defined to be the  $\mathrm{THH}(k)$ -de-completed  $\mathbb{T}$ -Tate construction  $\mathrm{THH}(\mathcal{C})^{\theta_{\mathrm{THH}(k)}\mathbb{T}}$  (resp. the  $\mathrm{THH}(k)$ -de-completed homotopy  $\mathbb{T}$ -fixed points  $\mathrm{THH}(\mathcal{C})^{\eta_{\mathrm{THH}(k)}\mathbb{T}}$ ).

Topological  $k$ -polynomial periodic (resp. negative) cyclic homology, even after  $p$ -completion, seems intractable in general, partially due to the global nature of its prismaticization. The situation is drastically simpler when the ring  $k = S$  is  $p$ -complete and perfectoid, thanks to the Bökstedt periodicity of  $\mathrm{THH}(S)_p^\wedge$ . We recollect some notations and computations in [BMS19, Prop 6.2 & 6.3]. However, we view  $\mathrm{THH}(S)^{tC_p}$  non-equivariantly as a  $\mathrm{TC}^-(S)$ -module, which follows more closely to the convention in [Rig25, Lem 2.1].

*Remark 4.10* ([BMS19, Prop 6.2 & 6.3]). Let  $S$  be a perfectoid ring,  $A := A_{\mathrm{inf}}(S) = W(S^\flat)$  with Frobenius endomorphism  $\varphi : A \rightarrow A$ , and  $\xi$  a chosen generator of the kernel  $\ker(\theta)$  of Fontaine's map  $\theta : A \rightarrow S$ . Then the commutative square

$$\begin{array}{ccc} \mathrm{TC}^-(S)_p^\wedge & \xrightarrow{\varphi_p^{h\mathbb{T}}} & \mathrm{TP}(S)_p^\wedge \\ \downarrow & & \downarrow \\ \mathrm{THH}(S)_p^\wedge & \xrightarrow{\varphi_p} & \mathrm{THH}(S)^{tC_p} \end{array}$$

is a pushout diagram of  $\mathbb{E}_\infty$ -rings, and its homotopy groups are given by

$$\begin{array}{ccc} A[u, v]/(uv - \xi) & \xrightarrow[v \mapsto \varphi(\xi)\sigma^{-1}]{u \mapsto \sigma} & A[\sigma^\pm] \\ \downarrow & & \downarrow \\ R[u] = (A/\xi)[u] & \xrightarrow{u \mapsto \sigma} & (A/\varphi(\xi))[\sigma^\pm] \end{array},$$

where  $|u| = |\sigma| = 2$  and  $|v| = -2$ , the vertical maps are  $A$ -linear, and the horizontal maps are  $\varphi$ -linear. The homotopy groups of the canonical map  $\mathrm{TC}^-(S)_p^\wedge \rightarrow \mathrm{TP}(S)_p^\wedge$  is given by the  $A$ -linear map

$$A[u, v]/(uv - \xi) \xrightarrow[v \mapsto \sigma^{-1}]{u \mapsto \xi\sigma} A[\sigma^\pm].$$

In particular, the  $\mathrm{THH}(S)$ -module  $S = \mathrm{THH}(S)/u$  in  $\mathrm{Sp}^{B\mathbb{T}}$  is perfect, which implies that

**Lemma 4.11.** *Let  $S$  be a perfectoid ring, and  $M$  a  $\mathrm{THH}(S)$ -module in  $\mathrm{Sp}^{g<\mathbb{T}}$ . Then the canonical map*

$$M^{\theta_{\mathrm{THH}(S)}\mathbb{T}} \otimes_{\mathrm{TP}(S)}^{\mathrm{L}} S^{t\mathbb{T}} \longrightarrow (M \otimes_{\mathrm{THH}(S)} S)^{\theta_S\mathbb{T}}$$

*is an equivalence after  $p$ -completion.*

**Corollary 4.12.** *Let  $S$  be a perfectoid ring, and  $\mathcal{C}$  a dualizable presentable stable  $S$ -linear  $\infty$ -category. Then the commutative diagram*

$$\begin{array}{ccc} \mathrm{TC}^{-,\mathrm{poly}/S}(\mathcal{C}) & \longrightarrow & \mathrm{TC}^-(\mathcal{C}) \\ \downarrow & & \downarrow \\ \mathrm{TP}^{\mathrm{poly}/S}(\mathcal{C}) & \longrightarrow & \mathrm{TP}(\mathcal{C}) \end{array}$$

*as an instance of (1) is Cartesian. If the assembly map  $\mathrm{HP}^{\mathrm{poly}}(\mathcal{C}/S) \rightarrow \mathrm{HP}(\mathcal{C}/S)$  is an equivalence after  $p$ -completion<sup>11</sup>, then the horizontal assembly maps are equivalences after  $(p, \ker(\theta))$ -completion.*

*Proof.* By Remark 2.2, this commutative diagram is Cartesian. When the assembly map  $\mathrm{HP}^{\mathrm{poly}}(\mathcal{C}/S) \rightarrow \mathrm{HP}(\mathcal{C}/S)$  is  $p$ -completely an equivalence, then by Remark 4.10, it is the  $(\mathrm{mod} \ker(\theta))$  reduction of the map  $\mathrm{TP}^{\mathrm{poly}/S}(\mathcal{C}) \rightarrow \mathrm{TP}(\mathcal{C})$ , thus the later is  $(p, \ker(\theta))$ -completely an equivalence, and the result follows.  $\square$

<sup>11</sup>By Proposition 4.4, this is the case when  $\mathcal{C} = D(R)$  for some  $p$ -completely smooth  $\mathbb{E}_1$ - $S$ -algebra  $R$ .

Similarly to Construction 4.8, we have

**Construction 4.13. (Motivic filtration)** *Let  $S$  be a perfectoid ring, and let  $A := A_{\text{inf}}(S)$ . Then by [BMS19], for every smooth  $S$ -algebra  $R$ , there is an exhaustive filtration  $\text{Fil}_M^*$ , functorial in  $R$ , on the canonical  $(p, \ker(\theta))$ -completed<sup>12</sup> map*

$$\text{TC}^-(R)_{(p, \ker(\theta))}^\wedge \longrightarrow \text{TP}(R)_{(p, \ker(\theta))}^\wedge$$

whose  $i$ -th associated graded piece  $\text{gr}_M^i$  is given by the canonical map

$$\text{Fil}_N^i \varphi_A^* \Delta_{R/A} \longrightarrow \varphi_A^* \Delta_{R/A}$$

for the Frobenius twisted prismatic cohomology. By Proposition 4.4 and Corollary 4.12, and sifted-colimit-preservation of the functors  $\text{CAlg}_S^{\text{an}} \rightarrow D(\text{TC}^-(S))$ ,  $R \mapsto \text{TP}^{\text{poly}/S}(R)$  (resp.  $R \mapsto \text{TC}^{-, \text{poly}/S}(R)$ ), for every animated  $k$ -algebra  $R$ , we get an exhaustive filtration  $\text{Fil}_M^*$  on the Cartesian square

$$\begin{array}{ccc} \text{TC}^{-, \text{poly}/S}(R)_{(p, \ker(\theta))}^\wedge & \longrightarrow & \text{TC}^-(R)_{(p, \ker(\theta))}^\wedge \\ \downarrow & & \downarrow \\ \text{TP}^{\text{poly}/S}(R)_{(p, \ker(\theta))}^\wedge & \longrightarrow & \text{TP}(R)_{(p, \ker(\theta))}^\wedge \end{array}$$

whose  $i$ -th associated graded piece is given by

$$\begin{array}{ccc} \text{Fil}_N^i \varphi_A^* \Delta_{R/A} & \longrightarrow & \text{Fil}_N^i \varphi_A^* \widehat{\Delta}_{R/A} \\ \downarrow & & \downarrow \\ \varphi_A^* \Delta_{R/A} & \longrightarrow & \varphi_A^* \widehat{\Delta}_{R/A} \end{array}.$$

*Remark 4.14.* The construction of HKR filtration and motivic filtration on polynomial cyclic theories is quite formal: one only needs a proposition similar to Proposition 4.4 and Corollary 4.12. In particular, the above construction also adapts to the Breuil–Kisin case (as in [BMS19, §11]) and the  $q$ -de Rham case.

*Remark 4.15.* The topological polynomial periodic cyclic homology should be comparable to  $\text{TC}^{(-1)}$  as introduced in [Man24, §1]. In Section 5, we will discuss and compare to the construction in the absolute case in an ongoing project of Devalapurkar–Hahn–Raksit–Yuan.

A natural question is whether the previous picture extends to relative prismatic cohomology over an arbitrary base prism? When the base prism is transversal, we have the following expectation.

*Remark 4.16.* In a companion paper [Mao24b], we defined *prismatic Hochschild homology*  $\text{HH}^\Delta(\mathcal{C}/A)$  for a transversal prism  $(A, I)$  and a dualizable presentable stable  $A/I$ -linear  $\infty$ -category  $\mathcal{C}$ , and formulated an HKR-type conjecture for  $p$ -completely smooth  $A/I$ -algebras. If that conjecture holds, then the  $p$ -completed  $(A, I)^{C_{p^r-1}}$ -de-completed  $\mathbb{T}/C_{p^r-1}$ -construction  $\left( \text{HH}^\Delta(R/A)^{C_{p^r-1}} \right)^{\theta_{(A, I)^{C_{p^r-1}} (\mathbb{T}/C_{p^r-1})}}$  for animated  $(A/I)$ -algebra would carry an exhaustive filtration with associated graded pieces equivalent to shifts of  $(\varphi_A^* \Delta_{R/A}) \otimes_A^{\mathbb{L}} (A/I_r)$ . We will address this in the future.

Up to our knowledge, it was not widely expected that the topological Hochschild homology as a cyclotomic spectrum contains enough information for a de-Nygaard-completion such as Definition 4.9. However, in view of Efimov’s rigidity of localizing motives, this is expected:

<sup>12</sup>The  $p$ -completed  $\text{TC}^-(R)_p^\wedge$  and  $\text{TP}(R)_p^\wedge$  are automatically  $\ker(\theta)$ -complete (since they are Nygaard-complete), but it is conceptually better to phrase it after  $(p, \ker(\theta))$ -completion, since the  $p$ -completed polynomial versions might not be  $\ker(\theta)$ -complete.

*Remark 4.17* (M. RAMZI). Let  $k$  be a commutative ring, and  $G$  a finite group<sup>13</sup>,  $A$  an  $\mathbb{E}_\infty$ -algebra in  $\mathrm{Sp}^{gG}$ , and  $E : \mathrm{Cat}_k^{\mathrm{perf}} \rightarrow \mathrm{Mod}_A(\mathrm{Sp}^{BG})$  a finitary symmetric monoidal localizing invariant. Then the symmetric monoidal functor  $E$  factors uniquely through the presentably stable symmetric monoidal  $\infty$ -category  $\mathrm{Mot}_{\mathrm{loc},k}$ , obtaining a functor  $\mathrm{Mot}_{\mathrm{loc},k} \rightarrow \mathrm{Mod}_A(D(k)^{BG})$  in  $\mathrm{CAlg}(\mathrm{Pr}_{\mathrm{St}}^L)$ . Efimov’s rigidity theorem, as mentioned in [Efi24, Rem 4.3], tells us that the presentably stable symmetric monoidal  $\infty$ -category  $\mathrm{Mot}_{\mathrm{loc},k}$  is rigid, thus we get a unique strongly continuous functor  $\mathrm{Mot}_{\mathrm{loc},k} \rightarrow \mathrm{Mod}_A(D(k)^{BG})^{\mathrm{rig}}$ .

If we are in addition given a finitary symmetric monoidal factorization

$$\mathrm{Cat}_k^{\mathrm{perf}} \xrightarrow{\tilde{E}} \mathrm{Mod}_A(\mathrm{Sp}^{gG}) \longrightarrow \mathrm{Mod}_A(D(k)^{BG})$$

of  $E$ , where the functor  $\tilde{E}$  is localizing as well, then by Remark 2.7, we see that the strongly continuous functor  $\mathrm{Mot}_{\mathrm{loc},k} \rightarrow \mathrm{Mod}_A(D(k)^{BG})^{\mathrm{rig}}$  coincides with the composite functor

$$\mathrm{Mot}_{\mathrm{loc},k} \xrightarrow{\tilde{E}} \mathrm{Mod}_A(\mathrm{Sp}^{gG}) \longrightarrow \mathrm{Mod}_A(D(k)^{BG})^{\mathrm{rig}}.$$

Informally, this tells us that  $\tilde{E}$  “knows” everything about refined  $E$ .

Now we apply this to the finitary symmetric monoidal localizing invariant

$$\tilde{E} := \mathrm{THH} : \mathrm{Cat}_k^{\mathrm{perf}} \longrightarrow \mathrm{CycSp}^{\mathrm{gen}} \longrightarrow \mathrm{Mod}_{\mathrm{THH}(k)}(\mathrm{Sp}^{gG})$$

for any finite cyclic group  $G$ . It follows that the refinement of  $\mathrm{Sp}^{BG}$ -valued  $\mathrm{THH}$  is completely determined by the functor  $\tilde{E}$ , thus by the (genuine) cyclotomic  $\mathrm{THH}$ . Roughly speaking, this implies that the cyclotomic  $\mathrm{THH}$  already contains “all” profinite or  $p$ -adic formal information, including any de-Nygaard-completion.

## 5. THE ABSOLUTE CASE

In an ongoing work of Devalapurkar–Hahn–Raksit–Yuan, they propose another de-completion  $\mathrm{THH}_\Delta$  of topological negative cyclic homology for animated rings. In this section, we show that our techniques apply to this situation as well, which leads to *de-completed Borel completion* (Definition 5.4), giving rise to a localizing invariant on  $\mathbb{Z}$ -linear categories corresponding to *absolute prismatic cohomology*. We also compare de-completed Borel completion with de-completed homotopy fixed points and de-completed Tate construction (Remarks 5.7 and 5.8, Proposition 5.23, and Corollary 5.26), and the construction in this section with previous constructions (Corollaries 5.31 and 5.35). We first recall their definition. Actually, our argument shows that  $\mathrm{THH}_\Delta$  extends to a localizing invariant for  $\mathrm{BP}\langle n \rangle$ -linear categories (Remark 5.32), which is independent of choice of  $n$ .

**Notation 5.1.** Let  $G$  be a finite group (resp.  $\mathbb{T}$ ). We will denote by

$$(-)^h : \mathrm{Sp}^{gG} \longrightarrow \mathrm{Sp}^{gG}$$

(resp.

$$(-)^h : \mathrm{Sp}^{g<\mathbb{T}} \longrightarrow \mathrm{Sp}^{g<\mathbb{T}}$$

) the Borel completion, which has a lax symmetric monoidal structure.

**Construction 5.2. (Devalapurkar–Hahn–Raksit–Yuan)** The functor

$$\mathrm{THH}_\Delta : \mathrm{CAlg}_{\mathbb{Z}}^{\mathrm{an}} \longrightarrow \mathrm{Mod}_{\mathrm{THH}(\mathbb{Z})^h}(\mathrm{Sp}^{g<\mathbb{T}})$$

---

<sup>13</sup>As in Remark 2.5, this does not work for  $G = \mathbb{T}$ , as the symmetric monoidal  $\infty$ -category  $\mathrm{Sp}^{g<\mathbb{T}}$  is not rigid, but only locally rigid.

is defined to be the left derived functor of the functor

$$\begin{aligned} \text{Poly}_{\mathbb{Z}}^{\text{fg}} &\longrightarrow \text{Mod}_{\text{THH}(\mathbb{Z})^h}(\text{Sp}^{g^{<\mathbb{T}}}) \\ R &\longmapsto \text{THH}(R)^h. \end{aligned}$$

By construction, we have a canonical map

$$\text{THH}_{\Delta}(R) \longrightarrow \text{THH}(R)^h$$

in  $\text{Mod}_{\text{THH}(\mathbb{Z})^h}(\text{Sp}^{g^{<\mathbb{T}}})$ , which is an equivalence when  $R$  is a finitely generated polynomial ring.

**Remark 5.3** (Devalapurkar–Hahn–Raksit–Yuan). It follows immediately from the discussion of [BL22, §6.2] (which explains the result in [BMS19] for animated rings) that there exists a motivic filtration on  $\text{THH}(R)^{C_{p^\infty}}$  (resp.  $(\text{THH}(R)^{\Phi C_p})^{C_{p^\infty}/C_p}$ ) whose associated graded pieces are given by twisted absolute Nygaard cohomology  $\text{Fil}_N^* \Delta_R\{*\}[2*]$  (resp. twisted absolute prismatic cohomology  $\Delta_R\{*\}[2*]$ ).

To extend  $\text{THH}_{\Delta}$  to a localizing invariant, as in Definition 3.3, the key is to de-complete Borel completion.

**Definition 5.4.** Let  $G$  be a finite group, and  $A$  an  $\mathbb{E}_1$ -algebra in the symmetric monoidal  $\infty$ -category  $\text{Sp}^{gG}$  of  $G$ -spectra. Then the  $A$ -de-completed Borel completion, denoted by  $(-)^{\eta_A}$ , is the filtered-colimit-preserving approximation

$$\text{RMod}_A(\text{Sp}^{gG}) \longrightarrow \text{RMod}_{A^h}(\text{Sp}^{gG})$$

of the Borel completion. The same if we replace  $G$  by  $\mathbb{T}$ , and the symmetric monoidal  $\infty$ -category  $\text{Sp}^{gG}$  of  $G$ -spectra by that  $\text{Sp}^{g^{<\mathbb{T}}}$  of cyclonic spectra.

**Remark 5.5.** Let  $G$  be a finite group (resp.  $\mathbb{T}$ ), and  $A$  an  $\mathbb{E}_1$ -algebra in the symmetric monoidal  $\infty$ -category  $\text{Sp}^{gG}$  (resp.  $\text{Sp}^{g^{<\mathbb{T}}}$ ) of  $G$ -spectra (resp. cyclonic spectra). As in Remark 2.4, the  $A$ -de-completed Borel completion factors canonically through the base change along the Borel completion map  $A \rightarrow A^h$ .

Actually, the de-completed Borel completion is simply a base change.

**Lemma 5.6.** Let  $G$  be a finite group, and  $A$  an  $\mathbb{E}_1$ -algebra in the symmetric monoidal  $\infty$ -category  $\text{Sp}^{gG}$  of  $G$ -spectra. Then the  $A$ -de-completed Borel completion  $(-)^{\eta_A}$  coincides with the base change functor

$$(-) \otimes_A^{\mathbb{L}} A^h : \text{RMod}_A(\text{Sp}^{gG}) \longrightarrow \text{RMod}_{A^h}(\text{Sp}^{gG}).$$

The same if we replace  $G$  by  $\mathbb{T}$ , and  $\text{Sp}^{gG}$  by  $\text{Sp}^{g^{<\mathbb{T}}}$ .

*Proof.* We start with the case that  $G$  is a finite group. By Remark 5.5, without loss of generality, we may assume that  $A$  is Borel complete, and by definition, it suffices to check that, for every  $G$ -orbit  $[G/H]$ , the tensor product  $[G/H] \otimes A$  is Borel complete. This follows from the dualizability of  $[G/H]$  in  $\text{Sp}^{gG}$ , cf. the proof of [BCN21, Ex 2.15].

The case for  $G = \mathbb{T}$  is similar: we have to show that  $[\mathbb{T}/C_n] \otimes A$  is Borel complete, and it suffices to restrict to  $\text{Sp}^{gC_m}$  for every  $m \in \mathbb{N}_{>0}$ , and  $[\mathbb{T}/C_n]$  is dualizable there as well.  $\square$

It then follows from Remark 2.4 that de-completed homotopy fixed points and de-completed Tate construction factor through de-completed Borel completion, thus the latter contains more information than the former. When the group is finite, this factorization takes a very simple form:

**Remark 5.7.** Let  $G$  be a finite group, and  $A$  an  $\mathbb{E}_1$ -algebra in the symmetric monoidal  $\infty$ -category  $\text{Sp}^{gG}$  of  $G$ -spectra. By restricting to compact objects in  $\text{RMod}_A(\text{Sp}^{gG})$ , we see that the  $A$ -de-completed homotopy  $G$ -fixed points  $(-)^{\eta_{AG}}$  factors as

$$\text{RMod}_A(\text{Sp}^{gG}) \xrightarrow{(-)^{\eta_A}} \text{RMod}_{A^h}(\text{Sp}^{gG}) \xrightarrow{(-)^G} \text{RMod}_{A^hG}(\text{Sp}).$$

*Remark 5.8.* Let  $G$  be a finite group, and  $A$  an  $\mathbb{E}_1$ -algebra in the symmetric monoidal  $\infty$ -category  $\mathrm{Sp}^{gG}$  of  $G$ -spectra. By restricting to compact objects in  $\mathrm{RMod}_A(\mathrm{Sp}^{gG})$ , we see that the  $A$ -de-completed  $G$ -Tate construction  $(-)^{\theta_A^G}$  factors as

$$\mathrm{RMod}_A(\mathrm{Sp}^{gG}) \xrightarrow{(-)^{\eta_A}} \mathrm{RMod}_{A^h}(\mathrm{Sp}^{gG}) \xrightarrow{(-)^{\Phi_{\{e\}}^G}} \mathrm{RMod}_{A^h G}(\mathrm{Sp}),$$

where  $(-)^{\Phi_{\{e\}}^G}$  is the geometric fixed points with respect to the family  $\{e\}$  of subgroups [AMR21, Def 5.7]. When  $G$  is a cyclic  $p$ -group, we have  $(-)^{\Phi_{\{e\}}^G} = (-)^{\Phi_{C_p}}$ .

Actually, this factorization also holds when  $G = \mathbb{T}$  for certain bases which we establish in Proposition 5.23 and Corollary 5.26. Now we formulate the main comparison theorem

**Theorem 5.9.** *Let  $R$  be a smooth  $\mathbb{E}_1$ - $\mathbb{Z}$ -algebra of bounded Tor-amplitude<sup>14</sup>. Then the assembly map*

$$\mathrm{THH}(R)^{\eta_{\mathrm{THH}(\mathbb{Z})}} \longrightarrow \mathrm{THH}(R)^h$$

*in  $\mathrm{Mod}_{\mathrm{THH}(\mathbb{Z})^h}(\mathrm{Sp}^{g<\mathbb{T}})$  is an equivalence after  $p$ -completion.*

Before giving its proof, we mention an immediate consequence: since finitely generated polynomial rings are smooth, it follows that

**Corollary 5.10.** *After  $p$ -completion, the functor  $\mathrm{THH}_{\Delta}$  factors as*

$$\mathrm{CAlg}_{\mathbb{Z}}^{\mathrm{an}} \xrightarrow{\mathrm{Perf}(-)} \mathrm{Cat}_{\mathbb{Z}}^{\mathrm{perf}} \longrightarrow \mathrm{Mod}_{\mathrm{THH}(\mathbb{Z})^h}(\mathrm{Sp}^{g<\mathbb{T}})_p^{\wedge},$$

*where the second functor is the composite functor*

$$\mathrm{Cat}_{\mathbb{Z}}^{\mathrm{perf}} \xrightarrow{\mathrm{THH}} \mathrm{Mod}_{\mathrm{THH}(\mathbb{Z})}(\mathrm{Sp}^{g<\mathbb{T}}) \xrightarrow{((-)^{\eta_{\mathrm{THH}(\mathbb{Z})}})_p^{\wedge}} \mathrm{Mod}_{\mathrm{THH}(\mathbb{Z})^h}(\mathrm{Sp}^{g<\mathbb{T}})_p^{\wedge},$$

*being the proposed localizing invariant extending  $\mathrm{THH}_{\Delta}$ .*

The strategy to prove Theorem 5.9 is similar to that for Corollary 4.12. However, the cyclonic spectrum  $\mathrm{THH}(\mathbb{Z})$  is more complicated than  $\mathrm{THH}(S)$  for perfectoid rings  $S$ . Nevertheless, the following two computational inputs, that we learnt from Robert BURKLUND and Jeremy HAHN, are sufficient.

**Lemma 5.11** ([BM94]). *Let  $p$  be a prime. Then we have*

$$\pi_*(\mathrm{THH}(\mathbb{Z})/p) \cong \mathbb{F}_p[\mu_1] \otimes \Lambda(\lambda_1)$$

*with  $\deg \mu_1 = 2p$  and  $\deg \lambda_1 = 2p - 1$ . Moreover, the cyclotomic Frobenius  $\mathrm{THH}(\mathbb{Z}) \rightarrow \mathrm{THH}(\mathbb{Z})^{tC_p}$  realizes  $\pi_*(\mathrm{THH}(\mathbb{Z})^{tC_p}/p)$  as  $\pi_*(\mathrm{THH}(\mathbb{Z})/p)[\mu_1^{-1}]$ .*

*Remark 5.12.* More generally, for every  $n \in \mathbb{Z}_{\geq -1}$ , [ACH24, Prop 2.7] shows that, for the  $n$ -th truncated Brown–Peterson spectrum  $\mathrm{BP}\langle n \rangle$ , we have

$$\pi_*(\mathrm{THH}(\mathrm{BP}\langle n \rangle; \mathbb{F}_p)) \cong \mathbb{F}_p[\mu_{n+1}] \otimes \Lambda(\lambda_1, \dots, \lambda_{n+1})$$

where  $\deg \mu_{n+1} = 2p^{n+1}$  and  $\deg \lambda_i = 2p^i - 1$  for  $i = 1, \dots, n+1$ . It follows from [HW22, Thm C] (explained in the paragraphs following [HRW22, Thm 6.1.4] as well) that the cyclotomic Frobenius  $\mathrm{THH}(\mathrm{BP}\langle n \rangle) \rightarrow \mathrm{THH}(\mathrm{BP}\langle n \rangle)^{tC_p}$  realizes the homotopy groups of the target modulo  $(p, v_1, \dots, v_n)$  as the localization of the source modulo  $(p, v_1, \dots, v_n)$  at  $\mu_{n+1}$ .

**Lemma 5.13** ([BM94]). *Let  $p$  be a prime. Then the class  $\mu_1 \in \pi_{2p}(\mathrm{THH}(\mathbb{Z})/p)$  lifts to a  $\mathbb{T}$ -equivariant class  $\tilde{\mu}_1$  satisfying the relation  $a_{\lambda} \tilde{\mu}_1 = v_1$ , where  $a_{\lambda}$  is the Euler class in [BL22, Ex 6.1.3], and  $v_1$  is the canonical class in  $\pi_{2p-2}(\mathrm{TC}^-(\mathbb{Z})/p)$ <sup>15</sup>.*

<sup>14</sup>Since the projective dimension of  $\mathbb{Z}$  is finite, this is the same as  $t$ -boundedness.

<sup>15</sup>When  $p$  is an odd prime, it comes from  $v_1 \in \pi_{2p-2}(\mathbb{S}/p)$ . When  $p = 2$ , over the sphere, only  $v_1^4 \in \pi_8(\mathbb{S}/2)$  is well-defined.

We give a proof of Proposition 5.15, since we need similar techniques to establish our main results.

**Lemma 5.14.** *Let  $M$  be a  $t$ -bounded  $\mathbb{T}$ -equivariant spectrum. Then for every composite number  $n \in \mathbb{N}_{>0}$ , the proper  $C_n$ -Tate construction<sup>16</sup>  $M^{\tau C_n}$  is contractible.*

*Proof.* We may assume that  $M$  is concentrated in degree 0. Then by the lax symmetric monoidal structure on  $(-)^{\tau C_n}$ , we may assume that  $M = \mathbb{Z}$ . Then this is established in [MNN19, Prop 5.24] (cf. [AMR17a, Rem 2.18]).  $\square$

**Proposition 5.15** (Devalapurkar–Hahn–Raksit–Yuan). *The spectrum  $(\mathrm{THH}(\mathbb{Z})^h)^{\Phi C_n}$  is contractible after profinite completion, for any composite number  $n \in \mathbb{N}_{>0}$ .*

*Proof.* Equivalently, we have to show that the proper  $C_n$ -Tate construction  $\mathrm{THH}(\mathbb{Z})^{\tau C_n}$  is contractible after profinite completion, for every composite number  $n \in \mathbb{N}_{>0}$ . Let  $p$  be a prime number. Note that  $\mathrm{THH}(\mathbb{Z})$  is equipped with proper Tate diagonal maps

$$\mathrm{THH}(\mathbb{Z}) \longrightarrow \mathrm{THH}(\mathbb{Z})^{\tau C_n}$$

as explained in [AMR17b, §0.4], thus  $\mathrm{THH}(\mathbb{Z})^{\tau C_n}$  acquires a  $\mathbb{Z}$ -module structure, and therefore we may check  $\mathrm{THH}(\mathbb{Z})^{\tau C_n} = 0$  after modulo  $(p, v_1)$ . It follows from Lemma 5.14 that

$$(\mathrm{THH}(\mathbb{Z})/(p, \tilde{\mu}_1))^{\tau C_n} = 0,$$

and by Lemma 5.13, we are done.  $\square$

**Corollary 5.16** (Devalapurkar–Hahn–Raksit–Yuan). *Let  $M$  be a  $\mathrm{THH}(\mathbb{Z})^h$ -module in  $\mathrm{Sp}^{g^{<\mathbb{T}}}$ . Then the genuine  $C_p$ -fixed points  $M^{C_p}$  and the geometric  $C_p$ -fixed points  $M^{\Phi C_p}$ , viewed as objects in  $\mathrm{Mod}_{\mathrm{THH}(\mathbb{Z})^{hC_p}}(\mathrm{Sp}^{g^{<(\mathbb{T}/C_p)})}$ , are Borel complete.*

Proposition 5.15 allows us to reduce to analyze de-completed  $C_p$ -Tate construction, by joint conservativity of  $\{(-)^{\Phi C_n} | n \in \mathbb{N}_{>0}\}$  and Remark 5.8:

**Corollary 5.17.** *Let  $M$  be a  $\mathrm{THH}(\mathbb{Z})^h$ -module in  $\mathrm{Sp}^{g^{<\mathbb{T}}}$ . Then the assembly map*

$$M^{\eta_{\mathrm{THH}(\mathbb{Z})^h}} \longrightarrow M^h$$

*is an equivalence after  $p$ -completion if and only if the assembly map*

$$M^{\eta_{\mathrm{THH}(\mathbb{Z})^h} C_p} \longrightarrow M^{tC_p}$$

*is an equivalence after  $p$ -completion.*

**Proof of Theorem 5.9** It follows from Corollary 5.17 and Lemma 5.18.  $\square$

It remains to establish the following version of Lemma 4.6 for  $\mathrm{THH}$ :

**Lemma 5.18.** *Let  $R$  be an  $\mathbb{E}_1$ - $\mathbb{Z}$ -algebra with bounded Tor-amplitude in  $D(\mathbb{Z})$ , and  $M$  a perfect  $R$ - $R$ -bimodule in  $D(\mathbb{Z})$ . Then the assembly map*

$$\mathrm{THH}(R; M^{\otimes_R^{\mathbb{L}} C_p})^{\theta_{\mathrm{THH}(\mathbb{Z})} C_p} \longrightarrow \mathrm{THH}(R; M^{\otimes_R^{\mathbb{L}} C_p})^{tC_p}$$

*is an equivalence after  $p$ -completion.*

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<sup>16</sup>See [AMR21, Def 4.5].

*Proof.* We mimic the proof of Lemma 4.6. By the exactness of both sides in  $M$ , we may assume that  $M$  is the free  $R$ - $R$ -bimodule  $R \otimes R$  of rank 1. Then the map in question becomes the assembly map

$$\mathrm{THH}(\mathbb{Z}; R^{\otimes_{\mathbb{Z}}^L p})^{\theta_{\mathrm{THH}(\mathbb{Z})} C_p} \longrightarrow \mathrm{THH}(\mathbb{Z}; R^{\otimes_{\mathbb{Z}}^L C_p})^{tC_p}$$

which follows from Lemma 5.19, being a polygonic THH version of Lemma 4.7.  $\square$

**Lemma 5.19.** *Let  $M$  be a  $\mathbb{Z}$ -module spectrum of bounded Tor-amplitude. Then the assembly map*

$$\mathrm{THH}(\mathbb{Z}; M^{\otimes_{\mathbb{Z}}^L p})^{\theta_{\mathrm{THH}(\mathbb{Z})} C_p} \longrightarrow \mathrm{THH}(\mathbb{Z}; M^{\otimes_{\mathbb{Z}}^L C_p})^{tC_p}$$

*is an equivalence after  $p$ -completion.*

*Proof.* Again, since both sides are exact in  $M$ , we may assume that  $M$  is a flat  $\mathbb{Z}$ -module, and the result is true for finite free  $\mathbb{Z}$ -modules  $M$ . By Lazard's theorem, the flat  $\mathbb{Z}$ -module  $M$  is a filtered colimit of finite free modules, thus it suffices to show that the functor

$$\begin{aligned} \mathrm{Mod}_{\mathbb{Z}}^b &\longrightarrow \mathrm{Sp}_p^\wedge \\ M &\longmapsto (\mathrm{THH}(\mathbb{Z}; M^{\otimes_{\mathbb{Z}}^L p})^{tC_p})_p^\wedge \end{aligned}$$

preserves filtered colimits, where  $\mathrm{Mod}_{\mathbb{Z}}^b$  is the category of flat  $\mathbb{Z}$ -modules. Note that  $\mathrm{THH}(\mathbb{Z}; M^{\otimes_{\mathbb{Z}}^L p})^{tC_p}$  acquires a  $\mathrm{THH}(\mathbb{Z})^{tC_p}$ -module structure, and in particular, a  $\mathbb{Z}$ -module structure, thus  $v_1 = 0$  on it. Now by Lemma 5.13, we have

$$\mathrm{THH}(\mathbb{Z}; M^{\otimes_{\mathbb{Z}}^L p})^{tC_p} / (p, v_1) \simeq (\mathrm{THH}(\mathbb{Z}; M^{\otimes_p^L p}) / (p, \tilde{\mu}_1))^{tC_p}$$

and as (non-equivariant) spectra, we have

$$\mathrm{THH}(\mathbb{Z}; M^{\otimes_{\mathbb{Z}}^L p}) / (p, \tilde{\mu}_1) \simeq (\mathrm{THH}(\mathbb{Z}) / (p, \widetilde{\mu}_1)) \otimes_{\mathbb{Z}}^L M^{\otimes_{\mathbb{Z}}^L p}$$

which is bounded, and preserves filtered colimits in  $M$ . The result follows from the fact that  $(-)^{tC_p}$  preserves uniformly bounded filtered colimits.  $\square$

Finally, we relate de-completed homotopy fixed points and de-completed Tate construction to de-completed Borel completion when the base is  $\mathrm{THH}(\mathbb{Z})^h$ . We would thank Georg TAMME for such a question. The point is that, as in the proofs of Proposition 5.15 and Lemma 5.18, after  $(p, v_1)$ -completion, the base  $\mathrm{THH}(\mathbb{Z})^h$  is close to be  $t$ -bounded. We start with some simple lemmas.

**Lemma 5.20.** *Let  $V \in \mathrm{Sp}^\omega$  be a finite spectrum, and  $A$  an  $\mathbb{E}_1$ -ring such that  $A \otimes V = 0$ . Then for every right  $A$ -module  $M$ , we have  $M \otimes V = 0$ .*

*Proof.* Note that  $M$  is a retract of  $M \otimes A$  as a spectrum, thus  $M \otimes V$  is a retract of  $M \otimes A \otimes V = 0$  as a spectrum.  $\square$

*Remark 5.21.* In our applications, we mainly take  $V$  to be  $\mathbb{S}/(p, v_1^4)$  (or equivalently,  $\mathbb{S}/(p, v_1)$  when  $p = 2$ ).

**Lemma 5.22.** *Let  $V \in \mathrm{Sp}^\omega$  be a finite spectrum, and  $A_0 \rightarrow A$  a map of  $\mathbb{E}_1$ -algebras in  $\mathrm{Sp}^{B\mathbb{T}}$  such that  $A_0 \otimes V$  is  $t$ -bounded below. Suppose that the spectrum  $A_0^{tC_p} \otimes V$ , which is in fact equivalent to  $(A_0 \otimes V)^{tC_p}$ , is contractible. Then*

- (1) *for every right  $A$ -module  $M$ , the spectrum  $(M^{t\mathbb{T}})_p^\wedge \otimes V$  is contractible; and*
- (2) *the functor*

$$((-)^{h\mathbb{T}})_p^\wedge \otimes V : \mathrm{RMod}_A(\mathrm{Sp}^{B\mathbb{T}}) \longrightarrow \mathrm{Sp}_p^\wedge$$

*preserves filtered colimits.*

*Proof.* For the contractibility of  $(M^{t\mathbb{T}})_p^\wedge \otimes V$ , or equivalently, that of  $M^{t\mathbb{T}} \otimes (V/p)$ , by Lemma 5.20 and the lax symmetric monoidal structure on  $(-)^{t\mathbb{T}}$ , it suffices to show the contractibility of  $(A_0^{t\mathbb{T}})_p^\wedge \otimes V$ . But then

$$\begin{aligned} (A_0^{t\mathbb{T}})_p^\wedge \otimes V &\simeq ((A_0 \otimes V)^{t\mathbb{T}})_p^\wedge \\ &\simeq ((A_0 \otimes V)^{tC_p})^{h(\mathbb{T}/C_p)} \\ &\simeq 0, \end{aligned}$$

where the second equivalence follows from the  $t$ -bounded-belowness of  $A_0 \otimes V$  and [NS18, Lem II.4.2].

The filtered-colimit-preservation of the functor  $((-)^{h\mathbb{T}})_p^\wedge \otimes V$  follows directly from the canonical fiber sequence

$$\Sigma(-)_{h\mathbb{T}} \xrightarrow{\text{Nm}_{\mathbb{T}}} (-)^{h\mathbb{T}} \longrightarrow (-)^{t\mathbb{T}}$$

and the Tate vanishing established above.  $\square$

**Proposition 5.23.** *Let  $A_0 \rightarrow A$  be a map of  $\mathbb{E}_1$ -algebras in  $\text{Sp}^{B\mathbb{T}}$ , and  $V \in \text{Sp}^\omega$  a finite spectrum such that  $A_0 \otimes V$  is  $t$ -bounded. Then*

- (1) *every  $A^{C_p}$ -module in  $\text{Sp}^{g^<(\mathbb{T}/C_p)}$  is  $p$ -completely<sup>17</sup> Borel-complete up to Bousfield  $V$ -localization; and*
- (2) *the  $p$ -completed de-completed homotopy fixed points functor*

$$((-)^{\eta_A \mathbb{T}})_p^\wedge : \text{RMod}_A(\text{Sp}^{g^<\mathbb{T}}) \longrightarrow \text{Sp}_p^\wedge$$

*coincides with the  $p$ -completed genuine  $C_{p^\infty}$ -fixed points functor up to Bousfield  $V$ -localization, and the  $p$ -completed de-completed Tate construction functor*

$$((-)^{\theta_A \mathbb{T}})_p^\wedge : \text{RMod}_A(\text{Sp}^{g^<\mathbb{T}}) \longrightarrow \text{Sp}_p^\wedge$$

*coincides with the composite functor*

$$(2) \quad \text{RMod}_A(\text{Sp}^{g^<\mathbb{T}}) \xrightarrow{(-)^{\Phi_{C_p}}} \text{RMod}_{A^{\Phi_{C_p}}}(\text{Sp}^{g^<(\mathbb{T}/C_p)}) \xrightarrow{(-)^{C_{p^\infty}/C_p}} \text{Sp}_p^\wedge$$

*up to Bousfield  $V$ -localization.*

*Proof.* We start with the first statement. Firstly, we check that, for every  $r \geq 1$ , we have  $(A_0^{C_p})^{\Phi_{C_{p^r}}} \otimes (V/\mathbb{L}p) \simeq 0$ . By “genuine” Tate-orbit lemma [AN21, Lem 5.3], the canonical map

$$(A_0^{C_p})^{\Phi_{C_{p^r}}} \otimes (V/p) = ((A_0 \otimes V/p)^{C_p})^{\Phi_{C_{p^r}}} \longrightarrow ((A_0 \otimes V/p)^{\Phi_{C_p}})^{\Phi_{C_{p^r}}} = (A_0 \otimes V/p)^{\Phi_{C_{p^{r+1}}}}$$

is an equivalence. By Lemma 5.14, we have  $(A_0 \otimes V/p)^{\Phi_{C_{p^{r+1}}}} \simeq 0$ , and by Lemma 5.20, we see that  $(A_0 \otimes V/p)^{\tau_{C_{p^{r+1}}}} \simeq 0$ .

It follows that  $A_0^{C_p}$  is  $p$ -completely Borel-complete up to Bousfield  $V$ -localization, as by  $p$ -completeness, it suffices to check the isomorphisms  $(-)^{\Phi_H} \rightarrow (-)^{\tau_H}$  on cyclic  $p$ -groups  $H$ . Then it follows from Lemma 5.20 and the lax symmetric monoidal structure on  $(-)^{C_m}$  and on the map  $(-)^{\Phi_{C_m}} \rightarrow (-)^{\tau_{C_m}}$  that

- (1)  $(A^{C_p})^{\Phi_{C_m}} \otimes V/p = 0$ , and thus
- (2) for every right  $A^{C_p}$ -module  $M$  in  $\text{Sp}^{g^<(\mathbb{T}/C_p)}$ , we have  $M^{\tau_{C_m}} \otimes V/p = M^{\Phi_{C_m}} \otimes V/p = 0$ .

In particular, every right  $A^{C_p}$ -module in  $\text{Sp}^{g^<(\mathbb{T}/C_p)}$  is  $p$ -completely Borel-complete up to Bousfield  $V$ -localization.

Now we prove the second statement. By construction, on compact objects, we have the desired results, essentially by [AMR21, Prop 5.9] and Propositions B.3 and B.4. It suffices to show that the functors  $(-)^{C_{p^\infty}}$  and  $((-)^{\Phi_{C_p}})^{C_{p^\infty}/C_p}$  preserve filtered colimits.

<sup>17</sup>Equivalently, it is saying that every  $A^{C_p}$ -module in  $\text{Sp}^{g^<(\mathbb{T}/C_p)}$  is Borel-complete up to Bousfield  $V/p$ -localization.

Note that  $(-)^{C_{p^\infty}} = ((-)^{C_p})^{C_{p^\infty}/C_p}$ , that is to say, we may write both functors  $(-)^{C_{p^\infty}}$  and  $((-)^{\Phi_{C_p}})^{C_{p^\infty}/C_p}$  as a composite

$$\mathrm{RMod}_A(\mathrm{Sp}^{g^{<\mathbb{T}}}) \longrightarrow \mathrm{RMod}_{A^{C_p}}(\mathrm{Sp}^{g^{<(\mathbb{T}/C_p)}}) \xrightarrow{(-)^{C_{p^\infty}/C_p}} \mathrm{Sp}_p^\wedge,$$

where we are tacitly using the lax symmetric monoidal structure on the natural transformation  $(-)^{C_p} \Rightarrow (-)^{\Phi_{C_p}}$ . Note that the first functor  $(-)^{\Phi_{C_p}}$  or  $(-)^{C_p}$  preserves filtered colimits, thus it suffices to see that the second functor  $(-)^{C_{p^\infty}/C_p}$  preserves filtered colimits up to Bousfield  $V$ -localization.

We are now reduced to show that the functor

$$\mathrm{RMod}_{A^{hC_p}}(\mathrm{Sp}^{B(\mathbb{T}/C_p)}) \xrightarrow{(-)^{h(\mathbb{T}/C_p) \otimes V}} \mathrm{Sp}_p^\wedge$$

preserves filtered colimits. By examining the fiber sequence

$$\Sigma(-)_{h(\mathbb{T}/C_p)} \xrightarrow{\mathrm{Nm}_{\mathbb{T}/C_p}} (-)^{h(\mathbb{T}/C_p)} \longrightarrow (-)^{t(\mathbb{T}/C_p)}$$

It remains to show that, for every right  $A^{hC_p}$ -module  $M$  in  $\mathrm{Sp}^{B(\mathbb{T}/C_p)}$ , we have  $(M^{t(\mathbb{T}/C_p)})_p^\wedge \otimes V = 0$ . Actually, for every right  $A_0^{hC_p}$ -module  $M$ , the lax symmetric monoidal structure on  $(-)^{t(\mathbb{T}/C_p)}$  gives rise to a  $((A_0^{hC_p})^{t(\mathbb{T}/C_p)})_p^\wedge$ -module structure on  $(M^{t(\mathbb{T}/C_p)})_p^\wedge$ , and since  $A_0 \otimes V$  is  $t$ -bounded, by Lemma 5.24, we have

$$\begin{aligned} ((A_0^{hC_p})^{t(\mathbb{T}/C_p)})_p^\wedge \otimes V &\simeq (((A_0 \otimes V)^{hC_p})^{t(\mathbb{T}/C_p)})_p^\wedge \\ &\simeq 0 \end{aligned}$$

and thus  $(M^{t(\mathbb{T}/C_p)})_p^\wedge \otimes V = 0$  by Lemma 5.20.  $\square$

We need the following consequence of the Tate fixed point lemma.

**Lemma 5.24.** *Let  $M$  be a  $t$ -bounded  $\mathbb{T}$ -equivariant spectrum. Then  $((M^{hC_p})^{t(\mathbb{T}/C_p)})_p^\wedge = 0$ .*

*Proof.* By dévissage, we reduce to the case that  $M$  is concentrated in degree 0. In this case, we have

$$\begin{aligned} ((M^{hC_p})^{t(\mathbb{T}/C_p)})/p &\simeq (M^{hC_p})^{t(C_{p^2}/C_p)} \\ &\simeq 0, \end{aligned}$$

by the Tate fixed point lemma and [NS18, Lem IV.4.12].  $\square$

*Example 5.25.* Let  $k$  be an animated ring, and  $\mathcal{C}$  a presentable stable  $k$ -linear  $\infty$ -category. Then by Proposition 5.23, we have

$$\begin{aligned} \mathrm{HC}^{-, \mathrm{poly}}(\mathcal{C}/k)_p^\wedge &= ((\mathrm{HH}(\mathcal{C}/\underline{k})^{\eta_{\underline{k}}})^{C_{p^\infty}})_p^\wedge, \\ \mathrm{HP}^{\mathrm{poly}}(\mathcal{C}/k)_p^\wedge &= \left( ((\mathrm{HH}(\mathcal{C}/\underline{k})^{\eta_{\underline{k}}})^{\Phi_{C_p}})^{h(C_{p^\infty}/C_p)} \right)_p^\wedge \\ &= ((\mathrm{THH}(\mathcal{C}) \otimes_{\mathrm{THH}(k)}^{\mathbb{L}} k^{tC_p})^{h(\mathbb{T}/C_p)})_p^\wedge. \end{aligned}$$

In retrospect, the object  $(\mathrm{THH}(\mathcal{C}) \otimes_{\mathrm{THH}(k)}^{\mathbb{L}} k^{tC_p})^{h(\mathbb{T}/C_p)}$  appeared [Dev23, Rem 3.5.6] in the even more general setup of  $k$  being an  $\mathbb{E}_\infty$ -ring.

Now we deduce the comparison over  $\mathrm{THH}(R)^h$  for every  $\mathbb{E}_\infty$ - $\mathbb{Z}$ -algebra  $R$ .

**Corollary 5.26.** *Let  $A$  be an  $\mathbb{E}_1$ - $\mathrm{THH}(\mathbb{Z})$ -algebra in  $\mathrm{Sp}^{B\mathbb{T}}$ . Then the  $(p, v_1)$ -completed de-completed homotopy fixed points functor*

$$((-)^{\eta_A \mathbb{T}})_{(p, v_1)}^\wedge : \mathrm{RMod}_A(\mathrm{Sp}^{g^{<\mathbb{T}}}) \longrightarrow \mathrm{Sp}_{(p, v_1)}^\wedge$$

coincides with the  $(p, v_1)$ -completed genuine  $C_{p^\infty}$ -fixed points functor, and the  $(p, v_1)$ -completed de-completed Tate construction functor

$$((-)^{\theta_A \mathbb{T}})_{(p, v_1)}^\wedge : \text{Mod}_A(\text{Sp}^{g^{< \mathbb{T}}}) \longrightarrow \text{Sp}_{(p, v_1)}^\wedge$$

coincides with the composite functor

$$\text{Mod}_A(\text{Sp}^{g^{< \mathbb{T}}}) \xrightarrow{(-)^{\Phi_{C_p}}} \text{Mod}_{A^{\Phi_{C_p}}}(\text{Sp}^{g^{< (\mathbb{T}/C_p)})} \xrightarrow{(-)^{C_{p^\infty}/C_p}} \text{Sp}_{(p, v_1)}^\wedge.$$

*Proof.* As a very special case of [HW22, Thm G], the spectrum  $\text{TR}(\mathbb{Z})/(p, v_1)$  is  $t$ -bounded. The result then follows from Proposition 5.23 by taking  $V = \mathbb{S}/(p, v_1^4)$  (or  $\mathbb{S}/(p, v_1)$  when  $p > 2$ ), and  $A_0 = \text{TR}(\mathbb{Z})^h$ .  $\square$

*Remark 5.27.* In Corollary 5.26, we can replace  $\mathbb{Z}$  by  $\text{BP}\langle n \rangle$ , and  $(p^{\alpha_0}, v_1^{\alpha_1}, \dots, v_{n+1}^{\alpha_{n+1}})$  in place of  $(p, v_1^4)$  for some  $(\alpha_0, \dots, \alpha_{n+1}) \in \mathbb{N}_{>0}^{n+1}$ , for which we need the full [HW22, Thm G].

*Remark 5.28.* Let  $S$  be a perfectoid ring. Then the image of  $\mu_1 \in \pi_{2p}(\text{THH}(\mathbb{Z})/p)$  under the map  $\text{THH}(\mathbb{Z}) \rightarrow \text{THH}(S)$  is  $\sigma^p$  up to a multiplier of a unit, where  $\sigma \in \pi_2(\text{THH}(S)_p^\wedge)$  is the Bökstedt element. To see this, when  $S = \mathbb{F}_p$ , this is established in [BM94], which implies the case for  $S$  being a perfect  $\mathbb{F}_p$ -algebra, since the map  $\text{THH}(\mathbb{Z}) \rightarrow \text{THH}(S)$  factors through  $\text{THH}(\mathbb{F}_p)$ . The general case follows by considering the composite

$$\text{THH}(\mathbb{Z}) \longrightarrow \text{THH}(S) \longrightarrow \text{THH}\left(S/\sqrt{pS}\right)$$

and note that  $S/\sqrt{pS}$  is a perfect  $\mathbb{F}_p$ -algebra.

*Example 5.29.* Let  $S$  be a perfectoid ring, and  $\mathcal{C}$  a dualizable presentable stable  $S$ -linear  $\infty$ -category. Then we have

$$\begin{aligned} \text{TC}^{-, \text{poly}}/S(\mathcal{C}/S)_{(p, \xi)}^\wedge &= ((\text{THH}(\mathcal{C})^{\eta_{\text{THH}(S)}})^{C_{p^\infty}})_{(p, \xi)}^\wedge, \\ \text{TP}^{\text{poly}}/S(\mathcal{C}/S)_{(p, \xi)}^\wedge &= (((\text{THH}(\mathcal{C})^{\eta_{\text{THH}(S)}})^{\Phi_{C_p}})^{C_{p^\infty}/C_p})_{(p, \xi)}^\wedge. \end{aligned}$$

Indeed, they follow from Corollary 5.26 and Remark 5.28 by taking  $A := \text{THH}(S)$  (where  $a_\lambda \tilde{\sigma} = \xi$  for the generator  $\tilde{\sigma} \in \pi_2(\text{TC}^-(S)_p^\wedge)$  up to a complex orientation, and  $a_\lambda \tilde{\mu}_1 = v_1$ ).

Finally, we show that de-completed Borel completion over  $\text{THH}(S)$  for perfectoid rings  $S$ , and over  $\mathbb{Z}$ , are “the same” as over  $\text{THH}(\mathbb{Z})$ .

**Proposition 5.30.** *The assembly map  $\text{THH}(S)^{\eta_{\text{THH}(\mathbb{Z})}} \rightarrow \text{THH}(S)^h$  in  $\text{Mod}_{\text{THH}(\mathbb{Z})^h}(\text{Sp}^{g^{< \mathbb{T}}})$  is an equivalence after  $p$ -completion.*

*Proof.* By Corollary 5.17 and Lemmas 2.10 and 5.6, it is equivalent to show that the map

$$\text{THH}(S) \otimes_{\text{THH}(\mathbb{Z})}^{\mathbb{L}} \text{THH}(\mathbb{Z})^{tC_p} \longrightarrow \text{THH}(S)^{tC_p}$$

induced by the cyclotomic Frobenius  $\text{THH}(S) \rightarrow \text{THH}(S)^{tC_p}$  is an equivalence after modulo  $p$ , but this follows immediately from Remark 5.28.  $\square$

It then follows from Lemma 5.6 that

**Corollary 5.31.** *Let  $M$  be a  $\text{THH}(S)$ -module in  $\text{Sp}^{g^{< \mathbb{T}}}$ . Then the assembly map*

$$M^{\eta_{\text{THH}(\mathbb{Z})}} \longrightarrow M^{\eta_{\text{THH}(S)}}$$

*in  $\text{Mod}_{\text{THH}(\mathbb{Z})^h}(\text{Sp}^{g^{< \mathbb{T}}})$  is an equivalence after  $p$ -completion.*

*Remark 5.32.* More generally, we may replace  $\mathbb{Z}$  by any truncated Brown–Peterson spectrum. Indeed, for  $-1 \leq n < m$ , the assembly map

$$\mathrm{THH}(\mathrm{BP}\langle n \rangle)^{\eta_{\mathrm{THH}(\mathrm{BP}\langle m \rangle)}} \longrightarrow \mathrm{THH}(\mathrm{BP}\langle n \rangle)^h$$

in  $\mathrm{RMod}_{\mathrm{THH}(\mathrm{BP}\langle m \rangle)^h}(\mathrm{Sp}^{g^{<\mathbb{T}}})$  is an equivalence after  $(p, v_1, \dots, v_n)$ -completion, as a consequence of Remark 5.12. In particular, this shows that  $\mathrm{THH}_{\Delta}$  extends to a localizing invariant

$$\mathrm{Cat}_{\mathrm{BP}\langle n \rangle}^{\mathrm{perf}} \xrightarrow{\mathrm{THH}} \mathrm{RMod}_{\mathrm{THH}(\mathrm{BP}\langle n \rangle)}(\mathrm{Sp}^{g^{<\mathbb{T}}}) \xrightarrow{(-)^{\eta_{\mathrm{THH}(\mathrm{BP}\langle n \rangle)}}} \mathrm{RMod}_{\mathrm{THH}(\mathrm{BP}\langle n \rangle)^h}(\mathrm{Sp}^{g^{<\mathbb{T}}})_{(p, v_1, \dots, v_n)}^{\wedge}$$

which is independent of choice of  $n \in \mathbb{Z}_{\geq -1}$  in the sense of compatibility with forgetful functors.

On the other hand, we can also deduce the result for  $\mathbb{Z}$  from the case  $S = \mathbb{F}_p$ :

**Proposition 5.33.** *The assembly map  $\underline{\mathbb{Z}}^{\eta_{\mathrm{THH}(\mathbb{Z})}} \longrightarrow \mathbb{Z}$  in  $\mathrm{Mod}_{\mathrm{THH}(\mathbb{Z})^h}(\mathrm{Sp}^{g^{<\mathbb{T}}})$  is an equivalence.*

*Proof.* Since  $\underline{\mathbb{Q}} \in \mathrm{CAlg}(\mathrm{Sp}^{g^{<\mathbb{T}}})$  is Borel, it suffices to show this equivalence after modulo every prime  $p$ . By Corollary 5.17 and Lemmas 2.10 and 5.6, we are reduced to show that the map

$$\tau_{\geq 0}(\mathbb{F}_p^{tC_p}) \otimes_{\mathrm{THH}(\mathbb{Z})}^{\mathbb{L}} \mathrm{THH}(\mathbb{Z})^{tC_p} \longrightarrow \mathbb{F}_p^{tC_p}$$

is an equivalence, where the  $\mathbb{E}_{\infty}$ -map  $\mathrm{THH}(\mathbb{Z}) \rightarrow \tau_{\geq 0}(\mathbb{F}_p^{tC_p})$  is the composite

$$\mathrm{THH}(\mathbb{Z}) \xrightarrow{\simeq} \tau_{\geq 0}(\mathrm{THH}(\mathbb{Z})^{tC_p}) \longrightarrow \tau_{\geq 0}(\mathbb{Z}^{tC_p}) \longrightarrow \tau_{\geq 0}(\mathbb{F}_p^{tC_p})$$

where the first equivalence is the Segal conjecture for  $\mathrm{THH}(\mathbb{Z})$ . Now this composite can be rewritten as the composite

$$\mathrm{THH}(\mathbb{Z}) \xrightarrow{\simeq} \tau_{\geq 0}(\mathrm{THH}(\mathbb{Z})^{tC_p}) \longrightarrow \tau_{\geq 0}(\mathrm{THH}(\mathbb{F}_p)^{tC_p}) \longrightarrow \tau_{\geq 0}(\mathbb{F}_p^{tC_p}),$$

and we have a composite pushout diagram

$$\begin{array}{ccccc} \tau_{\geq 0}(\mathrm{THH}(\mathbb{Z})^{tC_p}) & \longrightarrow & \tau_{\geq 0}(\mathrm{THH}(\mathbb{F}_p)^{tC_p}) & \longrightarrow & \tau_{\geq 0}(\mathbb{F}_p^{tC_p}) \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{THH}(\mathbb{Z})^{tC_p} & \longrightarrow & \mathrm{THH}(\mathbb{F}_p)^{tC_p} & \longrightarrow & \mathbb{F}_p^{tC_p} \end{array}$$

of  $\mathbb{E}_{\infty}$ -rings, where the left pushout is Proposition 5.30, and the right pushout is simply detected by both vertical maps are inverting the Bökstedt element  $\mu_0 \in \pi_2(\mathrm{THH}(\mathbb{F}_p)^{tC_p})$ .  $\square$

*Remark 5.34.* Let  $p$  be an odd prime. [DR25, Prop 5.2.1] implies that the map  $\mathrm{THH}(\mathbb{Z})^{tC_p} \rightarrow \mathbb{Z}^{tC_p}$  coincides with the map  $\mathrm{THH}(\mathbb{Z})^{tC_p} \rightarrow \mathrm{THH}(\mathbb{F}_p)^{tC_p}$  induced by the ring map  $\mathbb{Z} \rightarrow \mathbb{F}_p$ .

*Question 4.* Is the statement of Remark 5.34 true when  $p = 2$ ?

**Corollary 5.35.** *Let  $M$  be a  $\underline{\mathbb{Z}}$ -module in  $\mathrm{Sp}^{g^{<\mathbb{T}}}$ . Then the assembly map*

$$M^{\eta_{\mathrm{THH}(\mathbb{Z})}} \longrightarrow M^{\eta_{\underline{\mathbb{Z}}}}$$

*in  $\mathrm{Mod}_{\mathrm{THH}(\mathbb{Z})^h}(\mathrm{Sp}^{g^{<\mathbb{T}}})$  is an equivalence.*

## 6. NONCOMMUTATIVE CRYSTALLINE-DE RHAM COMPARISON REVISITED

In this section, we explain that original ideas in [PV19], along with an observation by A. RAKSIT, adapt to de-completed case, which leads to a comparison (Corollary 6.28), which “authentically” corresponds to the crystalline-de Rham comparison. As explained in Remark 6.27, the argument in [DR25] is enough for this comparison when  $p$  is an odd prime, so our result is stronger only when  $p = 2$ .

The key is to produce a map  $\mathrm{HH}(\mathbb{F}_p/\mathbb{Z})^{\theta_{\mathbb{Z}}C_p} \rightarrow \mathbb{Z}^{tC_p}$  (Construction 6.15). As A. RAKSIT observed, up to  $(-)^{tC_p}$ , we can replace  $\mathrm{HH}(\mathbb{Z}[t]/\mathbb{Z})$  by  $\mathrm{dR}_{\mathbb{Z}[t]/\mathbb{Z}}$ , and consequently, we can identify  $\mathrm{HH}(\mathbb{F}_p/\mathbb{Z})^{\theta_{\mathbb{Z}}C_p}$  with  $\mathrm{dR}_{\mathbb{F}_p/\mathbb{Z}}^{tC_p}$ . Combining with the map  $\mathrm{dR}_{\mathbb{F}_p/\mathbb{Z}} \rightarrow \mathbb{Z}_{(p)}$  induced by the PD-structure on  $(p) \subseteq \mathbb{Z}_{(p)}$ , we get the map we want. As a consequence, we get a map  $\mathrm{THH}(\mathbb{F}_p)^{tC_p} \rightarrow \mathrm{HH}(\mathbb{F}_p/\mathbb{Z})^{\theta_{\mathbb{Z}}C_p} \rightarrow \mathbb{Z}^{tC_p}$ . We then show that this is an equivalence (Proposition 6.25), the blueprint of whose proof can be found after Construction 6.16.

We start with A. RAKSIT’s observation. Let  $k$  be a commutative ring, and  $R$  a smooth commutative  $k$ -algebra. In general, we are only equipped with an HKR-filtration on the periodic cyclic homology whose associated graded pieces are equivalent to shifts of Hodge-completed derived de Rham cohomology of  $R/k$ . However, when  $R$  is of relative dimension  $\leq 1$ , the filtration in question, along with the commutative ring structure<sup>18</sup>, is simply extended from the filtration on  $k^{t\mathbb{T}}$ , which was observed by A. RAKSIT as in Construction 6.13. A slight improvement of his argument gives a stronger result: we compare associated  $[0, 1]$ -graded pieces of HKR-filtered Hochschild homology and Hodge-filtered de Rham cohomology in Corollary 6.12. For this purpose, we first review some facts on homotopy coherent cochain complexes with weights in  $[0, 1]$ , which is systematically studied in K. MAGIDSON’s work [Mag24].

**Notation 6.1.** Let  $k$  be a commutative ring. We denote by  $\mathrm{DG}_- \mathrm{DAlg}_k$  the  $\infty$ -category of  $h_-$ -differential graded derived commutative  $k$ -algebras [Rak20, Nota 5.3.1], and  $\mathrm{DG}_-^{[0,1]} \mathrm{DAlg}_k \subseteq \mathrm{DG}_- \mathrm{DAlg}_k$  the full subcategory of objects whose weights are concentrated in degrees  $[0, 1]$ .

*Remark 6.2* ([Mag24, §3.3]). Let  $k$  be a commutative ring. Then the  $\infty$ -category  $\mathrm{DG}_-^{[0,1]} \mathrm{DAlg}_k$  is equivalent to the  $\infty$ -category of triples  $(A, M, \eta)$  where

- $A$  is a derived commutative  $k$ -algebra;
- $M$  is an  $A$ -module; and
- $\eta : A \rightarrow M$  is a derivation (or equivalently, a map  $L_{A/k} \rightarrow M$  of  $A$ -modules).

Given a triple  $(A, M, \eta)$  as above, the corresponding object in  $\mathrm{DG}_-^{[0,1]} \mathrm{DAlg}_k$  is equivalent to the square-zero extension of  $A(0)$  by  $M(1)[-1]$  determined by the derivation  $\eta$ , whose underlying  $k$ -module spectrum is equivalent to  $\mathrm{fib}(A \rightarrow M)$ .

The key construction is the following:

**Construction 6.3.** Let  $k$  be a commutative ring. We construct a functor

$$\mathrm{DG}_-^{[0,1]} \mathrm{DAlg}_k \longrightarrow \mathrm{DAlg}(D(k)^{B\mathbb{T}})$$

as follows. Let  $(A, M, \eta)$  be a triple corresponding to an object of  $\mathrm{DG}_-^{[0,1]} \mathrm{DAlg}_k$  as in Remark 6.2. Equip  $A$  and  $M$  with trivial  $\mathbb{T}$ -action, we get a derivation  $\eta : A \rightarrow M$  in  $\mathrm{DAlg}(D(k)^{B\mathbb{T}})$ . Now we compose this derivation with the map  $M = M \otimes \mathbb{S} \xrightarrow{M \otimes a_\lambda} M \otimes \mathbb{S}^\lambda = \Sigma^\lambda M$ , obtaining another derivation  $\tau : A \rightarrow \Sigma^\lambda M$  in  $\mathrm{DAlg}(D(k)^{B\mathbb{T}})$ . We take the square-zero extension  $\left[ A \xrightarrow{\tau} \Sigma^\lambda M \right]$  of  $A$  by  $\Sigma^\lambda M[-1]$  determined by the

<sup>18</sup>The key is the commutative ring structure.

derivation  $\tau$ . This gives rise to a functor

$$\begin{aligned} \mathrm{DG}_{-}^{[0,1]} \mathrm{DAlg}_k &\longrightarrow \mathrm{DAlg}(D(k)^{B\mathbb{T}}) \\ (A, M, \eta) &\longmapsto [A \xrightarrow{\tau} \Sigma^\lambda M]. \end{aligned}$$

Moreover, the map  $M \otimes a_\lambda : M \rightarrow \Sigma^\lambda M$  induces a map  $[A \xrightarrow{\eta} M] \rightarrow [A \xrightarrow{\tau} \Sigma^\lambda M]$  of derived commutative algebras in  $D(k)^{B\mathbb{T}}$ , where  $[A \xrightarrow{\eta} M]$  is equipped with trivial  $\mathbb{T}$ -action.

*Remark 6.4.* In fact, in Construction 6.3, we can endow the target with a filtered circle action. Since this notion is not used in the text, we will not discuss this filtered enhancement.

We need the following categorification of Tate construction.

**Definition 6.5** ([PV19, §2]). Let  $k$  be a commutative ring spectrum. The  $\mathbb{T}$ -Tate category  $D(k)^{t\mathbb{T}}$  is the quotient category

$$D(k)^{B\mathbb{T}} / \langle M \otimes \mathbb{T} | M \in D(k) \rangle$$

where  $\langle M \otimes \mathbb{T} | M \in D(k) \rangle \subseteq D(k)^{B\mathbb{T}}$  is the stable subcategory generated by induced  $\mathbb{T}$ -representations.

The magic of Construction 6.3 is that it does not change the Tate construction, which is formally formulated as follows.

**Proposition 6.6.** Let  $k$  be a commutative ring, and  $(A, M, \eta)$  a triple corresponding to an object of  $\mathrm{DG}_{-}^{[0,1]} \mathrm{DAlg}_k$ . Then the map

$$[A \xrightarrow{\eta} M] \longrightarrow [A \xrightarrow{\tau} \Sigma^\lambda M]$$

in  $\mathrm{DAlg}(D(k)^{B\mathbb{T}})$  constructed in Construction 6.3, where  $[A \xrightarrow{\eta} M]$  is equipped with trivial  $\mathbb{T}$ -action, is an equivalence after passing along the quotient functor  $D(k)^{B\mathbb{T}} \rightarrow D(k)^{t\mathbb{T}}$ .

*Proof.* The point is that  $\mathrm{fib}(a_\lambda : \mathbb{S} \rightarrow \mathbb{S}^\lambda) \simeq \mathbb{S} \otimes \mathbb{T}$  is an induced  $\mathbb{T}$ -representation.  $\square$

Let  $k \rightarrow R$  be a smooth map of commutative algebras. The next goal, Corollary 6.12, is to show that, if we apply this construction to the truncated de Rham complex

$$0 \longrightarrow R \xrightarrow{d} \Omega_{R/k}^1 \longrightarrow 0$$

we get the associated  $[0, 1]$ -graded piece of the Hochschild homology  $\mathrm{HH}(R/k)$ . We explain this more formally for maps  $k \rightarrow R$  of derived commutative rings as follows.

*Remark 6.7.* Let  $k$  be a derived commutative ring. The forgetful functor  $D(k)^{B\mathbb{T}} = \mathrm{coAlg}_{k[\mathbb{T}]^\vee}(D(k)) \rightarrow D(k)$  maps derived commutative algebras in  $D(k)^{B\mathbb{T}}$  to derived commutative  $k$ -algebras, and it carries the adic filtration, as defined in [Mag24, Cons 3.2.24], on a map  $f$  in  $\mathrm{DAlg}(D(k)^{B\mathbb{T}})$  to the adic filtration on the underlying map  $f$  in  $\mathrm{DAlg}_k$ .

**Lemma 6.8.** Let  $k \rightarrow A$  be a map of derived commutative rings. Then the HKR-filtration on  $\mathrm{HH}(A/k)$ , as a filtered derived commutative algebra in  $D(k)^{B\mathbb{T}}$ , coincides with the adic filtration on the map  $\mathrm{HH}(A/k) \rightarrow A$  in  $\mathrm{DAlg}(D(k)^{B\mathbb{T}})$ .

*Proof.* By the universal property of adic filtration (defined as a left adjoint in [Mag24, Cons 3.2.24])<sup>19</sup>, we get a map

$$\mathrm{Fil}_{\mathrm{ad}}(\mathrm{HH}(A/k) \rightarrow A) \longrightarrow \mathrm{Fil}_{\mathrm{HKR}} \mathrm{HH}(A/k)$$

in  $\mathrm{FilDA}(\mathrm{g}(D(k)^{B\mathbb{T}}))$ . To show that this is an equivalence, by conservativity, it suffices to show the equivalence after passing along the forgetful functor  $\mathrm{FilDA}(\mathrm{g}(D(k)^{B\mathbb{T}})) \rightarrow \mathrm{FilDA}(\mathrm{g}_k)$ , and then it follows from Remark 6.7 that the left hand side is still the adic filtration of the map  $\mathrm{HH}(A/k) \rightarrow A$ .

As both the source and the target preserve geometric realizations in  $A \in \mathrm{DA}(\mathrm{g}_k)$ , and the  $\infty$ -category  $\mathrm{DA}(\mathrm{g}_k)$  is generated by  $\{\mathrm{LSym}_k(M) \mid M \in D(k)\}$  under geometric realizations, we reduce to the special case that  $A = \mathrm{LSym}_k(M)$  for some  $M \in D(k)$ . As in the proof of [Rak20, Lem 6.2.7], we have

$$\mathrm{Fil}_{\mathrm{HKR}} \mathrm{HH}(A/k) = \mathrm{LSym}_A^{\geq *}(M[1] \otimes_k^{\mathbb{L}} A)$$

which does coincide with the adic filtration of the map  $\mathrm{HH}(A/k) \rightarrow A$ .  $\square$

*Remark 6.9.* Let  $\mathcal{C}$  be  $D(k)$  or  $D(k)^{B\mathbb{T}}$ , and  $A \rightarrow B$  be a map of derived commutative algebras in  $\mathcal{C}$ . Then the derived commutative algebra  $\mathrm{gr}_{\mathrm{ad}}^{[0,1]}(A \rightarrow B)$  can be realized as the square-zero extension

$$\left[ B \xrightarrow{\mathrm{d}} L_B \xrightarrow{\mathrm{can}} L_{B/A} \right]$$

of  $B$  by  $L_{B/A}[-1]$  in  $\mathcal{C}$ .

**Lemma 6.10.** *Let  $k$  be a commutative ring, and  $A$  a derived commutative  $k$ -algebra. Then the canonical map*

$$L_{\mathrm{HH}(A/k)/k} \otimes_{\mathrm{HH}(A/k)}^{\mathbb{L}} A \longrightarrow L_{A/k}$$

*in  $\mathrm{Mod}_A(D(k)^{B\mathbb{T}})$  can be identified with the map*

$$L_{A/k} \otimes_k^{\mathbb{L}} k[\mathbb{T}] \longrightarrow L_{A/k}$$

*induced by the  $\mathbb{T}$ -equivariant projection  $\mathbb{T} \rightarrow *$ .*

*Proof.* Let  $X \in \mathrm{An}$  be an anima, and  $B_X := A^{\otimes_k^{\mathbb{L}} X}$  the  $X$ -th tensor power of  $A$ . Then we have an equivalence

$$L_{B_X/k} \otimes_{B_X}^{\mathbb{L}} A \simeq L_{A/k} \otimes_k^{\mathbb{L}} k[X]$$

in  $\mathrm{Mod}_A(D(k))$ , which is functorial in  $X \in \mathrm{An}$ . Now we restrict this to the full subcategory  $(B\mathbb{T})^{\triangleright} \subseteq \mathrm{An}$  spanned by  $\{\mathbb{T}, *\}$ , we obtained the identification that we want.  $\square$

**Proposition 6.11.** *Let  $k$  be a commutative ring, and  $A$  an animated commutative  $k$ -algebra. Then the derived commutative algebra  $\mathrm{gr}_{\mathrm{HKR}}^{[0,1]} \mathrm{HH}(A/k)$  can be identified with the square-zero extension*

$$\left[ A \xrightarrow{\mathrm{d}} L_{A/k} \xrightarrow{L_{A/k} \otimes^{\mathbb{L}} a_\lambda} \Sigma^\lambda L_{A/k} \right]$$

*of  $A$  by  $\Sigma^{\lambda-1} L_{A/k} \simeq L_{A/k}[1]$  (via a complex orientation of  $k$ ) in  $\mathrm{DA}(\mathrm{g}(D(k)^{B\mathbb{T}}))$ .*

*Proof.* By Remark 6.9 and Lemma 6.8, it suffices to identify the map

$$f : L_{A/k} \longrightarrow L_{A/\mathrm{HH}(A/k)}$$

in  $\mathrm{Mod}_A(D(k)^{B\mathbb{T}})$  with the map

$$L_{A/k} \xrightarrow{L_{A/k} \otimes^{\mathbb{L}} a_\lambda} \Sigma^\lambda L_{A/k}.$$

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<sup>19</sup>In the special case of a surjection of animated commutative rings, this universal property is explained in [CS26, Prop 13.3] as well.

We examine the transitivity sequence

$$L_{\mathrm{HH}(A/k)/k} \otimes_{\mathrm{HH}(A/k)}^{\mathbb{L}} A \longrightarrow L_{A/k} \xrightarrow{f} L_{A/\mathrm{HH}(A/k)}$$

in  $\mathrm{Mod}_A(D(k)^{B\mathbb{T}})$  associated to the composite map

$$k \longrightarrow \mathrm{HH}(A/k) \longrightarrow A$$

in  $\mathrm{DAlg}(D(k)^{B\mathbb{T}})$ . By Lemma 6.10, the map  $f$  fits into a fiber sequence

$$L_{A/k} \otimes_k^{\mathbb{L}} k[\mathbb{T}] \longrightarrow L_{A/k} \xrightarrow{f} L_{A/\mathrm{HH}(A/k)}$$

in  $\mathrm{Mod}_A(D(k)^{B\mathbb{T}})$ , and the result follows from the fiber sequence

$$\mathbb{S} \otimes \mathbb{T} \longrightarrow \mathbb{S} \xrightarrow{a_\lambda} \mathbb{S}^\lambda.$$

□

**Corollary 6.12.** *Let  $k$  be a commutative ring, and  $A$  an animated commutative  $k$ -algebra. Then the derived commutative algebra  $\mathrm{gr}_{\mathrm{HKR}}^{[0,1]} \mathrm{HH}(A/k) \in \mathrm{DAlg}(D(k)^{B\mathbb{T}})$  can be identified with the image of  $L\Omega_{A/k}^{[0,1]} \in \mathrm{DG}_-^{[0,1]} \mathrm{DAlg}_k$  under the functor in Construction 6.3.*

Now we restrict to smooth maps of dimension  $\leq 1$ . In this case, both HKR-filtration and Hodge-filtration are concentrated in weights  $[0, 1]$ , thus we have

**Construction 6.13. (Raksit)** *Let  $k$  be a commutative ring, and  $A$  a smooth  $k$ -algebra of dimension  $\leq 1$ . Then by Construction 6.3 and Corollary 6.12, we get a map*

$$\mathrm{dR}_{R/k} \longrightarrow \mathrm{HH}(R/k)$$

in  $\mathrm{DAlg}(D(k)^{B\mathbb{T}})$ , which becomes an equivalence in  $D(k)^{t\mathbb{T}}$  by Proposition 6.6. This construction is functorial in smooth maps  $k \rightarrow R$  of relative dimension  $\leq 1$ .

Now we want to apply this to analyze the de-completed  $C_p$ -Tate construction  $\mathrm{HH}(\mathbb{F}_p/\mathbb{Z})^{\theta_{\mathbb{Z}C_p}}$ .

*Remark 6.14.* The composite lax symmetric monoidal functor

$$D(\mathbb{Z})^{B\mathbb{T}} \xrightarrow{(-)^h} \mathrm{Mod}_{\mathbb{Z}}(\mathrm{Sp}^{g < \mathbb{T}}) \xrightarrow{(-)^{\theta_{\mathbb{Z}C_p}}} \mathrm{Mod}_{\mathbb{Z}^{tC_p}}(\mathrm{Sp}^{B(\mathbb{T}/C_p)})$$

factors canonically through  $D(\mathbb{Z})^{t\mathbb{T}}$ .

**Construction 6.15.** *Applying Construction 6.13 to the coCartesian diagram*

$$\begin{array}{ccc} k[t] & \xrightarrow{t \mapsto p} & k \\ \downarrow & & \downarrow \\ k & \longrightarrow & k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \end{array},$$

of animated  $k$ -algebras, we get a map

$$\mathrm{dR}_{(k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p)/k} \longrightarrow \mathrm{HH}(k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p/k)^{\eta_k}$$

of derived commutative algebras in  $\mathrm{Mod}_{\mathbb{Z}}(\mathrm{Sp}^{g < \mathbb{T}})$ , which becomes an equivalence after taking  $(-)^{\Phi_{C_p}}$  by Remark 6.14. In particular, when  $k = \mathbb{Z}_{(p)}$ , the map

$$\mathrm{dR}_{\mathbb{F}_p/\mathbb{Z}_{(p)}} = D_{\mathbb{Z}_{(p)}}(p) \longrightarrow \mathbb{Z}_{(p)}$$

of augmented (derived) commutative  $\mathbb{Z}_{(p)}$ -algebras (where  $D_{\mathbb{Z}_{(p)}}(p)$  is the PD-envelope of  $(p) \subseteq \mathbb{Z}_{(p)}$ ), induced by the PD-structure on  $(\mathbb{Z}_{(p)}, (p))$ , gives rise to a map

$$\mathrm{HH}(\mathbb{F}_p/\mathbb{Z})^{\theta_{\mathbb{Z}C_p}} \rightarrow \mathbb{Z}^{tC_p}$$

of  $\mathbb{E}_\infty$ - $\mathbb{Z}^{tC_p}$ -algebras in  $\mathrm{Sp}^{B(\mathbb{T}/C_p)}$ .

**Construction 6.16.** Composing with the map  $\mathrm{THH}(\mathbb{F}_p)^h \rightarrow \mathrm{HH}(\mathbb{F}_p/\mathbb{Z})^{\eta_{\mathbb{Z}}}$  of  $\mathbb{E}_\infty$ - $\mathrm{THH}(\mathbb{Z})^h$ -algebras in  $\mathrm{Sp}^{g^{<\mathbb{T}}}$ , we get a composite map

$$\mathrm{THH}(\mathbb{F}_p)^{tC_p} \longrightarrow \mathrm{HH}(\mathbb{F}_p/\mathbb{Z})^{\theta_{\mathbb{Z}C_p}} \longrightarrow \mathbb{Z}^{tC_p}$$

of  $\mathbb{E}_\infty$ - $\mathrm{THH}(\mathbb{Z})^{tC_p}$ -algebras in  $\mathrm{Sp}^{B(\mathbb{T}/C_p)}$ .

*Remark 6.17.* The map in Construction 6.16 lives in the Borel equivariant world, while our construction relies on genuine equivariant homotopy theory.

Our next goal is to show that the map in Construction 6.16 is an equivalence. We check it after modulo  $p$ . The point is that, although we do not know whether this map on the nose acquires a  $\mathbb{Z}$ -linear structures, its modulo  $p$  has an  $\mathbb{F}_p$ -linear structure. More precisely, our argument follows the following strategy<sup>20</sup> in [PVV18]:

- (1) Give the composite map  $\mathrm{THH}(\mathbb{F}_p)^{tC_p} \rightarrow \mathbb{Z}^{tC_p} \rightarrow \mathbb{F}_p^{tC_p}$  a  $\mathbb{Z}$ -linear structure (Construction 6.20);
- (2) Show that, under this  $\mathbb{Z}$ -linear structure, the induced map  $\mathrm{THH}(\mathbb{F}_p)^{tC_p} \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \rightarrow \mathbb{F}_p^{tC_p}$  is an equivalence (Lemma 6.21);
- (3) Show that the induced map above coincides with the modulo  $p$  reduction of the map  $\mathrm{THH}(\mathbb{F}_p)^{tC_p} \rightarrow \mathbb{Z}^{tC_p}$  in Construction 6.16 (Lemma 6.24).

*Remark 6.18.* Concretely, the map  $D_{\mathbb{Z}_{(p)}}(p) \rightarrow \mathbb{Z}_{(p)}$  is given by the map

$$\begin{aligned} \mathbb{Z}_{(p)} \left[ t, \frac{t^2}{2!}, \frac{t^3}{3!}, \dots \right] / (t-p) &\longrightarrow \mathbb{Z}_{(p)} \\ \frac{t^n}{n!} &\longmapsto \frac{p^n}{n!}, \end{aligned}$$

and therefore, after (derived) modulo  $p$ , it becomes the augmentation map

$$\begin{aligned} \mathbb{F}_p \left[ t, \frac{t^p}{p}, \frac{t^{p^2}}{p^{p+1}}, \dots \right] / (t) &\longrightarrow \mathbb{F}_p \\ \frac{t^{p^r}}{p^{p^{r-1}+\dots+1}} &\longmapsto 0. \end{aligned}$$

It follows that, the map  $\mathrm{dR}_{\mathbb{F}_p/\mathbb{Z}_{(p)}} \otimes_{\mathbb{Z}_{(p)}}^{\mathbb{L}} \mathbb{F}_p \rightarrow \mathbb{F}_p$  induced by the PD-structure on  $(p) \subseteq \mathbb{Z}_{(p)}$  coincides with the map  $\mathrm{dR}_{k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p/k} \rightarrow \mathrm{dR}_{k/k} = k$  induced by the multiplication map  $k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \rightarrow k$ , for derived commutative  $\mathbb{F}_p$ -algebras  $k$ .

We get a de-completed version of [PV19, Cor 2.7] (without restricting to odd primes):

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<sup>20</sup>We thank A. PETROV for explaining this.

**Corollary 6.19.** *The modulo  $p$  reduction of the map  $\mathrm{HH}(\mathbb{F}_p/\mathbb{Z})^{\Phi_{C_p}} \rightarrow \mathbb{Z}^{tC_p}$  of  $\mathbb{E}_\infty$ - $\mathbb{Z}^{tC_p}$ -algebras in  $\mathrm{Sp}^{B(\mathbb{T}/C_p)}$  in Construction 6.15 becomes the map  $\mathrm{HH}(k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p/k)^{\Phi_{C_p}} \rightarrow \mathrm{HH}(k/k)^{\Phi_{C_p}}$  of  $\mathbb{E}_\infty$ - $\mathbb{F}_p^{tC_p}$ -algebras in  $\mathrm{Sp}^{B(\mathbb{T}/C_p)}$  induced by the multiplication map  $k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \rightarrow k$ , for  $k = \mathbb{F}_p$ <sup>21</sup>.*

**Construction 6.20.** *Let  $k = \mathbb{F}_p$ . The composite map in Construction 6.16 fits into a commutative diagram*

$$\begin{array}{ccccc} \mathrm{THH}(k)^{tC_p} & \longrightarrow & \mathrm{HH}(k/\mathbb{Z})^{\theta_{\mathbb{Z}} C_p} & \longrightarrow & \mathbb{Z}^{tC_p} \\ & & \downarrow & & \downarrow \\ & & \mathrm{HH}(k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p/\mathbb{F}_p)^{\theta_{\mathbb{F}_p} C_p} & \longrightarrow & \mathbb{F}_p^{tC_p} \end{array}$$

*of  $\mathbb{E}_\infty$ - $\mathrm{THH}(\mathbb{Z})^{tC_p}$ -algebras in  $\mathrm{Sp}^{B(\mathbb{T}/C_p)}$ . It follows from Corollary 6.19 that the composite map  $\mathrm{THH}(k)^{tC_p} \rightarrow \mathbb{F}_p^{tC_p}$  is equivalent to applying  $(-)^{tC_p}$  to the map*

$$\mathrm{THH}(\mathbb{F}_p) \longrightarrow \mathrm{HH}(\mathbb{F}_p/\mathbb{F}_p)$$

*induced by the map  $(\mathbb{S} \rightarrow \mathbb{F}_p) \rightarrow (\mathbb{F}_p \rightarrow \mathbb{F}_p)$  of maps of  $\mathbb{E}_\infty$ -rings, which acquires a structure of maps of  $\mathbb{E}_\infty$ - $\mathbb{Z}^{tC_p}$ -algebras in  $\mathrm{Sp}^{B(\mathbb{T}/C_p)}$ . This map is also the same as the multiplication map  $\mathrm{THH}(\mathbb{F}_p) \rightarrow \mathbb{F}_p$ .*

**Lemma 6.21.** *The map  $\mathrm{THH}(\mathbb{F}_p)^{tC_p} \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \rightarrow \mathbb{F}_p^{tC_p}$  induced by the  $\mathbb{Z}^{tC_p}$ -linear map  $\mathrm{THH}(\mathbb{F}_p)^{tC_p} \rightarrow \mathbb{F}_p^{tC_p}$  in Construction 6.20 is an equivalence.*

*Proof.* The composite map  $\mathbb{Z} \rightarrow \mathrm{THH}(\mathbb{F}_p) \rightarrow \mathbb{F}_p$  is the canonical map since it is a map of commutative algebras in the heart of  $D(\mathbb{Z})^{B\mathbb{T}}$ . By [NS18, Cor IV.4.13], the first map becomes an equivalence after taking  $(-)^{tC_p}$ , and the result follows.  $\square$

Finally, we are in the following slightly tricky situation: given two  $\mathbb{Z}$ -modules  $M$  and  $N$ , along with a (non- $\mathbb{Z}$ -linear) map  $f : M \rightarrow N$ . Suppose that the composite map  $g : M \rightarrow N \rightarrow N \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$  carries a  $\mathbb{Z}$ -linear structure, which induces a map  $M \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \rightarrow N \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$ . We claim that this map coincides with the modulo  $p$  reduction of  $f$ . We break this into two steps. The first is to produce the map  $M/\mathbb{L}\mathbb{F}_p \rightarrow N \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$  without the  $\mathbb{Z}$ -linear structure on  $g$ .

**Construction 6.22.** *Let  $\mathcal{C}$  be a presentable stable  $\infty$ -category, and  $M \in \mathcal{C}$  an object, and  $L \in \mathrm{Mod}_{\mathbb{F}_p}(\mathcal{C})$  an  $\mathbb{F}_p$ -module in  $\mathcal{C}$ , and  $f : M \rightarrow L$  a map in  $\mathcal{C}$ . This gives rise to a composite map*

$$M/p \longrightarrow L/p \simeq L \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \xrightarrow{\mathrm{mult}} L.$$

**Lemma 6.23.** *Let  $\mathcal{C}$  be a presentable stable  $\infty$ -category, and  $M \in \mathrm{Mod}_{\mathbb{Z}}(\mathcal{C})$  a  $\mathbb{Z}$ -module in  $\mathcal{C}$ , and  $L \in \mathrm{Mod}_{\mathbb{F}_p}(\mathcal{C})$  an  $\mathbb{F}_p$ -module in  $\mathcal{C}$ , and  $f : M \rightarrow L$  a  $\mathbb{Z}$ -linear map in  $\mathcal{C}$ . Then the map  $M \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \rightarrow L$  in  $\mathrm{Mod}_{\mathbb{F}_p}(\mathcal{C})$ , as a map in  $\mathcal{C}$ , coincides with the map  $M/p \rightarrow L$  in Construction 6.22.*

*Proof.* The map  $f \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p : M \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \rightarrow L \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$  in  $\mathrm{Mod}_{\mathbb{F}_p}(\mathcal{C})$ , as a map in  $\mathcal{C}$ , coincides with the map  $f/p : M/p \rightarrow L/p$ . The result follows.  $\square$

**Lemma 6.24.** *Let  $\mathcal{C}$  be a presentable stable  $\infty$ -category, with  $M$  and  $N$  two  $\mathbb{Z}$ -modules in  $\mathcal{C}$ , and  $f : M \rightarrow N$  a map in  $\mathcal{C}$ . We give the composite map  $g : M \xrightarrow{f} N \rightarrow N \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$  a  $\mathbb{Z}$ -linear structure, which induces a map  $h : M \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \rightarrow N \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$  in  $\mathrm{Mod}_{\mathbb{F}_p}(\mathcal{C})$ . Then as a map in  $\mathcal{C}$ , the map  $h$  coincides with the map  $f/p$ .*

<sup>21</sup>Although we only apply to the case that  $k = \mathbb{F}_p$ , this notation allows us to distinguish different  $\mathbb{F}_p$ -algebra structures on  $\mathbb{F}_p \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$ .

*Proof.* By Lemma 6.23, the map  $h$ , as a map in  $\mathcal{C}$ , coincides with the composite

$$M/p \xrightarrow{g/p} (N \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p)/p \simeq N \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \xrightarrow{\text{id}_N \otimes \text{mult}} N \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p.$$

Thus it suffices to show that the composite

$$N/p \rightarrow (N \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p)/p \simeq N \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \xrightarrow{\text{id}_N \otimes \text{mult}} N \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$$

is equivalent to identity. Note that the first map can be identified with

$$N \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \xrightarrow{\text{id}_N \otimes 1 \otimes \text{id}_{\mathbb{F}_p}} N \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$$

and the result follows from the fact that the composite map

$$\mathbb{F}_p \xrightarrow{1 \otimes \text{id}_{\mathbb{F}_p}} \mathbb{F}_p \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \xrightarrow{\text{mult}} \mathbb{F}_p$$

of animated rings is identity.  $\square$

**Proposition 6.25.** *The map  $\text{THH}(\mathbb{F}_p)^{tC_p} \rightarrow \mathbb{Z}^{tC_p}$  of  $\mathbb{E}_{\infty}$ - $\text{THH}(\mathbb{Z})^{tC_p}$ -algebras in Construction 6.16 is an equivalence.*

*Proof.* Since both the source and the target is  $p$ -complete by [NS18, Lem I.2.9], it suffices to check that this map is an equivalence after modulo  $p$ . Now the result follows from Lemmas 6.21 and 6.24.  $\square$

**Corollary 6.26.** *Let  $M$  be a  $\text{THH}(\mathbb{Z})^h$ -module in  $\text{Sp}^{g < \mathbb{T}}$ . Then there exists a lax symmetric monoidal (in  $M$ ) equivalence*

$$\left( (M \otimes_{\text{THH}(\mathbb{Z})^h}^{\mathbb{L}} \text{THH}(\mathbb{F}_p)^h)^{\theta_{\text{THH}(\mathbb{F}_p)}^{\mathbb{T}}} \right)_p^{\wedge} \rightarrow ((M \otimes_{\text{THH}(\mathbb{Z})^h}^{\mathbb{L}} \mathbb{Z})^{\theta_{\mathbb{Z}}^{\mathbb{T}}})_p^{\wedge}$$

of  $\text{TP}(\mathbb{Z})$ -modules.

*Proof.* By Corollary 5.26, it suffices to produce a symmetric monoidal equivalence

$$(M \otimes_{\text{THH}(\mathbb{Z})^h}^{\mathbb{L}} \text{THH}(\mathbb{F}_p)^h)^{\Phi C_p} \longrightarrow (M \otimes_{\text{THH}(\mathbb{Z})^h}^{\mathbb{L}} \mathbb{Z})^{\Phi C_p}$$

in  $\text{Mod}_{\text{THH}(\mathbb{Z})^{tC_p}}(\text{Sp}^{B(\mathbb{T}/C_p)})$ , which is an immediate consequence of Proposition 6.25.  $\square$

*Remark 6.27.* When  $p$  is an odd prime, [DR25, Prop 5.2.1] shows that the map  $\mathbb{Z} \rightarrow \text{THH}(\mathbb{F}_p)$  induces an equivalence  $\mathbb{Z}^{tC_p} \rightarrow \text{THH}(\mathbb{F}_p)^{tC_p}$  of  $\mathbb{E}_{\infty}$ - $\text{THH}(\mathbb{Z})^{tC_p}$ -algebras, which implies Corollary 6.26 (although a priori, the map there might be different from the one here), and the map even has a  $\mathbb{Z}^{\mathbb{T}}$ -module structure.

**Corollary 6.28.** *Let  $\mathcal{C}$  be a dualizable presentable stable  $\mathbb{Z}$ -linear  $\infty$ -category. Then there exists a lax symmetric monoidal (in  $\mathcal{C}$ ) equivalence*

$$\text{TP}^{\text{poly}}/\mathbb{F}_p(\mathcal{C} \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p)_p^{\wedge} \longrightarrow \text{HP}^{\text{poly}}(\mathcal{C}/\mathbb{Z})_p^{\wedge}$$

of  $\text{TP}(\mathbb{Z})$ -modules.

*Proof.* Apply Corollary 6.26 to  $M = \text{THH}(\mathcal{C}) \otimes_{\text{THH}(\mathbb{Z})}^{\mathbb{L}} \text{THH}(\mathbb{Z})^h$ .  $\square$

## 7. COMPARISON TO THH

Recall that the *Cartier isomorphism* identifies algebraic de Rham cohomology groups of smooth  $\mathbb{F}_p$ -algebras with their algebraic differential forms. We give two noncommutative analogues. In this section, we discuss one of them, which compares polynomial co-periodic cyclic homology of dualizable presentable stable  $\mathbb{F}_p$ -linear  $\infty$ -categories with their topological Hochschild homology. This comparison was proved in [Kal20, Cor 11.15] for associative  $\mathbb{F}_p$ -algebras, using Goodwillie derivative of (Hochschild–)Witt trace theory. We give two arguments. Although the second argument is much shorter, the first argument gives us more information, which is used in Section 8.

The key to the first argument is the observation that the  $\mathbb{T}/C_p$ -equivariant  $\mathbb{E}_1$ - $\mathbb{Z}^{tC_p}$ -module  $\mathbb{F}_p^{tC_p}$  is co-induced.

**Construction 7.1.** *Let  $n \in \mathbb{N}_{>0}$ . There is a map*

$$\mathbb{Z}^{\mathbb{T}/C_n} \longrightarrow \mathbb{Z}/n$$

*in  $D(\mathbb{Z})^{B\mathbb{T}}$ , where  $\mathbb{Z}/n$  is equipped with the trivial action, i.e.  $\varpi_1^*(\mathbb{Z}/n)$ . Indeed, this map is taken to be represented<sup>22</sup> by the following surjective map*

$$\begin{array}{ccc} 0 & & -1 \\ \mathbb{Z} & \xrightleftharpoons[n]{0} & \mathbb{Z} \\ \downarrow & & \downarrow \\ \mathbb{Z}/n & \xrightleftharpoons[0]{0} & 0 \end{array}$$

*of mixed complexes concentrated in homological degrees  $[-1, 0]$ . Since it is surjective, it is a fibration in the projective model structure on mixed complexes, with kernel being the mixed complex  $\left(n\mathbb{Z} \xrightleftharpoons[n]{0} \mathbb{Z}\right)$ , which represents  $\mathbb{Z}^{\mathbb{T}} \in D(\mathbb{Z})^{B\mathbb{T}}$ . Consequently, we get a fiber sequence*

$$\mathbb{Z}^{\mathbb{T}} \longrightarrow \mathbb{Z}^{\mathbb{T}/C_n} \longrightarrow \mathbb{Z}/n$$

*in  $D(\mathbb{Z})^{B\mathbb{T}}$ .*

**Remark 7.2.** One can also construct the map and the fiber sequence in Construction 7.1 by the *gold relation*  $a_{\lambda^n} u_{\lambda^m} = (n/m) a_{\lambda^m} u_{\lambda^n}$  for  $(m, n) \in \mathbb{N}_{>0}^2$  with  $m \mid n$  in [HHR17, Lem 3.6] (where  $u_{\lambda^m}$ ’s become equivalence after forgetting to the Borel equivariant objects), where  $\lambda^n$  is the complex  $S^1$ -representation  $S^1 \xrightarrow{(-)^n} \mathbb{C}^\times$  viewed as a real representation, and  $a_{\lambda^n}$  is the *Euler class*  $\mathbb{S}^0 \rightarrow \mathbb{S}^{\lambda^n}$ , i.e. the map of Thom spectra of the inclusion  $\{0\} \subseteq \lambda^n$  of representations. Indeed, there is a fiber sequence

$$\Sigma_{\mathbb{T}}^\infty[\mathbb{T}/C_n]_+ \longrightarrow \mathbb{S}^0 \longrightarrow \mathbb{S}^{\lambda^n}$$

of cyclonic spectra (and thus, of  $\mathbb{T}$ -equivariant spectra), cf. [Sul20, Obs 2.32]. Taking  $\mathbb{Z}$ -linear dual, and applying the “octahedral axiom” to the gold relation  $a_{\lambda^n} \sim n a_{\lambda^1}$ , we get the fiber sequence in Construction 7.1.

This allows us to establish results beyond char  $p$ , as observed by Yuri SULYMA in [Sul20, Lem 4.9].

**Corollary 7.3.** *The map  $\mathbb{Z}^{\mathbb{T}/C_n} \rightarrow \mathbb{Z}/n$  in  $D(\mathbb{Z})^{B\mathbb{T}}$  as in Construction 7.1 induces an equivalence*

$$\mathbb{Z}^{tC_n} \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{Z}^{\mathbb{T}/C_n} \longrightarrow (\mathbb{Z}/n)^{tC_n}$$

*in  $\text{Mod}_{\mathbb{Z}^{tC_n}}(D(\mathbb{Z})^{B(\mathbb{T}/C_n)})$ .*

<sup>22</sup>The  $\mathbb{E}_1$ -monoidal equivalence of the monoidal  $\infty$ -category  $D(\mathbb{Z})^{B\mathbb{T}}$  and the monoidal  $\infty$ -category of mixed complexes up to quasi-isomorphism is explained in details in [Lei22, §5.4].

*Proof.* Applying  $(-)^{tC_n}$  to the fiber sequence in Construction 7.1, we see that the induced map  $(\mathbb{Z}^{\mathbb{T}/C_n})^{tC_n} \rightarrow (\mathbb{Z}/n)^{tC_n}$  in  $D(\mathbb{Z})^{B(\mathbb{T}/C_n)}$  is an equivalence (which even has an  $\mathbb{E}_1$ -structure). It suffices to establish an equivalence

$$\mathbb{Z}^{tC_n} \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{Z}^{\mathbb{T}/C_n} \longrightarrow (\mathbb{Z}^{\mathbb{T}/C_n})^{tC_n}$$

in  $\text{Mod}_{\mathbb{Z}^{tC_n}}(D(\mathbb{Z})^{B(\mathbb{T}/C_n)})$ . This is given by equivalences

$$\begin{aligned} \mathbb{Z}^{tC_n} \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{Z}^{\mathbb{T}/C_n} &\simeq (\mathbb{Z}^{tC_n} \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{Z}[\mathbb{T}/C_n])[-1] \\ &\simeq (\mathbb{Z}^{tC_n} \otimes (\mathbb{T}/C_n))[-1] \\ &\simeq (\mathbb{Z}[\mathbb{T}/C_n])^{tC_n}[-1] \\ &\simeq (\mathbb{Z}^{\mathbb{T}/C_n})^{tC_n} \end{aligned}$$

in  $\text{Mod}_{\mathbb{Z}^{tC_n}}(D(\mathbb{Z})^{B(\mathbb{T}/C_n)})$ .  $\square$

*Warning 7.4.* The map in Corollary 7.3 does not carry any  $\mathbb{E}_1$ -structure when  $n = 2$ . Indeed, on the left hand side, the square-zero class  $e \in \pi_{-1}(\mathbb{Z}^{\mathbb{T}/C_n})$  gives rise to a square-zero class  $e \in \pi_{-1}(\mathbb{Z}^{\mathbb{T}/C_n} \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{Z}^{tC_n})$ , while the homotopy ring  $\pi_*(\mathbb{F}_2^{tC_2})$  is isomorphic to  $\mathbb{F}_2((s))$  for a generator  $s \in \pi_1(\mathbb{F}_2^{tC_2})$ , which is integral, thus any square-zero elements is necessarily zero.

*Question 5.* Is there any version of  $\mathbb{E}_1$ -enhancement of Corollary 7.3?

Now we compare polynomial periodic cyclic homology of dualizable presentable stable  $\mathbb{F}_p$ -linear  $\infty$ -categories. Slightly more generally, we consider  $\text{THH}(\mathbb{F}_p)$ -modules in cyclonic spectra. By Lemma 2.13 and the symmetric monoidal structure on  $(-)^{\Phi_{C_p}}$ , we have

**Lemma 7.5.** *Let  $k$  be a commutative algebra. Then the composite functor*

$$\text{Mod}_{\text{THH}(k)}(\text{Sp}^{g < \mathbb{T}}) \xrightarrow{(-) \otimes_{\text{THH}(k)}^{\mathbb{L}} \underline{k}} \text{Mod}_{\underline{k}}(\text{Sp}^{g < \mathbb{T}}) \xrightarrow{((-)^{\theta_{\underline{k}}^{\mathbb{T}}})_p^{\wedge}} D(k^{t\mathbb{T}})_p^{\wedge}.$$

*is equivalent to the composite functor*

$$\begin{array}{ccc} \text{Mod}_{\text{THH}(k)}(\text{Sp}^{g < \mathbb{T}}) & \xrightarrow{(-)^{\Phi_{C_p}}} & \text{Mod}_{\text{THH}(k)^{\Phi_{C_p}}}(\text{Sp}^{g_p(\mathbb{T}/C_p)}) \\ & \downarrow & \\ & \text{Mod}_{k^{tC_p}}(\text{Sp}^{g_p(\mathbb{T}/C_p)}) & \xrightarrow{(-)^{\eta_{k^{tC_p}}(\mathbb{T}/C_p)}} D(k^{t\mathbb{T}}) \end{array}$$

*where the vertical arrow is the base change along the map  $\text{THH}(k)^{\Phi_{C_p}} \rightarrow k^{tC_p}$  of  $\mathbb{T}/C_p$ - $\mathbb{E}_{\infty}$ -rings.*

*Remark 7.6.* Recall that both maps in the composite map  $\underline{\mathbb{Z}} \rightarrow \underline{\mathbb{Z}}_p = \text{TR}(\mathbb{F}_p) \rightarrow \text{THH}(\mathbb{F}_p)$  of  $\mathbb{T}$ - $\mathbb{E}_{\infty}$ -rings become equivalences after taking geometric fixed points  $(-)^{\Phi_{C_p}}$  (cf. [AMR21, Rem 10.9]). It follows that we can identify the  $\mathbb{T}$ - $\mathbb{E}_{\infty}$ -ring  $\text{THH}(\mathbb{F}_p)$  with  $\underline{\mathbb{Z}}^{\Phi_{C_p}}$ . Moreover, by [HM97], there exists a Bökstedt element  $\sigma \in \pi_2 \text{TF}(\mathbb{F}_p)$  which maps to a Bökstedt element in  $\pi_2 \text{TR}^r(\mathbb{F}_p)$  for every  $r \in \mathbb{N}_{>0}$ , and by [NS18, Lem II.6.1] and computational results about  $\text{THH}(\mathbb{F}_p)$ , we can identify the  $\mathbb{T}$ - $\mathbb{E}_{\infty}$ -ring  $\text{THH}(\mathbb{F}_p)[\sigma^{-1}]$  with the Borel  $\mathbb{T}$ - $\mathbb{E}_{\infty}$ -ring  $\mathbb{Z}^{tC_p}$ . Consequently, for every commutative ring  $k$ , the map  $\underline{k}^{\Phi_{C_p}} \rightarrow k^{tC_p}$  of  $\mathbb{T}$ - $\mathbb{E}_{\infty}$ -rings is simply inverting  $\sigma$ .

**Lemma 7.7.** *The composite functor*

$$\text{Mod}_{\text{THH}(\mathbb{F}_p)^{tC_p}}(\text{Sp}^{g_p(\mathbb{T}/C_p)}) \longrightarrow \text{Mod}_{\mathbb{F}_p^{tC_p}}(\text{Sp}^{g_p(\mathbb{T}/C_p)}) \xrightarrow{(-)^{\eta_{\mathbb{F}_p^{tC_p}}(\mathbb{T}/C_p)}} D(\mathbb{Z}^{tC_p})$$

*coincides with the forgetful functor, where the first functor is the base change along the map  $\text{THH}(\mathbb{F}_p)^{tC_p} \rightarrow \mathbb{F}_p^{tC_p}$  of Borel  $\mathbb{E}_{\infty}$ - $p$ -cyclonic spectra.*

Recall that the map  $\mathbb{Z} \rightarrow \mathrm{THH}(\mathbb{F}_p)$  as in Remark 7.6 also becomes an equivalence after taking  $(-)^{tC_p}$  by [NS18, Cor IV.4.13], thus we can replace  $\mathrm{THH}(\mathbb{F}_p)^{tC_p}$  by  $\mathbb{Z}^{tC_p}$  in Lemma 7.7.

*Proof.* Since both functors in question preserve filtered colimits, it suffices to check on compact objects of  $\mathrm{Mod}_{\mathbb{Z}^{tC_p}}(\mathrm{Sp}^{g_p(\mathbb{T}/C_p)})$ . Note that the first base change functor preserves compact objects. The result follows from Corollary 7.3.  $\square$

*Remark 7.8.* The target of the composite functor in Lemma 7.7 has an  $\mathbb{F}_p^{t\mathbb{T}}$ -module structure, and such a structure is yet to explore.

Summarizing the above discussion, we get

**Proposition 7.9.** *The composite functor*

$$\mathrm{Mod}_{\mathrm{THH}(\mathbb{F}_p)}(\mathrm{Sp}^{g^{<\mathbb{T}}}) \xrightarrow{(-)^{\otimes_{\mathrm{THH}(\mathbb{F}_p)}^{\mathbb{F}_p}}} \mathrm{Mod}_{\mathbb{F}_p}(\mathrm{Sp}^{g^{<\mathbb{T}}}) \xrightarrow{(-)^{\theta_{\mathbb{F}_p}^{\mathbb{T}}}} D(\mathbb{Z}^{t\mathbb{T}})$$

*coincides with the composite functor*

$$\mathrm{Mod}_{\mathrm{THH}(\mathbb{F}_p)}(\mathrm{Sp}^{g^{<\mathbb{T}}}) \xrightarrow{(-)^{\Phi_{C_p}}} \mathrm{Mod}_{\mathrm{THH}(\mathbb{F}_p)^{\Phi_{C_p}}}(\mathrm{Sp}^{g_p(\mathbb{T}/C_p)}) \longrightarrow D(\mathbb{Z}^{t\mathbb{T}}),$$

*where the second functor is inverting  $\sigma$  to the underlying  $\mathrm{THH}(\mathbb{F}_p)^{\Phi_{C_p}}$ -module.*

Applying to the  $\mathrm{THH}(\mathbb{F}_p)$ -module spectrum  $\mathrm{THH}(\mathcal{C})$ , we get:

**Corollary 7.10.** *Let  $\mathcal{C}$  be a dualizable presentable stable  $\mathbb{F}_p$ -linear  $\infty$ -category. Then the polynomial periodic cyclic homology  $\mathrm{HP}^{\mathrm{poly}}(\mathcal{C}/\mathbb{F}_p)$  is equivalent to  $\mathrm{THH}(\mathcal{C})[\sigma^{-1}]$  as  $\mathbb{Z}^{t\mathbb{T}}$ -module spectra.*

*Remark 7.11.* We do not compare the multiplicative structures on polynomial periodic cyclic homology (as a lax symmetric monoidal functor) with topological Hochschild homology in Corollary 7.10.

Now we give a second proof of Lemma 7.7, inspired by the proof of [Mat20, Prop 2.15]. The key is the following lemma.

**Lemma 7.12.** *The composite functor*

$$\mathrm{Mod}_{\mathrm{THH}(\mathbb{F}_p)^{tC_p}}(\mathrm{Sp}^{B(\mathbb{T}/C_p)}) \longrightarrow \mathrm{Mod}_{\mathbb{F}_p^{tC_p}}(\mathrm{Sp}^{B(\mathbb{T}/C_p)}) \xrightarrow{(-)^{h(\mathbb{T}/C_p)}} D(\mathrm{TP}(\mathbb{F}_p))$$

*coincides with the forgetful functor.*

*Proof.* Since the image of  $u \in \pi_2 \mathrm{TC}^-(\mathbb{F}_p)$  under the canonical map  $\mathrm{TC}^-(\mathbb{F}_p) \rightarrow \mathrm{TP}(\mathbb{F}_p)$  is  $up \in \pi_2 \mathrm{TP}(\mathbb{F}_p)$  as reviewed in Remark 4.10, this composite functor coincides with the composite functor

$$\mathrm{Mod}_{\mathrm{THH}(\mathbb{F}_p)^{tC_p}}(\mathrm{Sp}^{B(\mathbb{T}/C_p)}) \xrightarrow{(-)^{h(\mathbb{T}/C_p)}} D(\mathrm{TP}(\mathbb{F}_p)) \xrightarrow{(-)^{/L_p}} D(\mathrm{TP}(\mathbb{F}_p)),$$

which is subsequently identified with the forgetful functor by the proof of [BMS19, Prop 6.4] (or more precisely, the first displayed formula there).  $\square$

Lemma 7.7 follows from the fact that the composite functor in Lemma 7.12, by virtue of identification with the forgetful functor, preserves filtered colimits, and that the first functor  $\mathrm{Mod}_{\mathrm{THH}(\mathbb{F}_p)^{tC_p}}(\mathrm{Sp}^{g_p(\mathbb{T}/C_p)}) \rightarrow \mathrm{Mod}_{\mathbb{F}_p^{tC_p}}(\mathrm{Sp}^{g_p(\mathbb{T}/C_p)})$  in Lemma 7.7 preserves compact objects.

## 8. CONJUGATE FILTRATION

As explained in the introduction of Section 7, another noncommutative analogue of the Cartier isomorphism is the conjugate filtration, which we will address in this section. As there, our version is constructed out of  $\mathrm{THH}(k)$ -modules for a base commutative ring  $k$ , and by Lemma 7.5, it is crucial to analyze the map  $\mathrm{THH}(k)^{\Phi C_p} \rightarrow k^{tC_p}$ , endowing the target  $k^{tC_p}$  a suitable filtration.

First, we note that, the homotopy  $C_p$ -fixed points  $k^{hC_p}$  of a commutative  $\mathbb{F}_p$ -algebra  $k$  is a direct summand of the  $C_p$ -Tate construction  $k^{tC_p}$ .

*Remark 8.1.* Let  $M$  be an  $\mathbb{F}_p$ -vector space. Then it follows from computations that the composite map

$$M^{hC_p} \longrightarrow M^{tC_p} \longrightarrow \tau_{\leq 0} M^{tC_p}$$

of  $\mathbb{F}_p^{hC_p}$ -module spectra is an equivalence. In particular, let  $k$  be a commutative  $\mathbb{F}_p$ -algebra. Then the composite map

$$k^{hC_p} \longrightarrow k^{tC_p} \longrightarrow \tau_{\leq 0} k^{tC_p}$$

of  $k^{hC_p}$ -module spectra is an equivalence, where the first map has an  $\mathbb{E}_\infty$ -structure, and the second map has a  $k^{tC_p}$ -module structure.

Recall that, in Section 7, a key fact is that the  $\mathbb{T}/C_p$ -equivariant  $\mathbb{Z}^{tC_p}$ -module spectrum  $\mathbb{F}_p^{tC_p}$  is co-induced. It is natural to ask whether the same holds for  $k^{tC_p}$  for any commutative  $\mathbb{F}_p$ -algebra  $k$ . We do not know the answer, but expect it to be false, at least functorially in  $k$  (see Remark 8.6). However, note that the  $\mathbb{E}_\infty$ -ring  $k^{tC_p}$  is 2-periodic, and we show that a “fundamental region” of  $k^{tC_p}$  with respect to its 2-periodicity is  $\mathbb{T}/C_p$ -equivariantly  $k$ -linearly co-induced.

**Notation 8.2.** Let  $n$  be a positive integer. Let  $\mathrm{CoInd}_e^{\mathbb{T}/C_n}$  denote right adjoint  $D(\mathbb{Z}) \rightarrow D(\mathbb{Z})^{B(\mathbb{T}/C_n)}$  to the (symmetric monoidal) forgetful functor  $D(\mathbb{Z})^{B(\mathbb{T}/C_n)} \rightarrow D(\mathbb{Z})$ .

**Construction 8.3.** Let  $M$  be a spectrum. Endow  $M$  with trivial  $\mathbb{T}$ -action, we get a map  $\varpi_p^* M \rightarrow M^{hC_p}$  in  $\mathrm{Fun}(B(\mathbb{T}/C_p), \mathrm{Sp})$ , where  $\varpi_p : \mathbb{T}/C_p \rightarrow *$  is the quotient map, which gives rise to a lax symmetric monoidal lax transformation

$$\varpi_p^*(-) \longrightarrow (-)^{hC_p}$$

between lax symmetric monoidal functors  $\mathrm{Sp} \rightarrow \mathrm{Fun}(B(\mathbb{T}/C_p), \mathrm{Sp})$ . On the other hand, there is a non- $\mathbb{T}/C_p$ -equivariant map  $M^{hC_p} \rightarrow M$  which, by adjunction, gives rise to a map  $M^{hC_p} \rightarrow \mathrm{CoInd}_e^{\mathbb{T}/C_p} M$  in  $\mathrm{Fun}(B(\mathbb{T}/C_p), \mathrm{Sp})$  which is functorial in  $M \in \mathrm{Sp}$ , and this has a lax symmetric monoidal structure. In summary, we have a composite lax symmetric natural transformation

$$\varpi_p^*(-) \longrightarrow (-)^{hC_p} \longrightarrow \mathrm{CoInd}_e^{\mathbb{T}/C_p}(-)$$

of lax symmetric monoidal functors  $\mathrm{Sp} \rightarrow \mathrm{Fun}(B(\mathbb{T}/C_p), \mathrm{Sp})$ .

**Lemma 8.4.** Let  $M$  be an  $\mathbb{F}_p$ -vector space. Then the composite map

$$\tau_{\geq -1} M^{hC_p} \longrightarrow M^{hC_p} \longrightarrow \mathrm{CoInd}_e^{\mathbb{T}/C_p} M$$

of  $\mathbb{T}/C_p$ -equivariant  $\mathbb{F}_p$ -module<sup>23</sup> spectra is an equivalence, where the second map is as in Construction 8.3.

*Proof.* Since the maps in question are already constructed, to check that it is an equivalence, we could pick a free  $\mathbb{Z}$ -lift  $\tilde{M}$  of  $M$ , namely, a free abelian group  $\tilde{M} = \mathbb{Z}^{\oplus I}$  with  $\tilde{M}/p \cong M$ . Then taking  $I$ -direct sum<sup>24</sup> of the equivalence in Corollary 7.3 and truncating at  $[-1, 0]$ , we get an equivalence as an inverse to the composite map in question.  $\square$

<sup>23</sup>The  $\mathbb{F}_p$ -module structure comes from lax symmetric monoidal structures in Construction 8.3.

<sup>24</sup>Tate construction preserves infinite direct sum of uniformly  $t$ -bounded objects.

It follows from lax symmetric monoidal structures in Construction 8.3 and Lemma 8.4 that

**Corollary 8.5.** *Let  $k$  be a commutative  $\mathbb{F}_p$ -algebra. Then there is a canonical equivalence*

$$\tau_{\geq -1} k^{hC_p} \longrightarrow \mathrm{CoInd}_e^{\mathbb{T}/C_p} k$$

of  $\mathbb{T}/C_p$ -equivariant  $k$ -modules.

*Remark 8.6.* Let  $M$  be an  $\mathbb{F}_p$ -vector space. It is unclear whether we can functorially identify  $M^{tC_p}$  with the co-induced  $\mathbb{T}/C_p$ -equivariant  $\mathbb{Z}$ -module spectrum  $\mathrm{CoInd}_e^{\mathbb{T}/C_p}(M^{t\mathbb{T}})$  (although we can do it functorially in free  $\mathbb{Z}$ -lifts  $\tilde{M}$ ), let alone  $\mathbb{F}_p$ -linearly. The multiplicative structure is even more complicated.

Now we describe the filtration on  $k^{hC_p}$  and  $k^{tC_p}$  as promised.

**Construction 8.7.** *Let  $k$  be a commutative  $\mathbb{F}_p$ -algebra. Then we consider the odd filtration on  $\mathbb{T}/C_p$ -equivariant  $\tau_{\geq 0}(k^{tC_p})$ -module spectra  $k^{hC_p}$ ,  $k^{tC_p}$ , and the canonical  $\mathbb{T}/C_p$ -equivariant  $\tau_{\geq 0}(k^{tC_p})$ -module map  $k^{hC_p} \rightarrow k^{tC_p}$ , given by the odd parts of the Whitehead filtrations, i.e. for  $i \in \mathbb{Z}$ , we have*

$$\begin{aligned} \mathrm{Fil}_{\mathrm{odd}}^i k^{hC_p} &:= \tau_{\geq 2i-1} k^{hC_p}; \\ \mathrm{Fil}_{\mathrm{odd}}^i k^{tC_p} &:= \tau_{\geq 2i-1} k^{tC_p}. \end{aligned}$$

We note that  $\mathrm{Fil}_{\mathrm{odd}}^{>0} k^{hC_p} = 0$ , and the maps  $\mathrm{gr}_{\mathrm{odd}}^i k^{hC_p} \rightarrow \mathrm{gr}_{\mathrm{odd}}^i k^{tC_p}$  are equivalences for  $i \leq 0$ .

*Remark 8.8.* Note that  $\mathrm{gr}_{\mathrm{odd}}^0 k^{tC_p} = \tau_{\geq -1} k^{hC_p}$  is concentrated in degree  $[-1, 0]$ , by [Lur17, Prop 2.2.1.8], the  $\tau_{\geq 0}(k^{tC_p})$ -module structure on  $\mathrm{gr}_{\mathrm{odd}}^0 k^{tC_p}$  descends canonically to a  $\tau_{[0,1]}(k^{tC_p})$ -module structure. We note that the  $\mathbb{T}/C_p$ -equivariant  $\mathbb{E}_\infty$ - $\mathbb{Z}$ -algebra  $\tau_{[0,1]}(k^{tC_p})$  coincides with the  $\mathbb{E}_\infty$ - $\mathbb{Z}$ -algebra  $k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$  with trivial  $\mathbb{T}/C_p$ -action<sup>25</sup>.

We now produce the conjugate filtration for  $\mathrm{THH}(k)$ -modules. Before this, we need a base-independence result, which says that the de-completed Tate construction over  $k^{tC_p}$  coincides with that over  $\tau_{\geq 0}(k^{tC_p})$ .

**Lemma 8.9.** *Let  $k$  be a commutative  $\mathbb{F}_p$ -algebra. Then the natural transformation*

$$(-)^{\eta_{\tau_{\geq 0}(k^{tC_p})}(\mathbb{T}/C_p)} \longrightarrow (-)^{\eta_{k^{tC_p}}(\mathbb{T}/C_p)}$$

of functors  $\mathrm{Mod}_{k^{tC_p}}(\mathrm{Sp}^{g_p^{\mathbb{T}}}) \rightarrow D(k)$ , induced by the map  $\tau_{\geq 0}(k^{tC_p}) \rightarrow k^{tC_p}$  in  $\mathrm{CAlg}(D(k)^{B(\mathbb{T}/C_p)})$ , is an equivalence.

*Proof.* It suffices to show that, for every compact generator  $M = k^{tC_p} \otimes [\mathbb{T}/C_{p^r}] \in \mathrm{Mod}_{k^{tC_p}}(\mathrm{Sp}^{g_p^{\mathbb{T}}})^{\aleph_0}$ , the assembly map

$$M^{\eta_{\tau_{\geq 0}(k^{tC_p})}(\mathbb{T}/C_p)} \longrightarrow M^{h(\mathbb{T}/C_p)}$$

is an equivalence. Recall that the Whitehead tower of  $k^{tC_p} = (\tau_{\geq 0} k^{tC_p})[\sigma^{-1}]$  is a sequential colimit of shifts of  $\tau_{\geq 0}(k^{tC_p})$ , where  $\sigma \in \pi_2(k^{tC_p})$  is a generator. Then the result follows from the fact that  $(-)^{h(\mathbb{T}/C_p)}$  preserves weakly Whitehead towers (cf. [NS18, Lem I.2.6], or the dual of [BMS19, Lem 3.3]).  $\square$

<sup>25</sup>This can be seen as follows. Let  $M \in D(\mathbb{Z})_{\geq 0}$  be a connective  $\mathbb{Z}$ -module spectrum. The trivial action on  $M$  gives rise to a  $\mathbb{T}/C_p$ -equivariant map  $M \rightarrow M^{hC_p} \rightarrow M^{tC_p}$ , and thus a map  $M \rightarrow \tau_{[0,1]}(M^{tC_p})$  in  $D(\mathbb{Z})^{B(\mathbb{T}/C_p)}$ . This map gives rise to a lax symmetric monoidal natural transformation

$$(-) \longrightarrow \tau_{[0,1]}((-)^{tC_p}),$$

and subsequently a lax symmetric monoidal natural transformation

$$(-) \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \longrightarrow \tau_{[0,1]}((-)^{tC_p})$$

as  $\tau_{[0,1]}(\mathbb{Z}^{tC_p}) = \mathbb{F}_p$ , which is an equivalence on abelian groups (where we could use the usual resolution for cyclic groups to compute  $(-)^{tC_p}$ ). We apply this to the commutative  $\mathbb{F}_p$ -algebra  $k$ . Alternatively, as the anonymous referee pointed out, one could show that  $k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$  is the connective cover of  $\tau_{[0,1]}(k^{tC_p})^{h(\mathbb{T}/C_p)}$ .

**Construction 8.10. (Conjugate filtration)** Let  $k$  be a commutative  $\mathbb{F}_p$ -algebra, and  $M$  a  $\mathrm{THH}(k)$ -module in  $\mathrm{Sp}^{g^{\leq \mathbb{T}}}$ . Then by Lemma 7.5 and Construction 8.7, the  $\mathbb{Z}^{\mathbb{T}}$ -module spectrum

$$(M \otimes_{\mathrm{THH}(k)}^{\mathbb{L}} \underline{k})^{\theta_{\mathbb{Z}^{\mathbb{T}}}} \simeq \left( M^{\Phi_{C_p}} \otimes_{\mathrm{THH}(k)^{\Phi_{C_p}}}^{\mathbb{L}} \tau_{\geq 0}(k^{tC_p}) \otimes_{\tau_{\geq 0}(k^{tC_p})}^{\mathbb{L}} k^{tC_p} \right)^{\eta_{\tau_{\geq 0}(k^{tC_p})}(\mathbb{T}/C_p)}$$

admits a filtration  $\mathrm{Fil}_{\mathrm{conj}}^*$  induced by the odd filtration on  $k^{tC_p}$ .

We now analyze the associated graded pieces of the conjugate filtration. It suffices to analyze the zeroth associated piece:

*Remark 8.11.* By 2-periodicity of  $k^{tC_p}$ , all  $\mathrm{gr}_{\mathrm{odd}}^i k^{tC_p}$ 's, as  $k^{\Phi_{C_p}}$ -modules, are isomorphic up to a shift. Thus to study associated graded pieces of odd filtered  $k^{tC_p}$ , it suffices to study  $\mathrm{gr}_{\mathrm{odd}}^0 k^{tC_p} \simeq \mathrm{gr}_{\mathrm{odd}}^i k^{tC_p}$ , thus the same for the conjugate filtration.

Since  $k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p = (\tau_{\geq 0} k^{tC_p})/\sigma$  is a perfect  $\tau_{\geq 0} k^{tC_p}$ -module in  $\mathrm{Sp}^{B(\mathbb{T}/C_p)}$ , we have

**Lemma 8.12.** Let  $k$  be a commutative  $\mathbb{F}_p$ -algebra. Then the natural transformation

$$(-)^{\eta_{\tau_{\geq 0}(k^{tC_p})}(\mathbb{T}/C_p)} \longrightarrow (-)^{\eta_{k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p}(\mathbb{T}/C_p)}$$

of functors  $\mathrm{Mod}_{k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p}(\mathrm{Sp}^{g^{\mathbb{T}}}) \rightarrow D(k)$ , induced by the map  $\tau_{\geq 0}(k^{tC_p}) \rightarrow k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$  in  $\mathrm{CAlg}(D(k)^{B(\mathbb{T}/C_p)})$ , is an equivalence (we tacitly used Remark 2.4).

Similarly, by perfectness of the  $k$ -module  $k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$ , we have

**Lemma 8.13.** Let  $k$  be a commutative  $\mathbb{F}_p$ -algebra. Then the natural transformation

$$(-)^{\eta_k(\mathbb{T}/C_p)} \longrightarrow (-)^{\eta_{k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p}(\mathbb{T}/C_p)}$$

of functors  $\mathrm{Mod}_{k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p}(\mathrm{Sp}^{g^{\mathbb{T}}}) \rightarrow D(k)$ , induced by the map  $k \rightarrow k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$  in  $\mathrm{CAlg}(D(k)^{B(\mathbb{T}/C_p)})$ , is an equivalence (we tacitly used Remark 2.4).

It follows from Remark 8.8 and Lemmas 8.12 and 8.13 that

**Lemma 8.14.** Let  $k$  be a commutative  $\mathbb{F}_p$ -algebra, and  $M$  a  $\mathrm{THH}(k)$ -module in  $\mathrm{Sp}^{g^{\leq \mathbb{T}}}$ . Then the 0-th associated graded piece  $\mathrm{gr}_{\mathrm{conj}}^0(M \otimes_{\mathrm{THH}(k)}^{\mathbb{L}} \underline{k})^{\theta_{\mathbb{Z}^{\mathbb{T}}}}$  of the conjugate filtration is equivalent to

$$\left( \left( M^{\Phi_{C_p}} \otimes_{\mathrm{THH}(k)^{\Phi_{C_p}}}^{\mathbb{L}} (k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p) \right) \otimes_{k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p}^{\mathbb{L}} \mathrm{gr}_{\mathrm{odd}}^0 k^{tC_p} \right)^{\eta_k(\mathbb{T}/C_p)}$$

in  $D(k)$  (again, we tacitly used Remark 2.4).

Now we embark to understand the composite map  $\mathrm{THH}(k) \xrightarrow{\simeq} \mathrm{THH}(k)^{\Phi_{C_p}} \rightarrow k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$  of  $\mathbb{T}$ - $\mathbb{E}_{\infty}$ -rings<sup>26</sup>. In fact, this could be understood for any commutative ring, not necessarily over  $\mathbb{F}_p$ .

*Remark 8.15.* Let  $k$  be an animated ring. By construction (cf. [NS18, §IV.2]), we have a commutative diagram

$$\begin{array}{ccccc} k & \longrightarrow & \mathrm{THH}(k) & & \\ \downarrow \simeq & & \downarrow \simeq & & \\ (N_e^{C_p} k)^{\Phi_{C_p}} & \longrightarrow & \mathrm{THH}(k)^{\Phi_{C_p}} & \longrightarrow & \underline{k}^{\Phi_{C_p}} \\ \downarrow & & \downarrow & & \downarrow \\ (k^{\otimes_{\mathrm{sp}}} k)^{tC_p} & \longrightarrow & \mathrm{THH}(k)^{tC_p} & \longrightarrow & k^{tC_p} \end{array}$$

<sup>26</sup>For our purposes, thanks to Remark 2.4, it suffices to understand the underlying  $\mathbb{T}$ -equivariant  $\mathbb{E}_{\infty}$ -ring. However, here we deduce slightly stronger results.

where the leftmost row are maps of  $\mathbb{E}_\infty$ -rings while the rest are  $\mathbb{T}$ - $\mathbb{E}_\infty$ -rings. The composite map  $k \rightarrow k^{tC_p}$  from the top left to the bottom right is the Tate-valued Frobenius.

*Remark 8.16.* Let  $k$  be an animated ring. Note that we have a commutative diagram

$$\begin{array}{ccc} k & \longrightarrow & \underline{k}^{\Phi C_p} \\ \downarrow & & \downarrow \\ \underline{k} & \xrightarrow{\tilde{\varphi}_k} & \underline{k} \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \end{array}$$

in  $\mathrm{CAlg}(D(\mathbb{Z}))$ , where the top arrow is found in Remark 8.15, the bottom arrow is induced by the “extended” Frobenius map  $\tilde{\varphi}_k : k \rightarrow k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \rightarrow k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$  of animated rings, and except the top left  $k$ , other terms are  $\mathbb{T}$ - $\mathbb{E}_\infty$ -rings, and the arrows between  $\mathbb{T}$ - $\mathbb{E}_\infty$ -rings have  $\mathbb{T}$ - $\mathbb{E}_\infty$ -structures. By the universal property of  $\mathrm{THH}$  in [ABG+18], we get a commutative diagram

$$\begin{array}{ccc} \mathrm{THH}(k) & \longrightarrow & \underline{k}^{\Phi C_p} \\ \downarrow & & \downarrow \\ \underline{k} & \xrightarrow{\tilde{\varphi}_k} & \underline{k} \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \end{array}$$

of  $\mathbb{T}$ - $\mathbb{E}_\infty$ -rings<sup>27</sup>.

So far, we have analyzed the part  $M^{\Phi C_p} \otimes_{\mathrm{THH}(k)^{\Phi C_p}}^{\mathbb{L}} (k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p)$  of the expression in Lemma 8.14. Now we analyze the part  $((-) \otimes_{k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p}^{\mathbb{L}} \mathrm{gr}_{\mathrm{odd}}^0 k^{tC_p})^{\eta_k(\mathbb{T}/C_p)}$  for commutative  $\mathbb{F}_p$ -algebras  $k$ . It follows from Corollary 8.5 that this simplifies to  $(-) \otimes_{k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p}^{\mathbb{L}} k$ , where the  $k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p$ -module structure on  $k$  is simply given by the multiplication map  $k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \rightarrow k$ . Recall that, for animated  $\mathbb{F}_p$ -algebras  $k$ , the composite map

$$k \xrightarrow{\tilde{\varphi}_k} k \otimes_{\mathbb{Z}}^{\mathbb{L}} \mathbb{F}_p \longrightarrow k$$

coincides with the usual Frobenius map  $\varphi_k : k \rightarrow k$ . The above discussion implies that

**Proposition 8.17.** *Let  $k$  be a commutative  $\mathbb{F}_p$ -algebra, and  $M$  a  $\mathrm{THH}(k)$ -module in  $\mathrm{Sp}^{g^{<\mathbb{T}}}$ . Then the 0-th associated graded piece  $\mathrm{gr}_{\mathrm{conj}}^0(M \otimes_{\mathrm{THH}(k)}^{\mathbb{L}} \underline{k})^{\theta_k \mathbb{T}}$  of the conjugate filtration is equivalent to the Frobenius twist*

$$(M^{\Phi C_p} \otimes_{\mathrm{THH}(k)}^{\mathbb{L}} k) \otimes_{k, \varphi_k}^{\mathbb{L}} k$$

in  $D(k)$ , where the  $\mathrm{THH}(k)$ -module structure on  $M^{\Phi C_p}$  is induced by the equivalence  $\mathrm{THH}(k) \xrightarrow{\sim} \mathrm{THH}(k)^{\Phi C_p}$ .

**Corollary 8.18.** *Let  $k$  be a commutative  $\mathbb{F}_p$ -algebra, and  $\mathcal{C}$  a dualizable presentable stable  $\mathbb{F}_p$ -linear  $\infty$ -category. Then the 0-th associated graded piece  $\mathrm{gr}_{\mathrm{conj}}^0 \mathrm{HP}^{\mathrm{poly}}(\mathcal{C}/k)$  is equivalent to  $\mathrm{HH}(\mathcal{C}/k) \otimes_{k, \varphi_k}^{\mathbb{L}} k$  in  $D(k)$ .*

Finally, we show that the conjugate filtration is complete. It suffices to establish the following connectivity result.

**Lemma 8.19.** *Let  $k$  be a commutative  $\mathbb{F}_p$ -algebra, and  $M$  a  $\mathrm{THH}(k)^{\Phi C_p}$ -module in  $\mathrm{Sp}^{g^{<(\mathbb{T}/C_p)}}$ . Suppose that  $M$  is connective. Then the spectrum*

$$\left( M \otimes_{\mathrm{THH}(k)^{\Phi C_p}}^{\mathbb{L}} \tau_{\geq -1} k^{tC_p} \right)^{\eta_{\tau_{\geq 0}(k^{tC_p})}(\mathbb{T}/C_p)}$$

is connective.

<sup>27</sup>Those who are not familiar with  $\mathbb{T}$ - $\mathbb{E}_\infty$ -rings can simply replace them by  $\mathbb{T}$ -equivariant  $\mathbb{E}_\infty$ -rings, and the  $\mathrm{THH}$  also has an analogous universal property by McClure–Schwänzl–Vogt, cf. [NS18, Prop IV.2.2].

Note that  $k^{tC_p}$  is co-induced in  $D(\mathbb{Z})$ . Although it might not be compatible with the  $\mathrm{THH}(k)^{\Phi_{C_p}}$ -module structure, this is already enough for us to prove the connectivity.

*Proof.* By the bar resolution of the relative tensor product, and the fact that the de-completed Tate construction preserves small colimits, it suffices to show that, for every  $n \in \mathbb{N}$ , the spectrum

$$(M \otimes_{\mathbb{Z}}^{\mathbb{L}} (\mathrm{THH}(k)^{\Phi_{C_p}})^{\otimes_{\mathbb{Z}}^{\mathbb{L}} n} \otimes_{\mathbb{Z}}^{\mathbb{L}} \tau_{\geq -1} k^{tC_p})^{\eta_{\tau_{\geq 0}(k^{tC_p})}(\mathbb{T}/C_p)}$$

is connective. We write  $k$  as a direct sum  $\mathbb{F}_p^{\oplus I}$ , and applying Corollary 7.3, we see that this spectrum is equivalent to the direct sum of  $I$ 's copies of

$$M \otimes_{\mathbb{Z}}^{\mathbb{L}} (\mathrm{THH}(k)^{\Phi_{C_p}})^{\otimes_{\mathbb{Z}}^{\mathbb{L}} n} \otimes_{\mathbb{Z}}^{\mathbb{L}} \tau_{\geq 0} \mathbb{Z}^{tC_p}$$

which is connective.  $\square$

By 2-periodicity of  $k^{tC_p}$ , and the fact that  $(-)^{\Phi_{C_p}}$  preserves connectivity, we deduce from Lemma 8.19 that

**Corollary 8.20.** *Let  $k$  be a commutative  $\mathbb{F}_p$ -algebra, and  $M$  a connective  $\mathrm{THH}(k)$ -module in  $\mathrm{Sp}^{g < \mathbb{T}}$ . Then the conjugate filtration on  $(M \otimes_{\mathrm{THH}(k)}^{\mathbb{L}} k)^{\eta_{\mathbb{E}^{\mathbb{T}}}}$  is complete.*

**Corollary 8.21.** *Let  $k$  be a commutative  $\mathbb{F}_p$ -algebra, and  $\mathcal{C}$  a dualizable presentable stable  $\mathbb{F}_p$ -linear  $\infty$ -category. Suppose that its topological Hochschild homology  $\mathrm{THH}(\mathcal{C})$  is bounded below<sup>28</sup>. Then the conjugate filtration on its polynomial periodic cyclic homology  $\mathrm{HP}^{\mathrm{poly}}(\mathcal{C}/k)$  is complete.*

## APPENDIX A. TATE COHOMOLOGY COMPLEX

We briefly show that our de-completed Tate cohomology on Lazard-semi-flat chain complexes of cohomological Mackey functors can be computed by *Tate cohomology complex* in [Kal15, §6.2], or [PVV18, §2.1], thus it is a homotopy invariant version of the latter.

Let  $k$  be a commutative ring. Recall that a *finitely generated permutation  $G$ -module* is a (left)  $k[G]$ -module of the form  $k[X]$  for some *finite*  $G$ -set  $X$ . We denote by  $\mathrm{Perm}_G(k) \subseteq \mathrm{LMod}_{k[G]}$  the full subcategory spanned by finitely generated permutation  $G$ -modules. We refer to [BG21] for comparison of different characterizations of cohomological Mackey functors, and [BCN21, Ex 2.5] for the equivalence  $D(\mathrm{Mack}_G^{\mathrm{coh}}(k)) = \mathrm{Mod}_{\mathbb{E}}(\mathrm{Sp}^{gG})$  between derived cohomological  $G$ -Mackey functors and  $\mathbb{E}$ -modules in genuine  $G$ -spectra.

Now we start with a flatness of cohomological Mackey functors.

**Definition A.1.** Let  $k$  be a commutative ring, and  $G$  a finite group.

- We say that a cohomological Mackey functor is *Lazard-flat* if it is a filtered colimit of finitely generated permutation modules.
- We say that an (unbounded) chain complex of cohomological Mackey functors is *Lazard-semi-flat* if it is a filtered colimit of bounded chain complexes of finitely generated permutation modules. Compare with [CH15, Thm 1.1].

*Remark A.2.* Let  $k$  be a commutative ring, and  $G$  a finite group. Then permutation  $k[G]$ -modules are compact in the category  $\mathrm{Mack}_G^{\mathrm{coh}}(k)$  of cohomological  $G$ -Mackey functors. Consequently, we have a canonical fully faithful functor

$$\mathrm{Ind}(\mathrm{Perm}_G) \hookrightarrow \mathrm{Mack}_G^{\mathrm{coh}}(k).$$

<sup>28</sup>This is the case when  $\mathcal{C} = D(R)$  for a  $(-1)$ -connective  $\mathbb{E}_1$ - $k$ -algebra  $R$ , or  $\mathcal{C} = D(X)$  for a quasicompact quasiseparated  $k$ -scheme  $X$ .

In particular, we can identify Lazard-flat (derived) cohomological Mackey functors as objects of  $\text{Ind}(\text{Perm}_G(k))$ .

Similarly, recall that a chain complex in an additive category is compact if and only if it is bounded and degreewise compact [CH15, Thm 4.5]. The preceding argument shows that we have a canonical fully faithful functor

$$\text{Ind}(\text{Ch}^b(\text{Perm}_G(k))) \hookrightarrow \text{Ch}(\text{Mack}_G^{\text{coh}}(k))$$

and thus we can identify Lazard-semi-flat chain complexes of cohomological Mackey functors as objects of  $\text{Ind}(\text{Ch}^b(\text{Perm}_G(k)))$ .

Lazard-flat cohomological Mackey functors (resp. Lazard-semi-flat chain complexes of cohomological Mackey functors) are in fact  $k[G]$ -modules (resp. chain complexes of  $k[G]$ -modules):

*Remark A.3.* Let  $k$  be a commutative ring, and  $G$  a finite group. Then finitely generated permutation  $k[G]$ -modules are compact in the category  $\text{LMod}_{k[G]}$  of left  $G$ -modules, thus the fully faithful functor

$$\text{Ind}(\text{Perm}_G(k)) \hookrightarrow \text{Mack}_G^{\text{coh}}(k)$$

in Remark A.2 factors through the inclusion  $\text{LMod}_{k[G]} \hookrightarrow \text{Mack}_G^{\text{coh}}(k)$ , and the fully faithful functor

$$\text{Ind}(\text{Ch}^b(\text{Perm}_G(k))) \hookrightarrow \text{Ch}(\text{Mack}_G^{\text{coh}}(k))$$

factors through the inclusion  $\text{Ch}(\text{LMod}_{k[G]}) \hookrightarrow \text{Ch}(\text{Mack}_G^{\text{coh}}(k))$ .

Now we review the Tate cohomology complex. Let  $k$  be a commutative ring, and  $G$  a finite group. A *complete resolution* [PVV18, §2.1] of the left  $k[G]$ -module  $k$  is an acyclic complex  $P_* \in \text{Ch}(\text{LMod}_{k[G]}^{\text{free}})$  of free left  $k[G]$ -modules along with an isomorphism  $\varepsilon : \mathbb{Z} \rightarrow \ker(d : P_0 \rightarrow P_{-1})$  of left  $k[G]$ -modules.

**Definition A.4.** Let  $k$  be a commutative ring,  $G$  a finite group, and  $P_*$  a complete resolution of  $k$ . The functor of the *Tate cohomology complex* is defined to be the functor

$$\begin{aligned} \text{Ch}(\text{Mack}_G^{\text{coh}}(k)) &\longrightarrow \text{Ch}(k), \\ M_* &\longmapsto (\text{Tot}^\oplus(M_* \otimes_k P_*))^G. \end{aligned}$$

On Lazard-semi-flat chain complexes of cohomological Mackey functors, the Tate cohomology complex represents the de-completed Tate cohomology. More precisely, we have

**Proposition A.5.** *Let  $k$  be a commutative ring, and  $G$  a finite group. Then we have a commutative diagram*

$$\begin{array}{ccc} \text{Ind}(\text{Ch}^b(\text{Perm}_G(k))) & \longrightarrow & \text{Ch}(k) \\ \downarrow & & \downarrow \\ D(\text{Mack}_G^{\text{coh}}(k)) = \text{Mod}_{\underline{k}}(\text{Sp}^{gG}) & \xrightarrow{(-)^{\theta_{\underline{k}}^G}} & D(k) \end{array}$$

of additive  $\infty$ -categories, where the top horizontal arrow is the composite of the inclusion in Remark A.2 and the Tate cohomology complex, and the vertical arrows are localizations at quasi-isomorphisms.

**Sketch of proof** Since all functors in question preserve filtered colimits, we can replace the top left term  $\text{Ind}(\text{Ch}^b(\text{Perm}_G(k)))$  by  $\text{Ch}^b(\text{Perm}_G(k))$ . By stability, we could further restrict to the full subcategory  $\text{Ch}_{\geq 0}^b(\text{Perm}_G(k))$  of bounded chain complexes concentrated in non-negative degrees. This is the same as freely adjoining finitary-geometric-realizations to  $\text{Perm}_G(k)$ . All functors in question preserve finitary geometric realizations, thus we can replace the top left term simply by  $\text{Perm}_G(k)$ .

In this case, the left vertical arrow is fully faithful, with essential image being compact in  $\text{Mod}_{\underline{k}}(\text{Sp}^{gG})$ . Therefore we may replace  $(-)^{\theta_{\underline{k}}^G}$  by  $(-)^{tG}$ . It remains to show that, for every finitely generated permutation  $G$ -module  $M$ , the complex  $(M \otimes_k P_*)^G$  represents  $M^{tG}$ , which follows from definition.  $\square$

APPENDIX B. TATE CONSTRUCTION WITH RESPECT TO CYCLIC  $p$ -GROUPS

Recall that in [NS18, Lem II.4.1 & II.4.2], they showed that, for every  $t$ -bounded below  $\mathbb{T}$ -equivariant spectrum  $X$ , the canonical map

$$X^{t\mathbb{T}} \longrightarrow \lim_r X^{tC_{p^r}} \longrightarrow \lim_r (X^{tC_p})^{h(C_{p^r}/C_p)}$$

exhibits the target as the  $p$ -completion of the source. We need a variant of this, which allows us to establish the equivalence for non- $t$ -bounded objects which are linear over a  $t$ -bounded object. We start with two simple observations.

**Lemma B.1.** *Let  $r \in \mathbb{N}_{>0}$  be a natural number, and  $X \in D(\mathbb{Z})^{BC_{p^r}}$  a  $C_{p^r}$ -equivariant  $\mathbb{Z}$ -module spectrum. Then the canonical map*

$$X^{tC_{p^r}} \longrightarrow (X^{tC_p})^{h(C_{p^r}/C_p)}$$

*is an equivalence, and that  $X^{tC_{p^r}}$  is  $p$ -complete.*

*Proof.* As in the proof of [NS18, Lem II.4.1], we examine the map

$$\begin{array}{ccccc} X_{hC_{p^r}} & \longrightarrow & X^{hC_{p^r}} & \longrightarrow & X^{tC_{p^r}} \\ \downarrow & & \downarrow & & \downarrow \\ (X_{hC_p})^{h(C_{p^r}/C_p)} & \longrightarrow & (X^{hC_p})^{h(C_{p^r}/C_p)} & \longrightarrow & (X^{tC_p})^{hC_{p^r-1}} \end{array}$$

of horizontal fiber sequences, where the middle vertical map is an equivalence. The cofiber of the leftmost vertical map is  $(X_{hC_p})^{t(C_{p^r}/C_p)}$ , which is zero by (an iterated application of) the Tate orbit lemma for  $\mathbb{T}$ -equivariant  $\mathbb{Z}$ -module spectra, [NS18, the footnote in Ex I.2.3].

For the  $p$ -completeness of  $X^{tC_{p^r}}$ , by the first part, it suffices to show the  $p$ -completeness of  $X^{tC_p}$ . Note that it is a  $\mathbb{Z}^{tC_p}$ -module by lax symmetric monoidal structure on  $(-)^{tC_p}$  as in [NS18, §I.3], and we know that there is an equivalence  $\mathrm{THH}(\mathbb{F}_p) \simeq \tau_{\geq 0}(\mathbb{Z}^{tC_p})$  of  $\mathbb{E}_\infty$ -rings by [NS18, Cor IV.4.16]. This gives rise to an  $\mathbb{F}_p$ -module structure on the spectrum  $X^{tC_p}$  (which might not be compatible with the  $\mathbb{Z}$ -module structure), thus the spectrum  $X^{tC_p}$  is  $p$ -complete.  $\square$

**Lemma B.2.** *Let  $X \in D(\mathbb{Z})^{B\mathbb{T}}$  be a  $\mathbb{T}$ -equivariant  $\mathbb{Z}$ -module spectrum. Then the canonical map*

$$X^{t\mathbb{T}} \longrightarrow \lim_r X^{tC_{p^r}}$$

*exhibits the target as the  $p$ -completion of the source.*

*Proof.* The proof of [NS18, Lem IV.4.12] shows that this map, as a map of spectra, can be rewritten as

$$X^{t\mathbb{T}} \longrightarrow \lim_r (X^{t\mathbb{T}}/p^r)$$

and the result follows immediately.  $\square$

Now we extend these result to modules over a  $t$ -bounded ring.

**Proposition B.3.** *Let  $V \in \mathrm{Sp}^\omega$  be a finite spectrum,  $r \in \mathbb{N}_{>0}$  a natural number, and  $A \in \mathrm{Sp}^{BC_{p^r}}$  a  $C_{p^r}$ -equivariant  $\mathbb{E}_1$ -ring such that  $A \otimes V$  is  $t$ -bounded above, and  $M \in \mathrm{RMod}_A(\mathrm{Sp}^{BC_{p^r}})$  a  $C_{p^r}$ -equivariant right  $A$ -module. Then the canonical map*

$$M^{tC_{p^r}} \longrightarrow (M^{tC_p})^{h(C_{p^r}/C_p)}$$

*is an equivalence after Bousfield  $V$ -localization, and  $M^{tC_{p^r}}$  is  $p$ -complete after Bousfield  $V$ -localization.*

*Proof.* First, we examine the fiber sequence

$$(\tau_{\geq 0}A) \otimes V \longrightarrow A \otimes V \longrightarrow (\tau_{< 0}A) \otimes V.$$

The second term is  $t$ -bounded above by assumption, and so is the third term by finiteness of  $V$ . It follows that the first term is  $t$ -bounded above and thus  $t$ -bounded. We may replace  $A$  by  $\tau_{\geq 0}A$ , and without loss of generality, we may assume that  $A$  is connective.

Let  $\mathcal{F} \subseteq \mathrm{Sp}^{BC_{p^r}}$  be the collection of  $X \in \mathrm{Sp}^{BC_{p^r}}$  such that the canonical map

$$X^{tC_{p^r}} \longrightarrow (X^{tC_p})^{h(C_{p^r}/C_p)}$$

is an equivalence, and that  $M^{tC_{p^r}}$  is  $p$ -complete. Since functors  $(-)^{tC_{p^r}}$ ,  $((-)^{tC_p})^{h(C_{p^r}/C_p)}$ , and  $(-)_p^\wedge$  are exact, the collection  $\mathcal{F}$  is closed under taking cofibers, translations, and retracts. By Lemma B.1, the collection  $\mathcal{F}$  contains every  $C_{p^r}$ -equivariant spectrum which admits a  $\mathbb{Z}$ -module structure.

We now show that  $M \otimes V \in \mathcal{F}$ . As in the proof of Lemma 5.20, the  $C_{p^r}$ -equivariant spectrum  $M \otimes V$  is a retract of  $M \otimes A \otimes V$ , thus it suffices to show that  $M \otimes A \otimes V \in \mathcal{F}$ . Now  $A \otimes V$  is  $t$ -bounded, thus it is a finite extension of  $C_{p^r}$ -equivariant  $\mathbb{Z}$ -modules. The result then follows.  $\square$

**Proposition B.4.** *Let  $V \in \mathrm{Sp}^\omega$  be a finite spectrum, and  $A \in \mathrm{Sp}^{B\mathbb{T}}$  a  $\mathbb{T}$ -equivariant  $\mathbb{E}_1$ -ring such that  $A \otimes V$  is  $t$ -bounded above, and  $M \in \mathrm{RMod}_A(\mathrm{Sp}^{B\mathbb{T}})$  a  $\mathbb{T}$ -equivariant right  $A$ -module. Then the canonical map*

$$M^{t\mathbb{T}} \longrightarrow \lim_r M^{tC_{p^r}}$$

*is exhibits the target as a  $p$ -completion of the source after Bousfield  $V$ -localization.*

*Proof.* As in the proof of Proposition B.3, we may assume that  $A$  is connective. Moreover, by Proposition B.3, the limit  $\lim_r (M \otimes V)^{tC_{p^r}}$  is  $p$ -complete. Let  $\mathcal{F} \subseteq \mathrm{Sp}^{B\mathbb{T}}$  be the collection of  $X \in \mathrm{Sp}^{B\mathbb{T}}$  such that the canonical map

$$(M \otimes V)^{t\mathbb{T}}/p \longrightarrow \lim_r (M \otimes V)^{tC_{p^r}}/p$$

is an equivalence. Then  $\mathcal{F}$  is closed under taking cofibers, translations, and retracts. By Lemma B.2, the collection  $\mathcal{F}$  contains every  $\mathbb{T}$ -equivariant spectrum which admits a  $\mathbb{Z}$ -module structure.

Now by assumption, the  $\mathbb{T}$ -equivariant spectrum  $M \otimes A \otimes V$  is a finite extension of  $\mathbb{T}$ -equivariant  $\mathbb{Z}$ -modules, and as  $M \otimes V$  is a retract of  $M \otimes A \otimes V$ , we deduce that  $M \otimes V \in \mathcal{F}$ .  $\square$

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